

To my family.

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Chapter 1

Introduction

1.1 Topic and motivation

Generative diffusion models have become the dominant paradigm for synthesising high-dimensional data: they power state-of-the-art systems for images, audio, video, and molecular structures (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021; Karras et al., 2022). The same construction underlies the image and video generators deployed at scale by the large technology companies, from latent diffusion (Rombach et al., 2022) onwards, and reaches beyond media: structure prediction in AlphaFold 3 places atomic coordinates with a diffusion module (Abramson et al., 2024). Their central object is the *score function*, the gradient of the log-density of progressively noised data,

$$S(x, t) = \nabla_x \log p_t(x), \tag{1.1}$$

which is all a model needs to know in order to reverse a noising process and turn pure noise back into data (Anderson, 1982). In practice the score is approximated by a neural network trained by denoising score matching (Hyvärinen, 2005; Vincent, 2011); the network is a black box, and the structure of the function it has to learn remains poorly understood.

A complementary research programme, initiated in the statistical-physics community, replaces the black box with *exactly solvable models*: data distributions simple enough that

the score can be computed in closed form, yet rich enough to exhibit the phenomena observed in real systems, such as speciation, memorisation, and dynamical phase transitions along the generation trajectory (Biroli and Mézard, 2023; Biroli et al., 2024; Raya and Ambrogioni, 2023; Ghio et al., 2024; Achilli et al., 2026). Exact scores act as a microscope: every regime, transition time, and failure mode can be traced back to explicit formulas.

This thesis works inside that programme, in a direction that is natural for data with *temporal or sequential structure*: the data are not a cloud of independent points but a *trajectory* $a_0, a_1, \dots, a_{K-1}$ of a Markov process, the prototype of a video, a time series, or any dynamic object. Each frame of the trajectory is corrupted independently by an Ornstein–Uhlenbeck (OU) channel, and the object of study is the *joint* score of the whole noisy sequence. Two classical bodies of theory then meet the modern one. First, the joint score of a Gaussian Markov chain is governed by the algebra of structured matrices: Toeplitz covariances, tridiagonal precisions, and their common eigenbasis. Second, by an identity that holds for any data distribution, computing a score is solving a Bayesian inference problem, and for chain-structured data that problem lives on a tree-shaped factor graph, the domain of *belief propagation* (BP) and of the Kalman filter, together with the Gaussian message closures of statistical physics (TAP, AMP) (Pearl, 1988; Mézard and Montanari, 2009; Kalman, 1960; Donoho et al., 2009). The thesis develops both routes in full, proves where they coincide, and measures what is lost when they are forced apart.

1.2 Positioning and gap

The literature this work builds on can be organised in three strands; Chapters 2–5 review them in depth.

Statistical physics of learning machines. Machine learning and statistical mechanics share a genealogy that runs from spin glasses through the Hopfield network and the Boltzmann machine to today’s generative models, a lineage recognised by the 2024 Nobel Prize in Physics (Hopfield, 1982; Ackley et al., 1985; The Royal Swedish Academy of Sci-

ences, 2024). Its modern branch produced closed-form analyses of diffusion generation for Gaussian mixtures, ferromagnets, and high-dimensional asymptotics: the regimes of the backward dynamics and their transitions (Biroli and Mézard, 2023; Biroli et al., 2024), the general theory of speciation (Achilli et al., 2026; Sclocchi et al., 2025), the dynamics of memorisation and generalisation in training (Bonnaire et al., 2025; Garnier-Brun et al., 2026), and, on the training side, the demonstration that energy-based models learn through a cascade of phase transitions (Bachtis et al., 2024). These works treat the data as unstructured collections of samples; the time axis they study is the diffusion (or training) time. The internal time of a *sequence*, the fact that a video frame is correlated with its neighbours through a dynamical rule, is absent from these analyses.

Inference on chains. Exact posterior inference on Gauss–Markov chains is classical: the Kalman filter and smoother (Kalman, 1960; Rauch et al., 1965), hidden Markov models (Rabiner, 1989), and their unifying formulation as sum–product message passing on factor graphs (Kschischang et al., 2001; Mézard and Montanari, 2009). What this literature does not ask is the question that matters for generative modelling: how does the exact posterior, and therefore the exact score, deform as a function of the diffusion time, and what does that deformation imply for anything, algorithm or network, that must compute it at every noise level?

Message passing with continuous variables. BP is exact on trees for discrete variables, where messages are finite probability vectors. For continuous variables the messages become probability densities, and practical algorithms must close them within a parametric family, almost always the Gaussian one; this is the step that connects BP to AMP and to the TAP equations of statistical physics (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016; Genovese and Piana, 2026). The classical justification of the Gaussian closure is a central-limit argument valid when every node receives many messages. On a chain each node receives exactly two: the central-limit argument fails by construction, and we are not aware of a quantitative assessment of the Gaussian message

closure on chain-structured diffusion posteriors.

The gap, in one sentence: there is no end-to-end, exactly solvable account of the joint diffusion score of a structured sequence, covering its matrix structure, its message-passing computation, and the price of the Gaussian closure when the sequence is not Gaussian.

1.3 Research questions

The thesis addresses four questions.

- RQ1.** For a stationary Gaussian AR(1) chain observed through coordinatewise Ornstein–Uhlenbeck noise, what is the joint score, and how do its spatial couplings vary with the diffusion time?
- RQ2.** How can the same Gaussian-chain score be computed by belief propagation, and how does the resulting recursion relate to Kalman filtering and smoothing?
- RQ3.** For selected non-Gaussian innovation models, how accurate is a Gaussian-message approximation relative to a validated numerical belief-propagation reference, over the parameter ranges studied?
- RQ4.** In the Gaussian model, how rapidly does the influence of remote observations on the score decay, and what limited guidance does this provide for local approximations of the score?

These questions are deliberately confined to the models named in them. No claim is made about general Markov processes, about universal properties of learned diffusion scores, or about practical neural architectures.

1.4 Contributions

1. For the stationary Gaussian AR(1) chain under OU corruption, we derive the joint score in closed form, diagonalise it for all diffusion times in a single eigenbasis,

and characterise the evolution of its coupling structure: a band-fill law $(Q_t)_{i,i+d} = (-1)^{d-1}(2t)^{d-1}(Q_0^d)_{i,i+d} + O(t^d)$ for small t , a return to isotropy at rate e^{-2t} , and exact bulk formulas for the posterior variance and the geometric decay rate $q(\alpha, t)$ of posterior correlations. The band-fill law corrects a prefactor error in an earlier working note of this project; the correction is verified numerically (Chapter 7).

2. We derive the belief-propagation computation of the same score on the chain's factor graph, show algebraically that Gaussian messages are closed under the updates, identify the resulting recursion, update by update, with the Kalman filter and the Rauch–Tung–Striebel smoother, and verify that it reproduces the matrix score to within double-precision rounding across a parameter grid (Chapter 7).
3. For Laplace-innovation chains we establish which identities remain exact, locate the precise step at which closed forms stop, and measure the accuracy of Gaussian-projected BP against a numerically validated grid implementation of the functional recursion: the median relative score error is 0.39 at $t = 0.02$ and decays below 10^{-3} for $t \gtrsim 0.9$, with the score direction substantially more accurate than its magnitude, in the tested regimes (Chapter 8).
4. In the Gaussian model we prove that the error of a radius- r local estimator of the score decays as $q(\alpha, t)^r$, and we measure that truncating the inference (windowed message passing) is more accurate than truncating the score matrix (banding) at small diffusion times (Chapter 7).
5. As a separate, self-contained result, we characterise the per-node (AMP/TAP) closure on the same model: it returns the exact score but a different posterior variance, with an existence boundary in closed form, a breakdown time $t_c(\alpha)$, and a critical coupling $\alpha_c = \sqrt{2} - 1$. All formulas are independently re-verified; the material is placed in Appendix B because it branches off the main line (statement of results in Section 10.1).

1.5 Methodology

The methodology is deliberate: no neural networks, no black boxes; closed forms first, numerics as audit. Every claim is either proved by explicit algebra (with the longer passages isolated in numbered *toolboxes*) or verified by deterministic numerical checks against ground truth computed by an independent method, in the spirit of the exactly solvable programme of Biroli and Mézard (2023). Where an approximation is introduced (the Gaussian closure of BP messages, local truncations), its error is *measured* against an exact reference (matrix algebra, or dense grid quadrature validated on the solvable case). All numbers in the thesis are regenerated by deterministic audit scripts, documented in Appendix C; we regard this reproducibility as a strength of the method, not as a contribution in itself.

1.6 Scope and limitations

The exact results concern one-dimensional state per frame, linear AR(1) dynamics, and stationary initial conditions. The non-Gaussian results are numerical, valid within the validated resolution regime of the reference solver and the tested parameter ranges. The locality results are proved for the Gaussian bulk only. No learning experiments are performed, so any implication for trained score networks is a hypothesis suggested by a controlled model, not an established property of practical systems.

1.7 Structure of the thesis

Chapters 2–5 contain the background and literature review; Chapters 6–8 contain the research. Throughout, each concept is developed once, in the chapter where it naturally belongs, and then used without repetition.

Chapter 2 develops the statistical mechanics this thesis uses (phase space and Liouville’s theorem, the Boltzmann–Gibbs distribution, free energy, the Ising model) and then follows the discipline into machine learning: spin glasses, the Hopfield network, the Boltz-

mann machine and its restricted variant, and the statistical mechanics of learning. Chapter 3 supplies the probabilistic dynamics: stochastic processes and stationarity, Markov chains, the AR(1) running example, sampling by dynamics, the Itô calculus, the Ornstein–Uhlenbeck corruption channel, the Fokker–Planck equation, and Anderson’s time-reversal theorem. Chapter 4 reviews modern generative modelling: energy-based models in LeCun’s formulation, score matching, the DDPM construction, the score-SDE synthesis, flow models, the score–posterior identity, the statistical-physics analyses of diffusion, and the cascade of phase transitions in energy-based-model training. Chapter 5 builds the inference machinery: the multivariate Gaussian and its precision-matrix reading, the spectral theorem, Markov random fields, factor graphs, the sum–product algorithm, Gaussian BP and the Kalman filter, the TAP/AMP closures, and the architectural mirror of locality in convolutional networks and U-Nets.

Chapter 6 records the preliminary work: the toy models, the first supervision models, and the corrections that produced the final research questions. Chapter 7 solves the Gaussian chain exactly (RQ1, RQ2, and the Gaussian half of RQ4), with the spectral form of the precision matrix as the organising object and belief propagation identified with the Kalman smoother. Chapter 8 leaves Gaussianity in the most controlled way possible, Laplace innovations, and measures what the Gaussian message family costs when the data are no longer Gaussian (RQ3). Chapter 10 states findings per research question, limitations, and the questions that follow. Three appendices collect the Gaussian identities with proofs, the AMP/TAP analysis with its exact phase boundary, and the reproducibility protocol.

Chapter 2

Statistical Mechanics and the Origins of Learning Machines

Statistical mechanics is the discipline that explains how probabilistic laws emerge from deterministic dynamics. It is also the field in which the neural network was invented. This chapter serves both facts, and its two halves have distinct jobs. The first half builds the small set of physical tools that the rest of the thesis uses (the Boltzmann distribution, the partition function and free energy, and the Ising energy), with the derivations carried out in full: the Boltzmann distribution is the stationary law of Langevin dynamics (Chapter 3), the free energy is the variational principle behind the diffusion-model training objective (Chapter 4), and sampling by simulating a physical process is statistical mechanics turned into an algorithm. The second half is the part of the literature review that most histories of machine learning skip: how spin glasses became the Hopfield network, how the Hopfield network became the Boltzmann machine, and how the statistical mechanics of these systems grew into the modern theory of learning. This is the tradition, recognised by the 2024 Nobel Prize in Physics, in which this thesis works. The physical presentation follows classical references (Goldstein et al., 2002; Landau and Lifshitz, 1980).

2.1 Hamiltonian mechanics and phase space

Consider a mechanical system with n degrees of freedom, described by generalised coordinates $q = (q_1, \dots, q_n)$ and conjugate momenta $p = (p_1, \dots, p_n)$. Its dynamics is generated by a single scalar function, the *Hamiltonian* $H(q, p)$, which for the systems considered here has the mechanical form

$$H(q, p) = \underbrace{\sum_{i=1}^n \frac{p_i^2}{2m_i}}_{\text{kinetic}} + \underbrace{U(q_1, \dots, q_n)}_{\text{potential}}, \quad (2.1)$$

and equals the total energy of the system. The state of the system at any instant is the pair (q, p) , a point in the $2n$ -dimensional *phase space* $\Gamma \subseteq \mathbb{R}^{2n}$. The evolution is given by *Hamilton's equations*,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad i = 1, \dots, n. \quad (2.2)$$

Two structural properties of (2.2) are the pillars on which statistical mechanics rests.

Energy conservation. Along any trajectory,

$$\frac{dH}{dt} = \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) = \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0. \quad (2.3)$$

The motion is therefore confined to the *energy shell* $\Gamma_E = \{(q, p) : H(q, p) = E\}$, a $(2n - 1)$ -dimensional hypersurface.

Incompressibility of the phase flow. The phase-space velocity field

$$v(q, p) = (\dot{q}, \dot{p}) = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right) \quad (2.4)$$

has identically vanishing divergence:

$$\nabla \cdot v = \sum_{i=1}^n \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right) = \sum_{i=1}^n \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0. \quad (2.5)$$

Hamiltonian flow behaves like an incompressible fluid in phase space: it can stretch and fold a region, but never change its volume. This identity is the seed of the probabilistic construction that follows.

2.2 Liouville's theorem

A single trajectory is useless for macroscopic physics: for a gas, $n \sim 10^{23}$ and neither the initial condition nor the solution of (2.2) is accessible. The move that founds statistical mechanics is to describe our *ignorance* of the microstate by a probability density $\rho(q, p, t)$ on phase space: $\rho(q, p, t) dq dp$ is the probability that the system is in the infinitesimal cell around (q, p) at time t . Liouville's theorem states how this density evolves.

Theorem 2.1 (Liouville). *Under the Hamiltonian flow (2.2), the phase-space density obeys*

$$\frac{\partial \rho}{\partial t} = -\{\rho, H\} = \sum_{i=1}^n \left(\frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial \rho}{\partial q_i} \right), \quad (2.6)$$

where $\{\cdot, \cdot\}$ is the Poisson bracket. Equivalently, the density is constant along trajectories: $d\rho/dt = 0$.

Toolbox 2.1 — Proof of Liouville's theorem, step by step

The density is locally conserved (probability mass is neither created nor destroyed, only transported by the flow), so it satisfies the continuity equation familiar from fluid dynamics:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0, \quad v = (\dot{q}, \dot{p}).$$

Expand the divergence of the product using $\nabla \cdot (\rho v) = v \cdot \nabla \rho + \rho (\nabla \cdot v)$:

$$\frac{\partial \rho}{\partial t} + \underbrace{\sum_{i=1}^n \left(\dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right)}_{v \cdot \nabla \rho} + \rho \underbrace{\sum_{i=1}^n \left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} \right)}_{\nabla \cdot v} = 0.$$

By the incompressibility identity (2.5) the last term vanishes. Substituting Hamilton's equations $\dot{q}_i = \partial H / \partial p_i$, $\dot{p}_i = -\partial H / \partial q_i$ into the middle term:

$$\frac{\partial \rho}{\partial t} + \sum_{i=1}^n \left(\frac{\partial H}{\partial p_i} \frac{\partial \rho}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial \rho}{\partial p_i} \right) = 0 \quad \Longleftrightarrow \quad \frac{\partial \rho}{\partial t} = -\{\rho, H\}.$$

Finally, the total (convective) derivative along a trajectory is

$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + v \cdot \nabla \rho = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0,$$

which is the second form of the statement: an observer riding a phase-space trajectory sees a constant local density. $\square$

Liouville's theorem has an immediate consequence: *any density that is a function of conserved quantities only is stationary*. If $\rho(q, p) = f(H(q, p))$ for some function f , then $\{\rho, H\} = f'(H) \{H, H\} = 0$, so $\partial \rho / \partial t = 0$. Equilibrium statistical mechanics is the study of which f nature chooses.

2.3 From dynamics to ensembles

An *ensemble* is a probability distribution over microstates that is stationary under the dynamics and compatible with the macroscopic constraints we impose (fixed energy, fixed temperature, and so on). Two ensembles matter for this thesis.

2.3.1 The microcanonical ensemble

For a perfectly isolated system, the energy is fixed at E and Liouville's theorem suggests the simplest choice compatible with it: a uniform density on the energy shell,

$$\rho_{\text{mc}}(q, p) = \frac{1}{\Omega(E)} \delta(H(q, p) - E), \quad \Omega(E) = \int_{\Gamma} \delta(H(q, p) - E) \, dq \, dp, \quad (2.7)$$

where $\Omega(E)$ counts the microstates at energy E . The postulate that all accessible microstates are equally likely (the *equal a priori probability* postulate) is motivated, not proved, by ergodicity: for ergodic dynamics, time averages along a single trajectory converge to averages over the shell, so the uniform measure is the one a long experiment effectively samples. Boltzmann's entropy attaches a number to this count,

$$S(E) = k_B \log \Omega(E), \quad (2.8)$$

with k_B the Boltzmann constant. Entropy is the logarithm of the number of microscopic ways a macroscopic state can be realised.

2.3.2 The canonical ensemble and the Boltzmann distribution

Isolated systems are an idealisation; the physically ubiquitous situation is a system exchanging energy with a much larger environment (a *heat bath*) at temperature T . The stationary distribution of the system alone is then no longer uniform: it is the *Boltzmann–Gibbs distribution*,

$$\boxed{\rho_{\beta}(x) = \frac{1}{Z(\beta)} e^{-\beta H(x)}, \quad Z(\beta) = \int e^{-\beta H(x)} \, dx, \quad \beta = \frac{1}{k_B T},} \quad (2.9)$$

where $x = (q, p)$ collects the microstate and $Z(\beta)$, the *partition function*, normalises the density. Equation (2.9) can be derived in two independent ways, and both derivations are instructive enough to be carried out in full: the first (physical) traces the exponential to the entropy of the bath; the second (information-theoretic) characterises it as the least-

committal distribution compatible with a fixed mean energy, a perspective due to Jaynes (1957) that returns, almost verbatim, in modern machine learning.

Toolbox 2.2 — The Boltzmann distribution from the heat bath

Let the total system (system S together with bath B) be isolated with total energy E_{tot} , and let the coupling be weak, so $H_{\text{tot}} = H_S + H_B$. Apply the microcanonical postulate to the total system: the probability that S is in a *specific* microstate x with energy $H_S(x) = \epsilon$ is proportional to the number of bath microstates carrying the remaining energy,

$$\mathbb{P}(x) \propto \Omega_B(E_{\text{tot}} - \epsilon) = \exp\left[\frac{1}{k_B} S_B(E_{\text{tot}} - \epsilon)\right],$$

using the definition (2.8) of the bath entropy. Since the bath is huge, $\epsilon \ll E_{\text{tot}}$, expand S_B to first order:

$$S_B(E_{\text{tot}} - \epsilon) = S_B(E_{\text{tot}}) - \epsilon \left. \frac{\partial S_B}{\partial E} \right|_{E_{\text{tot}}} + O(\epsilon^2/E_{\text{tot}}).$$

The derivative of the bath entropy with respect to its energy *defines* the bath temperature, $\partial S_B/\partial E = 1/T$. Substituting,

$$\mathbb{P}(x) \propto \underbrace{e^{S_B(E_{\text{tot}})/k_B}}_{\text{constant in } x} \cdot e^{-\epsilon/(k_B T)} \propto e^{-\beta H_S(x)}.$$

The quadratic remainder is suppressed by the bath's heat capacity and vanishes in the thermodynamic limit. Normalising yields (2.9). Note what happened: the exponential form is nothing but the first-order Taylor expansion of the bath's entropy. The Boltzmann factor is the price, in bath microstates, of lending energy ϵ to the system.

□

Toolbox 2.3 — The Boltzmann distribution from maximum entropy

Forget physics; ask, with Jaynes (1957): among all densities ρ with a given mean energy $\mathbb{E}_\rho[H] = U$, which one assumes the least beyond the constraint? “Least” is measured by the Gibbs–Shannon entropy $S[\rho] = -k_B \int \rho \log \rho$. Maximise $S[\rho]$ subject to $\int \rho = 1$ and $\int \rho H = U$ with Lagrange multipliers λ_0, λ_1 :

$$\mathcal{L}[\rho] = -k_B \int \rho \log \rho \, dx - \lambda_0 \left(\int \rho \, dx - 1 \right) - \lambda_1 \left(\int \rho H \, dx - U \right).$$

Take the functional derivative and set it to zero pointwise:

$$\frac{\delta \mathcal{L}}{\delta \rho(x)} = -k_B (\log \rho(x) + 1) - \lambda_0 - \lambda_1 H(x) = 0 \implies \rho(x) = \exp \left[-1 - \frac{\lambda_0}{k_B} - \frac{\lambda_1}{k_B} H(x) \right].$$

The density is an exponential of the constrained quantity. This is completely general: constraining $\mathbb{E}[\phi_j(x)]$ for features ϕ_j produces $\rho \propto \exp[-\sum_j \beta_j \phi_j(x)]$, the exponential family. The multiplier λ_0 is fixed by normalisation and produces Z ; renaming $\lambda_1/k_B = \beta$ and identifying $\beta = 1/(k_B T)$ through thermodynamics recovers (2.9) exactly. Second-order check: $\delta^2 \mathcal{L} / \delta \rho^2 = -k_B / \rho < 0$, so the stationary point is the unique maximum (the entropy is strictly concave). $\square$

The second derivation deserves a remark. The Boltzmann distribution is not a statement about molecules but about *inference under constraints*. Any time we posit a model of the form $p(x) \propto e^{-E(x)}$, as we will for Markov random fields (Chapter 5) and energy-based models (Chapter 4), we are doing statistical mechanics, whether the variables are spins, pixels, or the frames of a video.

2.4 The partition function and free energy

The partition function $Z(\beta)$ in (2.9) looks like a mere normaliser; it is in fact the central computational object of the theory, and the reason approximate inference exists as a field.

Its logarithm defines the *Helmholtz free energy*,

$$F(\beta) = -\frac{1}{\beta} \log Z(\beta). \quad (2.10)$$

Toolbox 2.4 — Everything is a derivative of $\log Z$

Differentiate $\log Z$ with respect to β , carrying the derivative through the integral:

$$-\frac{\partial \log Z}{\partial \beta} = -\frac{1}{Z} \frac{\partial}{\partial \beta} \int e^{-\beta H} = \frac{\int H e^{-\beta H}}{\int e^{-\beta H}} = \mathbb{E}_{\rho_\beta}[H] = U,$$

the mean energy. Differentiate once more:

$$\begin{aligned} \frac{\partial^2 \log Z}{\partial \beta^2} &= \frac{\partial}{\partial \beta} \left(-\frac{\int H e^{-\beta H}}{Z} \right) = \frac{\int H^2 e^{-\beta H}}{Z} - \left(\frac{\int H e^{-\beta H}}{Z} \right)^2 \\ &= \mathbb{E}[H^2] - \mathbb{E}[H]^2 = \text{Var}(H) \geq 0. \end{aligned}$$

So $\log Z$ is convex in β , its first derivative gives the mean energy, its second the energy fluctuations; the latter equals $k_B T^2 C_V$ with C_V the heat capacity. This is a *fluctuation–dissipation* relation: how a system responds to heating is determined by its spontaneous equilibrium fluctuations.

The same identity holds for any observable coupled linearly to a field. If the Gibbs weight carries a factor $e^{h \cdot a}$, then $Z(h) = \int \nu(a) e^{h \cdot a} da$ is a cumulant generating function and

$$\frac{\partial \langle a_k \rangle}{\partial h_j} = \frac{\partial^2 \log Z}{\partial h_j \partial h_k} = \text{Cov}(a_k, a_j), \quad (2.11)$$

so a susceptibility is a covariance. Susceptibilities defined this way are the standard probe of phase transitions, and they are the instrument used by Bachtis et al. (2024) to locate the transitions discussed in Section 2.9. Chapter 9 shows that the response of a diffusion score to its own observations is exactly of the form (2.11), with the observation playing the role of the field. Equation (2.11) is a static identity within one fixed measure, and is not the two-time dynamical fluctuation–dissipation theorem of

non-equilibrium statistical mechanics. Finally, rewriting the Gibbs–Shannon entropy of ρ_β :

$$S[\rho_\beta] = -k_B \mathbb{E}[\log \rho_\beta] = -k_B \mathbb{E}[-\beta H - \log Z] = \frac{U}{T} + k_B \log Z,$$

and multiplying by $-T$: $-TS = -U + F$, i.e.

$$F = U - TS.$$

The free energy is the exact trade-off between energy and entropy: at low T minimising F means minimising energy (ordered states win), at high T it means maximising entropy (disordered states win). Phase transitions are the non-analyticities of F where the winner changes. □

There is one more characterisation of F worth recording, because it is the statistical-mechanics form of the variational principle that underlies both the training objective of diffusion models (Chapter 4) and the Bethe free energies of message passing. For *any* trial density μ ,

$$\underbrace{\mathbb{E}_\mu[H] - \frac{1}{\beta} S[\mu]/k_B}_{\text{variational free energy } F[\mu]} = F + \frac{1}{\beta} D_{\text{KL}}(\mu \parallel \rho_\beta) \geq F, \quad (2.12)$$

with equality iff $\mu = \rho_\beta$: the Boltzmann distribution is the unique minimiser of the variational free energy, and the gap is exactly a Kullback–Leibler divergence. Every variational method in machine learning is a restriction of (2.12) to a tractable family of trial densities.

2.5 Interacting systems: the Ising paradigm

Everything so far holds for arbitrary H ; the physics (and the difficulty) enters when the energy couples many degrees of freedom. The canonical example is the *Ising model* (Ising,

1925): binary spins $\sigma_i \in \{-1, +1\}$ on the nodes of a graph $G = (V, \mathcal{E})$, with energy

$$H(\sigma) = - \sum_{(i,j) \in \mathcal{E}} J_{ij} \sigma_i \sigma_j - \sum_{i \in V} h_i \sigma_i, \quad (2.13)$$

where J_{ij} are pairwise couplings and h_i external fields. The Gibbs distribution $p(\sigma) \propto e^{-\beta H(\sigma)}$ makes neighbouring spins prefer alignment when $J_{ij} > 0$. Three lessons from this model shape the rest of the thesis.

(i) The partition function is the hard part. $Z = \sum_{\sigma} e^{-\beta H(\sigma)}$ runs over $2^{|V|}$ configurations. On two-dimensional lattices Onsager computed it exactly (Onsager, 1944); on general graphs the computation is #P-hard, and all of approximate inference (Monte Carlo sampling, variational methods, message passing) exists because of this wall. On *trees*, however, Z and all marginals are computable exactly in time linear in the system size by the transfer-matrix recursion, the same recursion that Chapter 5 presents as belief propagation. This tractable island is precisely where our research model lives.

(ii) Collective order emerges. For β beyond a critical value the two-dimensional model magnetises spontaneously: a macroscopic fraction of spins align even at $h = 0$, and the symmetric state splits into two ergodic components. This spontaneous symmetry breaking is the archetype of a *phase transition*, and its dynamical counterpart, a generation trajectory committing to one mode of a multimodal distribution, is exactly the *speciation* phenomenon of diffusion models (Biroli et al., 2024; Raya and Ambrogioni, 2023).

(iii) Disorder creates complexity. Making the couplings J_{ij} random (of both signs) yields spin glasses, and, as the next sections recount, spin glasses yielded neural networks. This is where the physics of this chapter stops being background and becomes the literature this thesis belongs to.

2.6 Disorder, frustration, and spin glasses

In a ferromagnet all couplings agree, and the two ground states (all spins up, all spins down) are found by inspection. Draw the couplings J_{ij} at random with both signs (Edwards and Anderson, 1975; Sherrington and Kirkpatrick, 1975) and this transparency collapses. A triangle with two positive bonds and one negative bond already cannot satisfy all three: whatever the spins do, at least one bond pays. This is *frustration*, and its large- N consequence is a free-energy landscape shattered into exponentially many valleys separated by extensive barriers: the system has many near-equally-good ways of being, none of them readable off the couplings. The mean-field theory of this phenomenon, developed through the replica and cavity methods (Mézard et al., 1987), and the Thouless–Anderson–Palmer (TAP) self-consistency equations (Thouless et al., 1977), took a decade to control and earned Parisi the 2021 Nobel Prize in Physics.

Two exports from this episode matter here. First, a language: rugged landscapes, metastable states, order parameters that measure similarity rather than magnetisation. Machine learning speaks this language every time it discusses loss surfaces and local minima. Second, a technology: the cavity and TAP equations are recursions on *local* quantities that solve a *global* inference problem, and they are the direct ancestors of the message-passing algorithms (Mézard and Montanari, 2009; Zdeborová and Krzakala, 2016; Krzakala and Zdeborová, 2024) that Chapter 5 develops and that the research chapters deploy on diffusion scores. The connection between disordered magnets and these algorithms is historical descent, not analogy.

2.7 The Hopfield network: memory as a phase of matter

Hopfield (1982) asked a physicist’s question about memory: can a system of simple binary units *store* and *retrieve* patterns as a collective effect, the way a magnet stores a magnetisation? His construction is an Ising system (2.13) in disguise. Take N binary neurons

$\sigma_i \in \{-1, +1\}$ and P patterns $\xi^\mu \in \{-1, +1\}^N$ to be memorised, and wire the couplings by the *Hebb rule*, “neurons that fire together wire together” (Hebb, 1949):

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu, \quad H(\sigma) = -\frac{1}{2} \sum_{i \neq j} J_{ij} \sigma_i \sigma_j. \quad (2.14)$$

The dynamics is the simplest imaginable: pick a neuron, align it with its local field, repeat,

$$\sigma_i \leftarrow \text{sign} \left(\sum_{j \neq i} J_{ij} \sigma_j \right). \quad (2.15)$$

Toolbox 2.5 — The Hopfield dynamics descends the energy

Let $h_i = \sum_{j \neq i} J_{ij} \sigma_j$ be the local field on neuron i and suppose the update (2.15) flips it, $\sigma'_i = -\sigma_i$. Only the terms of H containing σ_i change, so

$$\Delta H = H(\sigma') - H(\sigma) = -(\sigma'_i - \sigma_i) h_i = -2 \sigma'_i h_i.$$

The update sets $\sigma'_i = \text{sign}(h_i)$, hence $\sigma'_i h_i = |h_i| \geq 0$ and $\Delta H \leq 0$, with strict decrease whenever a flip actually occurs. Since H is bounded below on a finite configuration space, the dynamics must terminate in a local minimum: *every* initial condition falls into an attractor. Memory retrieval is this fall: initialise at a corrupted pattern, descend, and read off the fixed point. $\square$

The patterns are, by construction, near the bottoms of the energy (2.14): retrieval works because each memory is an attractor with a basin. How many memories fit is a genuine phase-transition question, answered with spin-glass tools by Amit et al. (1985): for random patterns with $P = \alpha N$, retrieval survives only below a critical load $\alpha_c \approx 0.138$, beyond which the memories are crushed into a useless glassy phase. In this model, memory is a phase of matter in the technical sense: it is entered and left through sharp transitions. The single-pattern case $P = 1$, known as the Mattis model (Mattis, 1976), is gauge-equivalent to the uniform ferromagnet: flipping $\sigma_i \rightarrow \xi_i \sigma_i$ maps the stored pattern onto the all-up state. This observation is worth keeping in mind: the Mattis model returns

in Section 4.8 as the exactly solvable teacher in a modern analysis of how energy-based models learn.

In 2024 the Nobel Prize in Physics went to John Hopfield and Geoffrey Hinton “for foundational discoveries and inventions that enable machine learning with artificial neural networks” (The Royal Swedish Academy of Sciences, 2024): formal recognition that the lineage traced in this section is physics, not just inspired by it.

2.8 From Hopfield to Boltzmann machines: learning enters

The Hopfield network stores what its designer wires into it. The decisive step towards machine *learning* was to make the couplings themselves the unknowns. Promote the zero-temperature dynamics (2.15) to stochastic updates at temperature $T = 1/\beta$ (each neuron aligns with its field only with probability given by the Boltzmann factor); the stationary law of this dynamics is exactly the Gibbs measure $p(\sigma) \propto e^{-\beta H(\sigma)}$ of the Ising energy. A *Boltzmann machine* (Ackley et al., 1985) is this object run backwards: fix a dataset, and ask for the couplings J that make the Gibbs measure reproduce it. Gradient ascent on the log-likelihood gives a rule of striking symmetry,

$$\frac{\partial}{\partial J_{ij}} \mathbb{E}_{\text{data}}[\log p(\sigma)] = \beta \left(\underbrace{\langle \sigma_i \sigma_j \rangle_{\text{data}}}_{\text{positive phase}} - \underbrace{\langle \sigma_i \sigma_j \rangle_{\text{model}}}_{\text{negative phase}} \right), \quad (2.16)$$

and learning stops when the model’s correlations match the data’s. This is the maximum-entropy principle of Toolbox 2.3 turned into an algorithm: matching prescribed moments with an exponential-family measure. The rule also contains, in miniature, the central obstruction of the field. The positive phase is an average over data, which is cheap. The negative phase is an average over the *model’s* Gibbs measure: a partition-function computation, the wall of Section 2.5, approachable only through sampling (Section 3.4).

The *restricted* Boltzmann machine (RBM) tames this cost by architecture (Smolensky, 1986): split the units into a visible layer v and a hidden layer h with couplings

only across layers, $E(v, h) = -v^\top W h - b^\top v - c^\top h$. On this bipartite graph each layer is conditionally independent given the other, so Gibbs sampling alternates two exact, fully parallel steps, and the contrastive-divergence approximation to (2.16) made training practical (Hinton, 2002). Stacked RBMs, the deep belief networks, delivered the first working deep architectures and relaunched neural networks as a field (Hinton et al., 2006). The lineage then split. The supervised branch, powered by backpropagation (Rumelhart et al., 1986), scaled through convolutional networks to the modern deep-learning era (Krizhevsky et al., 2012; LeCun et al., 2015) and largely left equilibrium physics behind. The *generative* branch, the one this thesis lives in, kept the Gibbsian core: a model is an energy, learning is moment matching, sampling is dynamics. Chapter 4 follows that branch from energy-based models to diffusion.

2.9 The statistical mechanics of learning

Physics did not stop contributing machines; it also built the theory of their behaviour. Gardner (1988) showed that learning itself can be treated as an equilibrium problem: fix a set of examples and compute, by replica methods, the volume of coupling space compatible with them; capacity, generalisation, and their transitions then come out as thermodynamics in *weight* space. This programme, extended through teacher–student settings, matured into the modern statistical physics of inference (Zdeborová and Krzakala, 2016), whose central lesson recurs throughout this thesis: high-dimensional inference problems have phases, and algorithms fail or succeed at sharp boundaries.

Two strands of this literature frame our research questions directly. The first is algorithmic: Mézard (2017) derived the mean-field message-passing (TAP) equations of the Hopfield model and its generalisations, exhibiting the equivalence between Hopfield networks and RBMs and showing that retrieval can be organised as a message-passing computation. It is the same machinery, transplanted from memories to diffusion scores, that the research chapters run on our model. The second is dynamical: training itself undergoes phase transitions. In linear networks, gradient descent learns the singular modes

of the data covariance sequentially (Saxe et al., 2014); in RBMs, the weight matrix’s singular vectors align with the data’s principal components in the early phase of learning (Decelle et al., 2017), and hidden units develop compositional receptive fields through condensation transitions (Tubiana and Monasson, 2017). The sharpest result in this line, Bachtis et al. (2024), examined in depth in Section 4.8, shows that RBM training is an entire *cascade* of second-order phase transitions, one per learned feature. Learning, like memory, turns out to be a sequence of symmetry breakings.

The order of a transition is the order of the derivative of the free energy that first misbehaves, and the distinction matters for reading these results. At a *first-order* transition the first derivative jumps: the order parameter is discontinuous, latent heat is absorbed, the two phases coexist at the transition, and correlation lengths stay finite. At a *second-order* transition the first derivative is continuous and the second diverges: the order parameter grows continuously from zero, the susceptibility (2.11) and the correlation length diverge, and observables acquire power laws. The cascade above is of the second kind, which is why it is detected through susceptibilities rather than through jumps. Both notions require the thermodynamic limit; in the finite models of Chapters 7 to 9 a correlation length that grows until it reaches the size of the system and then stops is a finite-size ceiling, not a divergence.

2.10 What this thesis takes from physics

A *complex system* is, operationally, a system whose macroscopic behaviour is not readable off its microscopic rules: interactions generate collective phenomena (order, phases, transitions, slow dynamics) that must be derived, not postulated. Statistical mechanics is the toolbox for that derivation, and after this chapter’s story its objects map one-to-one onto the objects of modern generative modelling:

Statistical mechanics	Generative modelling
microstate x	data point / latent configuration
energy $H(x)$	negative log-density $-\log p(x)$ (up to Z)
Boltzmann distribution $e^{-\beta H}/Z$	model distribution
partition function Z	normalising constant / evidence
free energy F	negative ELBO (up to constants)
Hebbian couplings, attractors	stored memories, retrieval
moment matching (2.16)	maximum-likelihood training
Langevin dynamics	sampling / reverse diffusion
phase transition	feature condensation, speciation
cavity / TAP equations	belief propagation / AMP

The next chapter supplies the probabilistic dynamics, Markov chains and their continuous-time limits, that turn the static ensembles of this chapter into processes that move probability around, which is what a diffusion model ultimately is.

Chapter 3

Stochastic Processes, SDEs, and the Ornstein–Uhlenbeck Channel

Chapter 2 answered the question “which distribution?”: Boltzmann’s. This chapter answers “how does probability move?”, in discrete and in continuous time. Its material enters the thesis in three places. Markov chains are the *data* whose diffusion score we compute exactly (the AR(1) sequence of Section 3.3 is the running example of every research chapter); stochastic differential equations are the *corruption and generation processes* of diffusion models (the Ornstein–Uhlenbeck channel of Section 3.6 appears on every page of the research chapters); and the Fokker–Planck and time-reversal theorems of Sections 3.7–3.8 are the mathematical basis of the generative programme reviewed in Chapter 4. Standard references are Robert and Casella (2004); Brockwell and Davis (1991); Øksendal (2003); Gardiner (2009); Risken (1996).

3.1 Stochastic processes and stationarity

A *stochastic process* is a collection of random variables indexed by time, $(X_k)_{k \geq 0}$ in discrete time or $(X_t)_{t \geq 0}$ in continuous time, defined on a common probability space; equivalently, a probability law on the space of trajectories. Two notions of time homogeneity are used constantly in this thesis and are worth stating precisely (Brockwell and Davis, 1991).

Definition 3.1 (Strict and weak stationarity). A process (X_k) is *strictly stationary* if all its finite-dimensional laws are invariant under time shifts: for every n , every set of indices $k_1, \dots, k_n$, and every shift h ,

$$(X_{k_1}, \dots, X_{k_n}) \stackrel{d}{=} (X_{k_1+h}, \dots, X_{k_n+h}).$$

It is *weakly (second-order) stationary* if its mean is constant and its autocovariance depends only on the lag, $\text{Cov}(X_k, X_{k+d}) = \gamma(d)$.

Strict stationarity implies weak stationarity whenever second moments exist; for *Gaussian* processes the two notions coincide, because a Gaussian law is determined by its first two moments. Stationarity has an immediate matrix consequence that the research chapters exploit: the covariance matrix of K consecutive observations of a weakly stationary process, $(\Sigma)_{ij} = \gamma(i-j)$, is constant along diagonals, a symmetric *Toeplitz* matrix. Whenever a covariance in this thesis is Toeplitz, stationarity is the reason.

3.2 Markov chains: kernels, stationarity, reversibility

Definition 3.2 (Markov chain). A sequence of random variables $(X_k)_{k \geq 0}$ with values in a state space $\mathcal{X}$ is a *Markov chain* if for every k and every measurable $A \subseteq \mathcal{X}$,

$$\mathbb{P}(X_{k+1} \in A \mid X_k, X_{k-1}, \dots, X_0) = \mathbb{P}(X_{k+1} \in A \mid X_k). \quad (3.1)$$

The conditional law is encoded by a *transition kernel* $M(x' \mid x)$: a density (or matrix, for discrete $\mathcal{X}$) with $\int M(x' \mid x) dx' = 1$ for every x .

The Markov property (3.1) is exactly a statement about *factorisation*: the joint law of the first K states is

$$p(x_0, x_1, \dots, x_{K-1}) = p_0(x_0) \prod_{k=0}^{K-2} M(x_{k+1} \mid x_k). \quad (3.2)$$

This factorisation is the most heavily used equation of the thesis. It says that the joint distribution of a Markov trajectory is a product of *local* terms, each touching two consecutive variables: the structure that makes the chain a tree in the factor-graph sense of Chapter 5, that makes its precision matrix tridiagonal in the Gaussian case of Chapter 7, and that makes belief propagation exact.

Marginal distributions evolve by integrating the kernel: $p_{k+1}(x') = \int M(x'|x) p_k(x) dx$, compactly $p_{k+1} = Mp_k$. A distribution π is *stationary* (invariant) for M if $\pi = M\pi$; a chain started in π stays in π forever, and the trajectory is then a strictly stationary process in the sense of Definition 3.1. The kernel satisfies *detailed balance* with respect to π if

$$\pi(x) M(x' | x) = \pi(x') M(x | x') \quad \text{for all } x, x' : \quad (3.3)$$

the equilibrium probability flux from x to x' equals the reverse flux, so the chain, watched in equilibrium, is statistically indistinguishable run forwards or backwards. Detailed balance implies stationarity (integrate (3.3) over x), and its power is practical: it is a local, pointwise condition, checkable without computing any integral.

Stationarity alone does not guarantee that the chain *finds* π ; for that one needs the chain to be able to reach everywhere (*irreducibility*) without getting trapped in cycles (*aperiodicity*). Under these conditions the fundamental convergence theorem holds (Robert and Casella, 2004): the marginal law of X_k converges to π from any initial condition, and time averages converge to expectations under π (the ergodic theorem),

$$\frac{1}{N} \sum_{k=1}^N f(X_k) \xrightarrow[N \rightarrow \infty]{\text{a.s.}} \mathbb{E}_\pi[f]. \quad (3.4)$$

Equation (3.4) is the rigorous form of the ergodic postulate invoked in Section 2.3, with the physical dynamics replaced by an algorithmic one.

3.3 The AR(1) chain: the running example of the thesis

We now introduce the specific Markov chain that the research chapters solve completely: the first-order autoregressive (AR(1)) process, the canonical Gaussian Markov chain (Brockwell and Davis, 1991). Let $\alpha \in (-1, 1)$ and let (η_k) be i.i.d. $\mathcal{N}(0, \sigma_\eta^2)$ innovations; define

$$a_{k+1} = \alpha a_k + \eta_k, \quad k = 0, 1, \dots, K-2. \quad (3.5)$$

The transition kernel is Gaussian, $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, and the joint law of the trajectory follows from (3.2). Throughout the thesis we use the stationary normalisation $\sigma_\eta^2 = 1 - \alpha^2$ and start the chain in its stationary law, $a_0 \sim \mathcal{N}(0, 1)$, so that every frame has unit marginal variance. A trajectory $a = (a_0, \dots, a_{K-1})$ is *one* data point, the prototype of a video or a time series; its frame index k is an internal time, to be kept distinct from the diffusion time t introduced in Section 3.6. Everything about this chain is computable in closed form; the next toolbox derives the three facts used throughout.

Toolbox 3.1 — Stationary distribution and autocovariance of AR(1)

(i) *Stationary variance.* Suppose $a_k \sim \mathcal{N}(0, \sigma^2)$ and impose that a_{k+1} has the same variance. Since η_k is independent of a_k ,

$$\text{Var}(a_{k+1}) = \alpha^2 \text{Var}(a_k) + \sigma_\eta^2 \stackrel{!}{=} \text{Var}(a_k) \implies \sigma_\infty^2 = \frac{\sigma_\eta^2}{1 - \alpha^2}.$$

The fixed point exists precisely because $|\alpha| < 1$; the map $\sigma^2 \mapsto \alpha^2 \sigma^2 + \sigma_\eta^2$ is a contraction, so *any* initial variance converges to σ_∞^2 geometrically. Since Gaussianity is preserved by the affine recursion, the stationary law is $\pi = \mathcal{N}(0, \sigma_\infty^2)$, which equals $\mathcal{N}(0, 1)$ under the stationary normalisation.

(ii) *Autocovariance.* In the stationary regime, for $d \geq 1$, unroll the recursion d times: $a_{k+d} = \alpha^d a_k + \sum_{j=0}^{d-1} \alpha^j \eta_{k+d-1-j}$. All innovation terms are independent of a_k , hence

$$\text{Cov}(a_k, a_{k+d}) = \alpha^d \text{Var}(a_k) = \alpha^d \sigma_\infty^2,$$

and by symmetry $\text{Cov}(a_i, a_j) = \alpha^{|i-j|} \sigma_\infty^2$ for all i, j : correlations decay *geometrically* with the lag, with rate α . The covariance matrix of K consecutive frames,

$$(\Sigma_0)_{ij} = \sigma_\infty^2 \alpha^{|i-j|},$$

is symmetric Toeplitz, as stationarity requires (Section 3.1).

(iii) *Detailed balance.* With $\pi = \mathcal{N}(0, \sigma_\infty^2)$ and $M(a'|a) = \mathcal{N}(a'; \alpha a, \sigma_\eta^2)$, both sides of (3.3) are proportional to

$$\exp \left[-\frac{a^2}{2\sigma_\infty^2} - \frac{(a' - \alpha a)^2}{2\sigma_\eta^2} \right] = \exp \left[-\frac{a^2 + (a')^2 - 2\alpha a a'}{2\sigma_\eta^2} \right],$$

where the exponent on the right is *symmetric* in (a, a') . Hence the stationary AR(1) chain is reversible: statistically, the movie of the chain looks the same played in either direction. □

The AR(1) chain is thus a working miniature of everything this chapter discusses: a Markov kernel with an explicit stationary law, geometric convergence, reversibility, and, decisive for us, exact Gaussian algebra at every step.

3.4 Sampling by dynamics: MCMC and Langevin

Sampling inverts the logic of the previous sections. There, a kernel M was given and we asked for its stationary law π . In sampling problems we are given π , typically a Boltzmann distribution $\pi(x) \propto e^{-\beta H(x)}$ known only up to its partition function, and we must *design* a kernel whose stationary law is π ; the ergodic theorem (3.4) then converts one long

trajectory into expectation values. This is how the negative phase of Boltzmann-machine learning (2.16) is estimated in practice, and the idea originates, fittingly, in computational statistical mechanics (Metropolis et al., 1953).

The *Metropolis–Hastings* chain (Metropolis et al., 1953; Hastings, 1970) moves from x by proposing $x' \sim g(\cdot|x)$ from any tractable proposal kernel and accepting the move with probability

$$A(x \rightarrow x') = \min \left\{ 1, \frac{\pi(x') g(x|x')}{\pi(x) g(x'|x)} \right\}, \quad (3.6)$$

staying at x otherwise. The target enters only through the *ratio* $\pi(x')/\pi(x)$, so the intractable partition function cancels: Metropolis–Hastings samples from $e^{-\beta H}/Z$ while only ever evaluating energy differences. The acceptance rule is engineered so that the resulting kernel satisfies detailed balance (3.3) with respect to π ; the verification is a short computation (Robert and Casella, 2004, Chapter 7). When x is high-dimensional but each conditional $\pi(x_i|x_{\setminus i})$ is tractable, the *Gibbs sampler* (Geman and Geman, 1984) cycles through the coordinates resampling each from its conditional, a Metropolis–Hastings move whose acceptance probability is identically one. Gibbs sampling thrives exactly when the model has sparse conditional structure: for a Markov chain, $\pi(x_k|x_{\setminus k})$ depends only on the two neighbours, and for the bipartite RBM of Section 2.8 each layer updates in one parallel step. Inference cost follows graph structure, a principle that Chapter 5 makes systematic.

For continuous x and differentiable energy, a different design principle uses gradients. Consider the discrete-time update

$$x_{k+1} = x_k + \epsilon \nabla \log \pi(x_k) + \sqrt{2\epsilon} \xi_k, \quad \xi_k \sim \mathcal{N}(0, I), \quad (3.7)$$

the (unadjusted) *Langevin algorithm*. The drift pushes towards high probability; the noise maintains exploration. Two observations prepare what follows. First, (3.7) requires only the *score* $\nabla \log \pi$; the normalising constant is never needed. This is precisely why the score is the natural object for generative modelling (Chapter 4), and why so much of this thesis is devoted to computing scores exactly. Second, as $\epsilon \rightarrow 0$ the recursion (3.7)

becomes a stochastic differential equation, the Langevin diffusion, whose stationary law is exactly π . Making that limit precise requires the Itô calculus and Fokker–Planck theory that occupy the rest of the chapter, where the claim is verified by direct computation.

3.5 From ODEs to SDEs: the Itô calculus

An ordinary differential equation $\dot{x} = f(x, t)$ moves a point along a deterministic velocity field. Physical systems, however, feel rapid, unresolved forces (collisions, thermal agitation) whose cumulative effect is well modelled by adding noise:

$$dX_t = \underbrace{f(X_t, t) dt}_{\text{drift}} + \underbrace{g(t) dW_t}_{\text{diffusion}}. \quad (3.8)$$

Here W_t is a *Wiener process* (standard Brownian motion): the unique process with $W_0 = 0$, independent increments, $W_t - W_s \sim \mathcal{N}(0, t - s)$, and continuous paths. Equation (3.8) is shorthand for the integral equation $X_t = X_0 + \int_0^t f(X_s, s) ds + \int_0^t g(s) dW_s$, and the whole subtlety lives in the last term: Brownian paths are nowhere differentiable, so $\int g dW$ cannot be a Riemann–Stieltjes integral. Itô’s construction (Itô, 1944) defines it as the limit of left-endpoint sums,

$$\int_0^t g(s) dW_s = \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} g(s_j) (W_{s_{j+1}} - W_{s_j}), \quad (3.9)$$

where the evaluation at the *left* endpoint s_j makes the integrand independent of the coming increment. Two consequences follow immediately: the Itô integral is a martingale (zero mean), and its variance is given by the *Itô isometry*, $\mathbb{E}[(\int_0^t g dW)^2] = \int_0^t g(s)^2 ds$, both because cross terms $\mathbb{E}[g(s_j)(W_{s_{j+1}} - W_{s_j})]$ vanish by independence.

It is worth recording why no convention-free definition is available, since the same two sums reappear in Chapter 9. Fix $[0, t]$ and a partition of mesh tending to zero. The *quadratic variation* $Q_n = \sum_j (W_{s_{j+1}} - W_{s_j})^2$ has mean $\sum_j \Delta s_j = t$ and variance $2 \sum_j (\Delta s_j)^2$, which vanishes in the limit, so $Q_n \rightarrow t$ in mean square. The *total variation*

$V_n = \sum_j |W_{s_{j+1}} - W_{s_j}|$ instead diverges, because $Q_n \leq (\max_j |W_{s_{j+1}} - W_{s_j}|) V_n$ and the maximum increment tends to zero by continuity: a bounded V_n would force $Q_n \rightarrow 0$, contradicting $Q_n \rightarrow t > 0$. A Riemann–Stieltjes integral $\int f dg$ requires g to have bounded variation, precisely so that the limit does not depend on where inside each subinterval f is evaluated. For Brownian motion that guarantee is absent, different evaluation points give genuinely different limits, and a convention must be fixed.

The price of the construction is a new chain rule. Its origin is the quadratic variation just computed: increments scale as $dW_t \sim \sqrt{dt}$, so second-order terms in dW are first-order in dt and cannot be discarded.

Toolbox 3.2 — Itô’s lemma, and where the extra term comes from

Let $Y_t = \phi(X_t, t)$ with ϕ twice differentiable and X_t solving (3.8). Taylor-expand ϕ to second order:

$$dY = \frac{\partial \phi}{\partial t} dt + \frac{\partial \phi}{\partial x} dX + \frac{1}{2} \frac{\partial^2 \phi}{\partial x^2} (dX)^2 + \dots$$

Substitute $dX = f dt + g dW$ and expand the square:

$$(dX)^2 = f^2(dt)^2 + 2fg dt dW + g^2(dW)^2.$$

Now use the multiplication table of Itô calculus, which is nothing but the scaling $dW \sim \sqrt{dt}$ made systematic:

$$(dt)^2 = 0, \quad dt dW = 0, \quad (dW)^2 = dt.$$

The last identity is the rigorous statement that the quadratic variation of W over $[0, t]$ equals t : for a partition of mesh $\rightarrow 0$, $\sum_j (W_{s_{j+1}} - W_{s_j})^2 \rightarrow t$ in mean square, since the sum has mean $\sum_j (s_{j+1} - s_j) = t$ and variance $2 \sum_j (s_{j+1} - s_j)^2 \rightarrow 0$. Keeping the surviving terms:

$$dY = \left(\frac{\partial \phi}{\partial t} + f \frac{\partial \phi}{\partial x} + \frac{g^2}{2} \frac{\partial^2 \phi}{\partial x^2} \right) dt + g \frac{\partial \phi}{\partial x} dW.$$

The half-Laplacian term, absent from the classical chain rule, is the engine of everything that follows: it is why densities spread, why the Fokker–Planck equation has a second-order term, and why exponential transformations of SDEs pick up variance corrections. $\square$

3.6 The Ornstein–Uhlenbeck process: the corruption channel

The noising process used by diffusion models, and by every result of this thesis, is the *Ornstein–Uhlenbeck process* (Uhlenbeck and Ornstein, 1930): Brownian motion pulled back to the origin by a linear spring,

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 = a. \quad (3.10)$$

(The drift and diffusion constants are a normalisation choice, the variance-preserving convention of the diffusion literature (Song et al., 2021); any other choice rescales time.) The OU process is the unique Gaussian, Markov, stationary process in continuous time, and it is exactly solvable.

Toolbox 3.3 — Exact solution of the OU SDE by integrating factor

Multiply (3.10) by e^t and consider $Y_t = e^t X_t$. By Itô’s lemma with $\phi(x, t) = e^t x$ (here $\partial^2 \phi / \partial x^2 = 0$, so no correction term arises):

$$dY_t = e^t X_t dt + e^t (-X_t dt + \sqrt{2} dW_t) = \sqrt{2} e^t dW_t.$$

The drift cancels exactly; that is what the integrating factor is for. Integrate from 0 to t and undo the substitution:

$$X_t = a e^{-t} + \sqrt{2} \int_0^t e^{-(t-s)} dW_s.$$

The integral is Gaussian (limit of Gaussian sums), with mean zero and variance given by the Itô isometry:

$$\text{Var}(X_t) = 2 \int_0^t e^{-2(t-s)} ds = 2 e^{-2t} \frac{e^{2t} - 1}{2} = 1 - e^{-2t}.$$

Hence the *transition kernel* of the OU channel is

$$p(x_t | X_0 = a) = \mathcal{N}(x_t; a e^{-t}, \Delta_t), \quad \boxed{\Delta_t = 1 - e^{-2t}}.$$

□

The boxed kernel is used on every page of the research chapters, so we fix its reading now. Corrupting a data point a for a time t produces

$$x = a e^{-t} + \sqrt{\Delta_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1) : \quad (3.11)$$

the signal is *contracted* by e^{-t} while isotropic noise of variance Δ_t fills in. At $t = 0$: $x = a$ (no corruption); as $t \rightarrow \infty$: $x \sim \mathcal{N}(0, 1)$ regardless of a (all information destroyed, the equilibrium of the spring). The signal-to-noise ratio e^{-2t}/Δ_t sweeps continuously from ∞ to 0, and it diverges as $1/2t$ for small t , a factor that reappears throughout the thesis as an error amplifier. The OU channel interpolates between the data distribution and a standard Gaussian, which is precisely what a diffusion model needs.

Two structural remarks, both load-bearing later. First, the OU kernel is Gaussian *in a as well as in x* : viewed as a function of the clean variable, $\mathcal{N}(x; a e^{-t}, \Delta_t) \propto \exp[-(e^{-2t}/2\Delta_t)(a - x e^t)^2]$, a Gaussian likelihood factor of precision e^{-2t}/Δ_t centred at the deconvolved observation $e^t x$. This is the fact that makes the posteriors of Chapter 7 jointly Gaussian and the messages of the BP computation closable. Second, if a random vector $a \sim \mathcal{N}(\mu, \Sigma_0)$ is pushed through (3.11) coordinate-wise with *independent* noises, the output is Gaussian with

$$\mu_t = e^{-t} \mu, \quad \Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I, \quad (3.12)$$

since means and covariances transform linearly and the added noise is white. Equation (3.12), shrink the covariance and add a multiple of the identity, is the entire forward process of this thesis in one line.

3.7 The Fokker–Planck equation: transporting densities

The SDE (3.8) moves *samples*; the corresponding motion of the *density* $p_t(x)$ is governed by the Fokker–Planck (or forward Kolmogorov) equation. This change of viewpoint, from trajectories to densities, is the same one Liouville’s theorem performed for Hamiltonian flow, and the derivation is parallel, with the Itô correction supplying a new, diffusive term.

Toolbox 3.4 — Derivation of the Fokker–Planck equation

Let ϕ be an arbitrary smooth test function of compact support, and compute $\frac{d}{dt} \mathbb{E}[\phi(X_t)]$ in two ways.

Way 1: through the samples. By Itô’s lemma,

$$d\phi(X_t) = \left(f \phi' + \frac{g^2}{2} \phi'' \right) dt + g \phi' dW_t,$$

and taking expectations kills the martingale term:

$$\frac{d}{dt} \mathbb{E}[\phi(X_t)] = \mathbb{E} \left[f(X_t, t) \phi'(X_t) \right] + \frac{g(t)^2}{2} \mathbb{E}[\phi''(X_t)].$$

Way 2: through the density. $\mathbb{E}[\phi(X_t)] = \int \phi(x) p_t(x) dx$, so

$$\frac{d}{dt} \mathbb{E}[\phi(X_t)] = \int \phi(x) \frac{\partial p_t(x)}{\partial t} dx.$$

Equate the two, and integrate Way 1 by parts to move all derivatives off ϕ (boundary terms vanish by compact support): once for the drift term, twice for the diffusion

term,

$$\int \phi \frac{\partial p_t}{\partial t} = \int \phi \left[-\frac{\partial}{\partial x} (f p_t) + \frac{g^2}{2} \frac{\partial^2 p_t}{\partial x^2} \right] dx.$$

Since ϕ is arbitrary, the integrands must agree pointwise:

$$\boxed{\frac{\partial p_t}{\partial t} = -\frac{\partial}{\partial x} (f(x, t) p_t(x)) + \frac{g(t)^2}{2} \frac{\partial^2 p_t(x)}{\partial x^2}.}$$

The first term transports probability along the drift (as in Liouville); the second spreads it diffusively. The diffusive term is the macroscopic footprint of the Itô correction. □

In vector form, and for later reference, the Fokker–Planck equation of (3.8) reads

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g^2 \nabla^2 p_t. \quad (3.13)$$

3.7.1 Stationary law of Langevin dynamics: closing the loop

Apply the Fokker–Planck equation to the *Langevin SDE* associated with a potential U (the continuous-time limit of the algorithm (3.7)):

$$dX_t = -\nabla U(X_t) dt + \sqrt{2\beta^{-1}} dW_t. \quad (3.14)$$

A stationary density solves $0 = \nabla \cdot (\nabla U p + \beta^{-1} \nabla p)$, i.e. the probability current $J = -(\nabla U) p - \beta^{-1} \nabla p$ is divergence-free; imposing the natural (decaying) boundary conditions forces $J \equiv 0$, hence $\nabla p/p = -\beta \nabla U$, hence

$$p_\infty(x) \propto e^{-\beta U(x)} : \quad (3.15)$$

Langevin dynamics equilibrates to the Boltzmann distribution of Chapter 2. This is the two-line proof of the claim left pending in Section 3.4, and it explains the deep kinship of sampling and physics: a Langevin sampler *is* an overdamped physical system at temperature $1/\beta$. Note also that the drift of (3.14) is (up to β) the *score* of the target,

$-\nabla U = \nabla \log p_\infty$: dynamics that sample a distribution are driven by its score.

For the OU process, $U(x) = x^2/2$ and $\beta = 1$: the stationary law is $\mathcal{N}(0, 1)$, confirming the $t \rightarrow \infty$ limit of Section 3.6.

3.8 Time reversal: Anderson’s theorem

One last piece of SDE theory converts everything above into a generative method. Suppose data $X_0 \sim p_0$ are corrupted by a forward SDE (3.8) up to time T , producing marginals p_t . Can the corruption be undone? Can we find an SDE which, run from $X_T \sim p_T$ backwards, reproduces the same marginals? The answer (Anderson, 1982) is yes, and the reverse drift needs exactly one unknown object:

$$dX_t = \left[f(X_t, t) - g(t)^2 \nabla_x \log p_t(X_t) \right] dt + g(t) d\bar{W}_t, \quad (3.16)$$

where time runs backwards from T to 0 and $\bar{W}$ is a reverse-time Wiener process. The correction to the drift is the *score* $\nabla_x \log p_t$ of the noised marginal: the object defined in (1.1) and computed exactly, for Markov-chain data, in the research chapters. A heuristic that makes (3.16) plausible follows from the Fokker–Planck equation: rewriting its diffusion term as a transport term,

$$\frac{g^2}{2} \frac{\partial^2 p_t}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{g^2}{2} p_t \frac{\partial \log p_t}{\partial x} \right),$$

one sees that the density evolution of (3.8) is *identical* to that of the noise-free ODE with velocity $f - \frac{g^2}{2} \nabla \log p_t$ (the *probability-flow ODE* of Song et al., 2021); running that transport backwards, and re-injecting noise consistently, yields (3.16). The rigorous statement and proof are in Anderson (1982).

3.9 From algorithmic chains to data chains

A closing change of perspective, to set up the research problem. In this chapter Markov chains began as *algorithms*: their stationary law was the target, their trajectory a means to an end. From Chapter 7 onwards the chain (3.5) plays the opposite role: it is the *data-generating process*, its trajectory $(a_0, \dots, a_{K-1})$ is one data point (a video of K frames), and the object of interest is the probability law of the whole trajectory after each frame has been independently corrupted by the OU channel (3.11). The factorisation (3.2) then stops being a computational convenience and becomes the structure to be discovered: the research programme of this thesis asks how much of (3.2) survives noising, and how an inference algorithm, or a neural network, can exploit what survives. In one sentence: the OU channel destroys data at a known rate, the Fokker–Planck equation describes the destruction at the level of densities, and Anderson’s theorem says the destruction is invertible by anyone who knows the score, which is why the rest of this thesis is about computing that score exactly when the data have Markov structure.

Chapter 4

From Energy-Based Models to Diffusion and Flows

Chapter 2 left the generative branch of machine learning at the Boltzmann machine: a model is an energy, learning is moment matching, sampling is dynamics. This chapter follows that branch to the present. It reviews, in the depth their role in this thesis demands, the three ideas the research chapters build on: the energy-based view of modelling (LeCun et al., 2006), the score-matching principle that removed the partition function from training (Hyvärinen, 2005; Vincent, 2011), and the diffusion construction that turned denoising into a complete generative method (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021), together with its deterministic sibling, the flow model. The chapter then reviews the statistical-physics literature on diffusion models, from the high-dimensional analyses of Biroli and Mézard (2023) to the speciation, memorisation, and generalisation results of the Biroli–Mézard line of work, and, on the training side, the cascade of phase transitions in energy-based-model learning (Bachtis et al., 2024). It closes by stating the gap this thesis addresses.

4.1 Energy-based models

Modern generative modelling asks a single question: given samples of an unknown distribution p_0 , build a machine that produces new ones. The last decade produced several answers: variational autoencoders (Kingma and Welling, 2014), adversarial networks (Goodfellow et al., 2014), normalizing flows (Rezende and Mohamed, 2015), diffusion models. It is easy to present them as unrelated tricks. The energy-based view of LeCun et al. (2006) is the antidote: it is the direct continuation of the Boltzmann-machine programme of Section 2.8, and, as this chapter unfolds, diffusion models will appear as the member of the energy-based family that finally solved its oldest problem.

An *energy-based model* (EBM) over $x \in \mathcal{X}$ assigns to each configuration a scalar energy $E_\theta(x)$ measuring how badly x fits the model’s idea of the data: low energy for plausible configurations, high for implausible ones. In LeCun et al.’s formulation the energy is the primitive object: inference means finding low-energy configurations, and learning means shaping the landscape, pushing the energy of observed data down and the energy of everything else up. A probabilistic model is recovered whenever one normalises with the Gibbs measure of Chapter 2,

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \quad Z(\theta) = \int_{\mathcal{X}} e^{-E_\theta(x)} dx, \quad (4.1)$$

the Boltzmann distribution (2.9) with β absorbed into the energy and the physics replaced by a parametric family. The appeal of (4.1) is that E_θ is unconstrained: any function defines a valid unnormalised density. The price is the partition function, and every basic operation hits it. *Evaluating* $p_\theta(x)$ requires $Z(\theta)$, intractable in general (Section 2.5). *Sampling* requires the MCMC machinery of Section 3.4, whose mixing time can be exponential in the presence of metastability. *Learning* by maximum likelihood requires the gradient

$$-\nabla_\theta \log p_\theta(x) = \nabla_\theta E_\theta(x) - \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta], \quad (4.2)$$

the “push down on data, pull up on the model’s own samples” rule, the continuous-

variable form of the Boltzmann-machine rule (2.16), whose second term again demands sampling from p_θ . LeCun et al. (2006) frame the alternatives systematically: either fight the negative phase with *contrastive* losses that pull up only on well-chosen offending configurations (contrastive divergence (Hinton, 2002) being the MCMC-flavoured member of the family), or design architectures and regularisers that bound the volume of low-energy space so that no pull-up is needed. The lesson of the tutorial is that the partition function is not an accounting detail; it is *the* design constraint of the field.

There are two classical escape routes, and this thesis walks both. The first is the *score*: since $\nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$, the gradient of the log-density in x is free of Z , because differentiating in x kills the constant. Sections 4.4–4.6 build the modern edifice standing on this observation. The second is *structure*: when the energy decomposes into local terms, marginals and even Z itself become computable by dynamic programming; that route (Markov random fields, factor graphs, message passing) is developed in Chapter 5.

4.2 The diffusion idea

Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) solve the generative problem with a two-process design inspired by non-equilibrium thermodynamics: a fixed *forward process* gradually corrupts data into pure noise, and a learned *reverse process* undoes the corruption step by step. A recent book-length survey of the family is Lai et al. (2025). The forward process used throughout this thesis is the OU channel of Section 3.6: at diffusion time t ,

$$x_t = a e^{-t} + \sqrt{\Delta_t} \varepsilon, \quad a \sim p_0, \quad \varepsilon \sim \mathcal{N}(0, I), \quad \Delta_t = 1 - e^{-2t}, \quad (4.3)$$

so that the noised marginal is the convolution

$$p_t(x) = \int p_0(a) \mathcal{N}(x; a e^{-t}, \Delta_t I) da. \quad (4.4)$$

As t grows, p_t interpolates from the data distribution to $\mathcal{N}(0, I)$. By Anderson's theorem (Section 3.8), the corruption is undone by the reverse SDE (3.16), whose only unknown ingredient is the score $\nabla_x \log p_t(x)$ of the noised marginals. Generative diffusion is therefore, at bottom, *score estimation*: whoever knows $\nabla \log p_t$ for all t can sample from p_0 . Note what this design buys relative to Section 4.1: no energy is ever normalised, no equilibrium is ever awaited. The Markov-chain convergence problem of MCMC is replaced by a *fixed-horizon* integration of an SDE whose drift is learned. The partition-function problem has not been solved; it has been side-stepped, by working only with objects (scores) from which Z has already been differentiated away.

4.2.1 The thermodynamic reading, and the direction of the arrow

The appeal to non-equilibrium thermodynamics in Sohl-Dickstein et al. (2015) is structural rather than decorative. Their forward process is a chain of small perturbations converting a structured distribution into a simple one, that is an annealing schedule, and the objective they derive is a variational bound of the same algebraic form as the work identities of non-equilibrium statistical mechanics: the log-likelihood is bounded by an average of log-ratios along the forward path, exactly as free-energy differences are bounded in Jarzynski-type equalities and estimated in annealed importance sampling. That analogy is what supplies the training objective, and it is the reason the field speaks of temperature, annealing and entropy at all.

The direction of the forward process is fixed by the second law, which for the Fokker-Planck equation (3.13) takes an exact form. Let p_t evolve with a gradient drift and stationary law $p_\infty \propto e^{-V}$, and consider the free energy $\mathcal{F}[p_t] = D_{\text{KL}}(p_t \| p_\infty)$, which is the mean energy minus the entropy up to a constant. Writing the dynamics as $\partial_t p = \nabla \cdot (p \nabla \log(p/p_\infty))$ and integrating by parts,

$$\frac{d\mathcal{F}}{dt} = \int \partial_t p \log \frac{p}{p_\infty} = - \int p \left\| \nabla \log \frac{p}{p_\infty} \right\|^2 \leq 0, \quad (4.5)$$

with equality only at $p_t = p_\infty$. This is the H-theorem: the free energy decreases monotonically and the relative entropy is consumed at a rate given by a relative Fisher information. The forward channel therefore destroys information about the data monotonically and irreversibly, and (4.5) identifies the rate. The reverse direction is not forbidden, but it is not free either: by Anderson’s theorem it requires the score of the intermediate laws to be supplied from outside. The entire modelling effort of a diffusion model is the estimation of that one object, and Chapters 7 to 9 are concerned with what that object looks like when the data are trajectories.

4.3 Denoising diffusion probabilistic models

The discrete-time formulation (Sohl-Dickstein et al., 2015; Ho et al., 2020) discretises (4.3) into a Markov chain $q(x_s|x_{s-1}) = \mathcal{N}(x_s; \sqrt{1-\beta_s}x_{s-1}, \beta_s I)$ with a variance schedule $\beta_1, \dots, \beta_S$, and learns a reverse chain $p_\theta(x_{s-1}|x_s)$. Training maximises a variational lower bound on the data log-likelihood; the physicist will recognise the Gibbs variational principle (2.12) in modern dress.

Toolbox 4.1 — The evidence lower bound of a diffusion model

For any latent-variable model with joint $p_\theta(x_{0:S})$ and any inference distribution $q(x_{1:S}|x_0)$, Jensen’s inequality gives

$$\begin{aligned} \log p_\theta(x_0) &= \log \int p_\theta(x_{0:S}) \, dx_{1:S} = \log \mathbb{E}_q \left[\frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)} \right] \\ &\geq \mathbb{E}_q \left[\log \frac{p_\theta(x_{0:S})}{q(x_{1:S}|x_0)} \right] =: \text{ELBO}. \end{aligned}$$

Insert the two Markov factorisations $p_\theta(x_{0:S}) = p(x_S) \prod_s p_\theta(x_{s-1}|x_s)$ and $q(x_{1:S}|x_0) = \prod_s q(x_s|x_{s-1})$, then regroup telescopically using Bayes’ rule $q(x_s|x_{s-1}) = q(x_{s-1}|x_s, x_0) q(x_s|x_0) / q(x_{s-1}|x_0)$:

$$-\text{ELBO} = D_{\text{KL}}(q(x_S|x_0) \parallel p(x_S)) - \mathbb{E}_q \log p_\theta(x_0|x_1)$$

$$+ \sum_{s=2}^S \mathbb{E}_q D_{\text{KL}}(q(x_{s-1}|x_s, x_0) \parallel p_\theta(x_{s-1}|x_s)),$$

one denoising step per summand. The decisive point: because the forward chain is Gaussian and conditioned on *both* endpoints, each target $q(x_{s-1}|x_s, x_0)$ is an explicit Gaussian. Matching it with a Gaussian p_θ reduces each KL term to a weighted regression of the posterior mean; concretely, to predicting the noise ε that was added (Ho et al., 2020). The thermodynamic reading of the ELBO gap as an entropy-production bound goes back to Sohl-Dickstein et al. (2015). $\square$

The practical upshot of Ho et al. (2020) is the loss

$$\mathcal{L}(\theta) = \mathbb{E}_{s, a \sim p_0, \varepsilon} \left\| \varepsilon - \varepsilon_\theta \left(\underbrace{a e^{-t_s} + \sqrt{\Delta_{t_s}} \varepsilon}_{x_{t_s}}, s \right) \right\|^2, \quad (4.6)$$

a denoising regression, and the empirical discovery that this simplified objective, with each noise level weighted equally and all the variational bookkeeping dropped, trains photorealistic image models. Its connection with the score is not incidental, as the next section shows: predicting the noise *is* estimating $\nabla \log p_t$ up to scale.

4.4 Score matching

Suppose we model the score directly by a function $s_\theta(x)$ and wish it to match $\nabla \log p(x)$ for data from p . The naive objective $\mathbb{E}_p \|s_\theta - \nabla \log p\|^2$ seems to require knowing the very score we lack. Two classical results remove the obstruction.

Toolbox 4.2 — Implicit score matching (Hyvärinen)

Expand the square and keep only θ -dependent terms:

$$\mathbb{E}_p \|s_\theta - \nabla \log p\|^2 = \mathbb{E}_p \|s_\theta\|^2 - 2 \mathbb{E}_p \langle s_\theta, \nabla \log p \rangle + \text{const.}$$

The cross term integrates by parts. Componentwise, using $p \partial_i \log p = \partial_i p$ and assuming $p s_{\theta,i} \rightarrow 0$ at infinity:

$$\int p s_{\theta,i} \partial_i \log p \, dx = \int s_{\theta,i} \partial_i p \, dx = - \int p \partial_i s_{\theta,i} \, dx.$$

Therefore

$$\mathbb{E}_p \|s_\theta - \nabla \log p\|^2 = \mathbb{E}_p \left[\|s_\theta(x)\|^2 + 2 \nabla \cdot s_\theta(x) \right] + \text{const} :$$

an objective involving only the model and data samples (Hyvärinen, 2005). Elegant, but the divergence term is expensive for large networks, which is why diffusion models use the denoising variant instead. $\square$

Toolbox 4.3 — Denoising score matching (Vincent)

Fix t and consider the noised marginal (4.4), written as $p_t(x) = \int p_0(a) q_t(x|a) \, da$ with Gaussian kernel $q_t(x|a) = \mathcal{N}(x; ae^{-t}, \Delta_t I)$. Then

$$\nabla_x \log p_t(x) = \frac{\int p_0(a) \nabla_x q_t(x|a) \, da}{p_t(x)} = \int \underbrace{\frac{p_0(a) q_t(x|a)}{p_t(x)}}_{=p(a|x)} \nabla_x \log q_t(x|a) \, da,$$

so that

$$\nabla_x \log p_t(x) = \mathbb{E} \left[\nabla_x \log q_t(x|a) \mid x \right] :$$

the score of the *marginal* is the posterior average of the score of the *kernel*, and the latter is trivial, $\nabla_x \log q_t(x|a) = -(x - ae^{-t})/\Delta_t$. A standard L^2 -projection argument turns this into a regression: for any s_θ ,

$$\mathbb{E}_{x \sim p_t} \|s_\theta(x) - \nabla \log p_t(x)\|^2 = \mathbb{E}_{a,x} \|s_\theta(x) - \nabla_x \log q_t(x|a)\|^2 + \text{const},$$

because conditional expectation is exactly the L^2 projection onto functions of x (Vincent, 2011). Substituting the Gaussian kernel score and $x = ae^{-t} + \sqrt{\Delta_t} \varepsilon$ gives $\nabla_x \log q_t = -\varepsilon/\sqrt{\Delta_t}$: *learning the score of noisy data is learning to predict the noise*,

and the DDPM loss (4.6) is denoising score matching in disguise. $\square$

The continuous-time synthesis (Song and Ermon, 2019; Song et al., 2021) trains one network $s_\theta(x, t)$ against denoising score matching at all noise levels and samples with the reverse SDE (3.16); an accessible exposition is Song (2021), and Karras et al. (2022) systematise the design space. This is the resolution, promised in Section 4.1, of the energy-based model’s oldest problem: Song and Ermon replace the energy by its gradient, the equilibrium sampler by a finite-time reverse SDE, and the intractable likelihood by a regression whose targets are exact.

4.5 Flows: the deterministic side of the same coin

Anderson’s theorem has a deterministic twin. As noted in Section 3.8, the marginals p_t of the forward SDE are also transported by the noise-free *probability-flow ODE* with velocity $f - \frac{g^2}{2} \nabla \log p_t$ (Song et al., 2021): the same score, consumed by an ODE instead of an SDE, turns generation into a smooth, invertible map from Gaussian noise to data. This places diffusion models inside an older family. *Normalizing flows* (Rezende and Mohamed, 2015; Papamakarios et al., 2021) construct an invertible map T_θ and evaluate likelihoods through the change-of-variables formula; their continuous-time version parametrises the velocity field of an ODE directly (Chen et al., 2018). What diffusion brought to this family is a *simulation-free training signal: flow matching* (Lipman et al., 2023) and stochastic interpolants (Albergo and Vanden-Eijnden, 2023) show that one can regress the velocity field of any prescribed interpolation between noise and data (the OU path (4.3) being one choice among many) with the same conditional trick as denoising score matching, never integrating the ODE during training; a systematic introduction is Holderrieth and Erives (2025). The unifying statement is worth recording: DDPMs, score SDEs, and flow-matching models all learn a vector field whose ground truth is a *conditional expectation under the noising channel*; they differ in the path of marginals and in whether sampling is stochastic or deterministic. Every exact result of this thesis therefore speaks to the

whole family at once: the object we compute in closed form, the conditional expectation behind the field, is the training target of all of them.

4.6 The score as a posterior expectation

The middle line of Vincent’s derivation deserves its own section, because every result of this thesis passes through it. Rearranging $\nabla \log p_t(x) = \mathbb{E}[-(x - ae^{-t})/\Delta_t \mid x]$:

$$\boxed{\nabla_x \log p_t(x) = \frac{e^{-t} \mathbb{E}[a \mid x] - x}{\Delta_t} \iff \mathbb{E}[a \mid x] = e^t \left(x + \Delta_t \nabla_x \log p_t(x) \right).} \quad (4.7)$$

We refer to (4.7) throughout as the *score-posterior identity*; in the empirical-Bayes literature it is known as Tweedie’s formula (Robbins, 1956; Miyasawa, 1961; Efron, 2011). The score of the noisy marginal and the Bayes-optimal denoiser $\mathbb{E}[a|x]$ determine each other exactly, with no assumption whatsoever on p_0 . Three readings will recur:

1. *Geometric*: the score points from x towards (the contracted image of) the posterior mean of the clean data; denoising direction and score direction coincide.
2. *Statistical*: estimating the score is solving a Bayesian inference problem, namely computing a posterior expectation. The score is not *like* an inference target; it *is* one.
3. *Algorithmic*: for sequence data with a Markov prior, that inference problem lives on a tree-shaped graph, so the score is computable by message passing, the programme made precise in Chapter 5 and executed in Chapter 7.

For *vector-valued* data $a \in \mathbb{R}^K$ noised coordinate-wise, (4.7) holds componentwise with the *full* posterior:

$$\left(\nabla_x \log p_t(x) \right)_k = \frac{e^{-t} \mathbb{E}[a_k \mid x_0, \dots, x_{K-1}] - x_k}{\Delta_t}. \quad (4.8)$$

The conditioning is on *all* coordinates: even though the noise is independent across frames,

the posterior mean of frame k , and hence the k -th score component, depends on every observation, because the prior couples the frames. Equation (4.8) is the precise reason the *joint* score differs from the per-frame marginal score, a distinction that is easy to blur and whose importance for structured data was impressed on this project in supervision (Chapter 6); Chapter 7 quantifies exactly how much coupling the conditioning induces, as a function of t .

4.7 The statistical physics of generative diffusion

A line of work initiated by Biroli, Mézard, and collaborators analyses diffusion models with the tools of Chapter 2, in the regime where the score is *exact*: either computed from a solvable p_0 or from the empirical measure of n training points. Because this is the literature our research extends, we review it in some detail.

High dimensions and the cost of asymmetry. Biroli and Mézard (2023) analyse generative diffusion in the high-dimensional limit for two controlled data models, a Gaussian model and the Curie–Weiss ferromagnet. In the latter, the reverse dynamics undergoes the symmetry-breaking mechanism of Section 2.5: at a critical diffusion time the trajectory commits to one of the two low-temperature states. Their sharpest quantitative point concerns data requirements: to reconstruct the *relative weights* of two asymmetric modes, the number of training samples must be much larger than the data dimension, and the paper characterises the scaling laws in dimension and sample size for efficient generation.

The three regimes of the backward dynamics. Biroli et al. (2024) study the backward process when the score has been trained optimally on n samples in dimension d , and identify three dynamical regimes separated by two transitions. Starting from pure noise, the trajectory first behaves as a free (Brownian-like) process; at the *speciation* time it commits to the gross structure of the data (a class, a mode), through a mechanism analogous to spontaneous symmetry breaking; at the later *collapse* time the trajectory is attracted to an individual memorised training point, through a mechanism analogous

to condensation in a glass. The speciation time is computable from a spectral analysis of the data correlation matrix, and the collapse time from an excess-entropy estimate; the dependence of the collapse time on n and d quantifies the curse of dimensionality for diffusion models. Related transition phenomena along the generation trajectory were exhibited by Raya and Ambrogioni (2023).

Speciation with general class structure. Achilli et al. (2026) develop the general theory of the speciation transition: where earlier analyses required classes separated by their first moments (Gaussian mixtures with distinct means), they characterise speciation times for arbitrary class structures through a free-entropy difference criterion, recovering the known Gaussian-mixture results and extending them to classes that differ only through higher-order or collective features, with successive speciation events for multiple classes. A hierarchical counterpart is Sclocchi et al. (2025): in a hierarchical generative model of data, the backward process started from time t exhibits a phase transition at a threshold time at which the probability of preserving the high-level class drops abruptly, while low-level features evolve smoothly; beyond the transition a regenerated sample can retain low-level elements of the original image while its class has changed. The statistical-physics treatment of generative diffusion in solvable models is developed at book length in the doctoral thesis of Achilli (2026).

Memorisation and generalisation. When the score is learned from finitely many samples rather than computed exactly, the question becomes when generation generalises and when it memorises. Bonnaire et al. (2025) identify two timescales in the training dynamics: an early time τ_{gen} at which the model begins to generate high-quality samples, and a later time τ_{mem} beyond which memorisation of training data emerges; τ_{mem} grows linearly with the training-set size n while τ_{gen} stays constant, opening a widening window in which overparameterised models generalise, a form of implicit dynamical regularisation. Garnier-Brun et al. (2026) refine the picture by identifying a phase of *biased* generalisation in between: the test loss still decreases while the model increasingly favours samples

anomalously close to training points, so the standard early-stopping criterion does not certify genuine novelty. Ghio et al. (2024) map generative samplers (flows, diffusions, autoregressive networks) onto statistical-physics sampling problems and delimit where each is efficient, importing the algorithmic-hardness picture of spin glasses (Zdeborová and Krzakala, 2016).

The exactly solvable programme, and what it holds fixed. The common method of these works is to choose p_0 rich enough to be interesting and simple enough that $\nabla \log p_t$ is available in closed form, then study the exact generation dynamics. What none of them vary is the *internal structure of a single data point*: their coordinates are exchangeable or i.i.d. conditional on a mode. There is no ordering and no dynamics *inside* the sample. That axis is precisely where this thesis works.

4.8 Learning as a cascade of phase transitions

The works above put phase transitions on the *sampling* side of generative modelling. A complementary result, Bachtis et al. (2024), which this section examines in detail, puts them on the *training* side, and completes the argument of Section 2.9 that learning itself is a sequence of symmetry breakings. The object of study is the restricted Boltzmann machine of Section 2.8, the minimal trainable EBM, with energy $E(v, h) = -v^\top W h - b^\top v - c^\top h$ and the moment-matching gradient (2.16). The central idea is to track training through the singular value decomposition of the weight matrix,

$$W = \sum_{\gamma=1}^r w_\gamma u_\gamma \bar{u}_\gamma^\top, \quad (4.9)$$

whose modes $(u_\gamma, \bar{u}_\gamma)$ act as candidate order parameters: the magnetisation of the visible layer along mode γ , $m_\gamma = u_\gamma \cdot v / \sqrt{N_v}$, measures whether the learned Gibbs measure has developed structure in that direction. At initialisation the weights are small and the model is a paramagnet: no direction in configuration space is preferred. The claim of Bachtis et al. (2024) is that training is an *effective cooling*: as the singular values grow,

the model crosses a sequence of genuine second-order phase transitions, one for each feature it learns, each signalled by a diverging susceptibility and falling in the mean-field (Curie–Weiss) universality class. The mechanism is visible in a two-line calculation.

Toolbox 4.4 — The retrieval transition of a one-mode RBM

Following Bachtis et al. (2024), take a single Gaussian hidden unit of variance $1/N_v$ coupled to N_v binary visibles through a weight vector w : the energy is $E(v, h) = -h w \cdot v + \frac{N_v}{2} h^2$. Integrating out the Gaussian h gives the visible marginal

$$p(v) \propto \exp\left[\frac{1}{2N_v} (w \cdot v)^2\right] :$$

a Mattis/Hopfield measure (Section 2.7) whose single pattern is w itself, made extensive by the $1/N_v$ variance scaling. Write $w = \bar{w} \xi$ with $\xi \in \{-1, +1\}^{N_v}$ and let $m = \frac{1}{N_v} \sum_i \xi_i v_i$ be the overlap; then $p(v) \propto \exp[\frac{N_v \bar{w}^2}{2} m^2]$. Counting configurations at fixed overlap ($\asymp \exp[N_v H(\frac{1+m}{2})]$ with H the binary entropy) gives the free-entropy density

$$\phi(m) = \frac{\bar{w}^2}{2} m^2 + H\left(\frac{1+m}{2}\right) = \log 2 + \frac{\bar{w}^2 - 1}{2} m^2 - \frac{m^4}{12} + O(m^6).$$

This is a Landau expansion: the quadratic coefficient changes sign at $\bar{w}^2 = \|w\|^2 / N_v = 1$. Below the threshold the maximiser is $m = 0$ (paramagnet); above it two symmetric maximisers $\pm m^*$ appear continuously (retrieval). The transition is second order with mean-field exponents, and the susceptibility along w diverges as $\|w\|^2 / N_v \uparrow 1$. Growing a singular value during training is therefore literally a quench through a Curie–Weiss critical point. $\square$

Why a cascade. With data generated by a Hopfield teacher whose two patterns are correlated combinations $\xi^{1,2} = \eta_1 \pm \eta_2$ of orthogonal directions, the data covariance is rank two with unequal strengths, and the early-time gradient flow of the student’s SVD amplitudes along η_1 and η_2 is exponential with *two different rates*. The stronger direction (the centre of mass of the clusters) reaches its critical magnitude first: the learned measure

splits into two lumps along $\pm\eta_1$. The weaker direction keeps growing at its own rate and crosses criticality later, splitting each lump again: four modes, the full teacher, retrieved in hierarchical order. Each crossing is a distinct second-order transition with its own critical time, its own diverging susceptibility, and hysteresis below it. Learning proceeds coarse-to-fine not by accident of optimisation but because the critical times are ordered by the spectrum of the data.

Evidence on real data. On MNIST, CelebA, and human-genome data, Bachtis et al. (2024) track the susceptibilities of individual SVD modes during training of binary–binary RBMs and observe the same signatures: successive susceptibility peaks that sharpen with system size, finite-size-scaling collapses consistent with mean-field exponents $(\gamma, \nu) = (1, \frac{1}{2})$ at upper critical dimension four, and hysteresis, the operational fingerprints of genuine transitions rather than smooth crossovers. This confirms and sharpens the earlier picture of RBM learning as progressive alignment with the data’s principal components (Decelle et al., 2017), of feature emergence as condensation (Tubiana and Monasson, 2017), and of sequential mode learning in linear networks (Saxe et al., 2014); a broad review of the EBM–physics interface is Carbone (2025).

What this means for the thesis. Phase transitions now appear on both faces of generative diffusion: in *sampling*, as speciation events along the reverse trajectory (Biroli et al., 2024; Achilli et al., 2026), and in *training*, as the cascade just described. Both are hierarchies of symmetry breakings ordered by the data’s spectrum. Our research adds the axis that both literatures hold fixed: the structure of the inference problem *inside a single data point*, and how the diffusion time t deforms it. The chain we solve is the minimal model in which that question is non-trivial and answerable exactly.

4.9 The gap, restated

Real sequential data (video, audio, trajectories) have exactly the structure the solvable programme leaves out: an internal (frame) time with a Markovian dependence along it.

For such data the score (4.8) is a smoothing posterior on a chain, an object with seventy years of inference theory behind it. Yet the questions a diffusion modeller asks about it are not answered by the classical literature, which fixes the noise level, nor by the diffusion literature, which ignores internal structure: how does the coupling structure of $\nabla \log p_t$ deform with t ? Can it be computed locally? What survives a Gaussian parametrisation of the messages when the data are not Gaussian? What limited guidance does that give for score architectures? Answering these questions, in the fully solvable setting of the AR(1) chain and its non-Gaussian perturbations, is the contribution of Chapters 7 and 8, with the inference tools prepared in Chapter 5 and the path that led to the precise formulation recorded in Chapter 6.

Chapter 5

Gaussian Vectors, Graphical Models, and Message Passing

The research chapters compute diffusion scores by Bayesian inference on a chain. This chapter builds the inference machinery itself, in the depth its central role deserves, and in one place: the multivariate Gaussian and its two parametrisations, the matrix tools (spectral theorem, Neumann series) that the Gaussian analysis runs on, Markov random fields and the Hammersley–Clifford theorem, factor graphs, the sum–product algorithm and its exactness on trees, the Gaussian message family and its classical incarnation, the Kalman filter, and the mean-field closures (TAP, AMP) of statistical physics. The chapter closes with the architectural side of the same story: convolutional networks, receptive fields, and the recent identification of U-Nets with belief propagation on hierarchical models. All of this material is standard; references are given section by section, and the research chapters use every part of it.

5.1 The multivariate Gaussian, in both parametrisations

Definition 5.1 (Multivariate Gaussian). A random vector $a \in \mathbb{R}^K$ is Gaussian with mean m and (positive definite) covariance Σ , written $a \sim \mathcal{N}(m, \Sigma)$, if its density is

$$\mathcal{N}(a; m, \Sigma) = \frac{1}{(2\pi)^{K/2} \sqrt{\det \Sigma}} \exp \left[-\frac{1}{2} (a - m)^\top \Sigma^{-1} (a - m) \right]. \quad (5.1)$$

Expanding the quadratic form shows that the same density can be written in *natural* (information) parameters,

$$\mathcal{N}(a; m, \Sigma) \propto \exp \left[-\frac{1}{2} a^\top J a + h^\top a \right], \quad J = \Sigma^{-1}, \quad h = \Sigma^{-1} m, \quad (5.2)$$

with J the *precision matrix* and h the precision-weighted mean. The two parametrisations are dual, and the duality organises everything that follows (Bishop, 2006; Koller and Friedman, 2009):

- *Linear maps.* If $a \sim \mathcal{N}(m, \Sigma)$ then $Aa + b \sim \mathcal{N}(Am + b, A\Sigma A^\top)$: Gaussianity is preserved by affine transformations, with moments transforming linearly. In particular a sum of independent Gaussians is Gaussian with means and covariances adding.
- *Marginalisation is easy in Σ .* The marginal of a sub-block a_1 of $a = (a_1, a_2)$ is Gaussian with mean m_1 and covariance Σ_{11} : simply read off the block.
- *Conditioning is easy in J .* Partitioning $J = \begin{pmatrix} J_{11} & J_{12} \\ J_{12}^\top & J_{22} \end{pmatrix}$, the conditional law of a_1 given a_2 is Gaussian with precision J_{11} (read off the block) and mean shifted by $-J_{11}^{-1} J_{12} a_2$. Eliminating a variable instead produces the *Schur complement* correction on its neighbours. Proofs of both statements are in Appendix A.
- *Products are easy in J .* Multiplying two Gaussian densities in the same variable adds their natural parameters, $(J, h) \mapsto (J_1 + J_2, h_1 + h_2)$. Bayesian updating with Gaussian likelihoods is addition.

The dual reading of zeros is used constantly in this thesis. A zero in the *covariance* means marginal independence of two coordinates. A zero in the *precision* means *conditional* independence given all the others: from the conditioning rule, if $J_{ij} = 0$ then a_i and a_j do not interact once the rest is fixed. A Gaussian vector can be, and in this thesis always is, globally correlated (dense Σ) while conditionally local (sparse J). Which matrix one should look at depends on the question, and confusing the two is a common source of error in reasoning about Gaussian dependence.

5.2 Matrix tools: the spectral theorem and matrix inversion

Since Gaussian computation is matrix computation, we record the two linear-algebra facts the research chapters use constantly.

Theorem 5.2 (Spectral theorem for symmetric matrices). *Every symmetric $A \in \mathbb{R}^{K \times K}$ has an orthonormal basis of eigenvectors: $A = U \operatorname{diag}(\omega_1, \dots, \omega_K) U^\top$ with $U^\top U = I$ and real eigenvalues ω_i .*

Three consequences matter here. First, analytic operations act on eigenvalues alone: if no eigenvalue vanishes,

$$A^{-1} = U \operatorname{diag}(1/\omega_i) U^\top, \quad \text{and more generally} \quad f(A) = U \operatorname{diag}(f(\omega_i)) U^\top.$$

Second, affine images share the eigenbasis: $(cA + dI)u_i = (c\omega_i + d)u_i$, so a whole family of matrices of the form $c(t)A + d(t)I$ is diagonalised once and for all by the U of A . Chapter 7 uses exactly this fact to diagonalise the diffusion problem at every noise level simultaneously. Third, for the symmetric Toeplitz covariances produced by stationary processes (Section 3.1), the eigenvectors are asymptotically the Fourier modes of the sequence (Brockwell and Davis, 1991): stationary Gaussian problems decouple, in the large- K limit, into independent scalar problems per frequency.

Inversion, when the inverse is not wanted eigenvalue by eigenvalue, has two workhorse expansions. The *Neumann series*

$$(I - B)^{-1} = \sum_{m \geq 0} B^m, \quad \text{convergent for } \|B\| < 1, \quad (5.3)$$

expresses the inverse of a near-identity matrix as a sum of powers; it is the tool behind the small- t expansions of Chapter 7, where powers of a banded matrix have growing but controlled bandwidth. The *Woodbury identity* relates the inverse of a matrix to the inverse of a low-rank or structured perturbation of it; its statement, proof, and the specific consequence used by the research chapters are collected in Appendix A. Finally, for the tridiagonal Toeplitz matrices that Markov chains produce, the inverse is available in closed form through a transfer-matrix (geometric-ansatz) computation, carried out where it is needed (Chapter 7).

5.3 Markov random fields

The Gaussian story above assigned structure to matrices. Graphical models assign it to distributions, for arbitrary (not necessarily Gaussian) variables.

Definition 5.3 (Markov random field). Let $G = (V, \mathcal{E})$ be an undirected graph with one random variable x_i per node. The collection $x = (x_i)_{i \in V}$ is a *Markov random field* (MRF) if it satisfies the local Markov property: for every node i ,

$$p(x_i \mid x_{V \setminus \{i\}}) = p(x_i \mid x_{\partial i}), \quad (5.4)$$

where ∂i is the set of neighbours of i in G .

The fundamental structure theorem connects this conditional-independence definition to the energy picture of Chapter 4: by the Hammersley–Clifford theorem (proved in Besag, 1974), a strictly positive density is an MRF on G if and only if it factorises over the *cliques*

of G ,

$$p(x) = \frac{1}{Z} \prod_{C \in \text{cliques}(G)} \psi_C(x_C) = \frac{1}{Z} \exp \left[- \sum_C E_C(x_C) \right]. \quad (5.5)$$

Conditional independence, graphical locality, and additive local energies are one and the same thing. The Ising model (2.13) is the textbook example. The decisive example for us is one-dimensional: the chain factorisation (3.2) makes a Markov chain an MRF on a line, with pairwise nearest-neighbour energies. For *Gaussian* MRFs the correspondence becomes matrix-concrete: the energy is the quadratic form $\frac{1}{2}a^\top J a - h^\top a$, its pairwise terms are the entries J_{ij} , and the graph of the MRF is exactly the sparsity pattern of the precision matrix. The dual reading of zeros of Section 5.1 and the Hammersley–Clifford theorem are the same statement, once in linear algebra and once in probability. A Gaussian Markov chain therefore has a tridiagonal precision, a fact Chapter 7 derives explicitly and exploits throughout.

One more structural remark that the research chapters rely on: if each variable of an MRF is observed through its own independent noisy channel, the likelihood factors touch one variable each, so no clique is enlarged. *The posterior of an MRF under per-node observations is an MRF on the same graph.* Observation changes the potentials, never the graph.

5.4 Factor graphs

MRFs record *which* variables interact but not *how* the joint factorises: a triangle of pairwise terms and a single triple term have the same graph. The representation that makes the factorisation itself graphical, and on which message passing is most naturally defined, is the *factor graph* (Kschischang et al., 2001; Koller and Friedman, 2009).

Definition 5.4 (Factor graph). Given a factorisation $p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a})$, its factor graph is the bipartite graph with a *variable node* for each x_i , a *factor node* for each ψ_a , and an edge (i, a) whenever $x_i \in x_{\partial a}$.

A connected factor graph with one fewer edge than nodes is a *tree*: it contains no

cycles, so any edge separates it into two disjoint components. Trees are the tractable case of inference: on them, marginals and partition functions are computable exactly by the recursion of the next section, whatever the alphabet of the variables. The factor graph of a Markov chain observed through per-node noise is a tree of a particularly simple shape (a spine of variables and transition factors, with one evidence leaf per variable), as Chapter 7 verifies by counting.

5.5 The sum-product algorithm

On a factor graph, sum-product belief propagation (BP) passes messages along every edge (Pearl, 1988; Kschischang et al., 2001; Mézard and Montanari, 2009). Variable-to-factor messages multiply the incoming factor messages from all *other* neighbours, and factor-to-variable messages integrate the factor against the incoming variable messages:

$$\nu_{i \rightarrow a}(x_i) \propto \prod_{b \in \partial i \setminus a} \hat{\nu}_{b \rightarrow i}(x_i), \quad \hat{\nu}_{a \rightarrow i}(x_i) \propto \int \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \nu_{j \rightarrow a}(x_j) dx_{\partial a \setminus i}, \quad (5.6)$$

with the belief at node i proportional to $\prod_{a \in \partial i} \hat{\nu}_{a \rightarrow i}(x_i)$. The “exclude the recipient” rule is the heart of the algorithm: the message $i \rightarrow a$ summarises everything node i knows *except* what it learned from a , so that no piece of evidence is ever counted twice along a path.

Theorem 5.5 (Tree exactness of sum-product). *On a tree-shaped factor graph, the equations (5.6) have a unique solution, reached after one inward and one outward pass, and the resulting beliefs are the exact marginals of p (Mézard and Montanari, 2009, Theorem 14.1).*

The reason is the separation property of trees: each edge splits the graph into two disjoint halves, and the message crossing it summarises exactly the half behind it. On graphs with cycles the same updates, iterated, define *loopy* BP, which is extraordinarily useful but approximate (Weiss and Freeman, 2001; Malioutov et al., 2006); none of that is needed in this thesis, because our graphs are trees.

The representation problem. For *discrete* variables each message is a finite probability vector and each update in (5.6) is finite arithmetic. For *continuous* variables each message is a probability *density*, an infinite-dimensional object, and the update integrals must be carried out in function space. Tree exactness (a property of the graph) says nothing about representability of the messages (a property of the model's function class); keeping the two apart is essential, and the distinction is where all the difficulty of the non-Gaussian research chapter lives. There are three ways to make continuous BP computable: restrict to models whose messages close in a finite-dimensional family (the Gaussian route, next section); represent messages numerically on a grid (the route our non-Gaussian reference solver takes, Chapter 8); or project each message onto a tractable family and accept the error (the route whose cost this thesis measures).

5.6 Gaussian belief propagation and the Kalman filter

The Gaussian family is the canonical closed message family, and its closure is pure algebra.

Toolbox 5.1 — The two closure lemmas of Gaussian message passing

Lemma 1 (products). For Gaussians in the same variable,

$$\mathcal{N}(a; m_1, v_1) \times \mathcal{N}(a; m_2, v_2) \propto \mathcal{N}(a; m, v), \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \quad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2} :$$

the sum of two quadratics in a is a quadratic in a ; precisions and precision-weighted means add, as the natural-parameter rule of Section 5.1 requires.

Lemma 2 (affine marginalisation). Pushing a Gaussian through a linear-Gaussian transition,

$$\int \mathcal{N}(a'; \alpha a, \sigma^2) \mathcal{N}(a; m, v) da = \mathcal{N}(a'; \alpha m, \alpha^2 v + \sigma^2),$$

because $a' = \alpha a + \eta$ is a sum of independent Gaussians. Read in the reverse direction the same computation gives $\mathcal{N}(a; m'/\alpha, (v' + \sigma^2)/\alpha^2)$.

A model whose factors are Gaussian in each variable performs, in (5.6), only these two operations. For such models every message is Gaussian, carried exactly by two numbers, and each update is a few arithmetic operations. $\square$

The oldest and most famous instance predates factor graphs by decades. For linear-Gaussian state-space models (a Gaussian Markov chain observed through linear Gaussian noise), the forward sum-product sweep is the *Kalman filter* (Kalman, 1960), alternating a *predict* step (Lemma 2, push the belief through the dynamics) and an *update* step (Lemma 1, absorb the new observation, with the correction weighted by the Kalman gain); the backward sweep combined with the forward one yields the fixed-interval smoother of Rauch et al. (1965), which conditions each state on *all* observations. That the classical filter and smoother are exactly the Gaussian specialisation of sum-product on the chain's factor graph was made explicit by Kschischang et al. (2001); textbook treatments are Bishop (2006); Koller and Friedman (2009). Chapter 7 derives the filter from the BP sweep, line by line, for our diffusion posterior, where the observation factors take a specific form (a pseudo-observation whose strength depends on the diffusion time), and verifies the equivalence numerically to double-precision rounding.

One remark, recorded here because it resolves a doubt raised repeatedly in supervision: the Gaussian closure of this section is *algebra*, valid at any node degree, and invokes no limit theorem. It must not be confused with the central-limit justification of the *approximate* Gaussian closures of the next section, which does require many incoming messages and which a chain, with two neighbours per node, does not provide.

5.7 From BP to TAP and AMP

Statistical physics developed its own message-passing calculus for *dense* graphs, where every node interacts weakly with many others. On such graphs the incoming messages to a node are numerous and nearly independent, so their product is governed by a central-limit argument: the cavity field is approximately Gaussian, and each message can be

summarised by a mean and a variance even when the underlying variables are not Gaussian. Historically the resulting self-consistency equations appeared first as the Thouless–Anderson–Palmer (TAP) equations of the Sherrington–Kirkpatrick spin glass (Thouless et al., 1977), whose distinctive ingredient is the Onsager reaction term, the correction that removes a node’s own influence from the field it feels. Three decades later the same structure re-emerged in compressed sensing as *approximate message passing* (AMP) (Donoho et al., 2009): a reduction of BP on a dense bipartite graph to iterations on means and variances, with the Onsager term reappearing as a memory correction. The derivation of AMP from BP by moment expansion is carried out for the ℓ_2 minimisation problem by Genovese and Piana (2026); the general technology, its state-evolution analysis, and its phase diagrams are reviewed by Zdeborová and Krzakala (2016) and, in lecture-note form, by Krzakala and Zdeborová (2024). Mézard (2017) derived the TAP equations of the Hopfield model, connecting this calculus back to the networks of Chapter 2.

The logic of the CLT justification should be kept explicit, because this thesis probes exactly its boundary. On a dense graph with N weak interactions per node, each message removes one interaction out of N , message and marginal differ at order $1/N$, and sums of many weak, nearly independent terms are Gaussian: the per-node Gaussian closure is controlled. On a chain every node has two neighbours: no aggregation takes place, each message carries the entire influence of half the system, and nothing justifies a Gaussian form for it beyond the special cases where it is exact. What happens when one imposes the closure anyway is a quantitative question. This thesis answers it on the chain in two complementary ways: exactly, for the per-node (AMP/TAP) closure on the Gaussian chain, where the breakdown admits a closed-form phase boundary (Appendix B); and numerically, for the Gaussian-projected messages on genuinely non-Gaussian chains (Chapter 8).

5.8 Locality in learned architectures: convolutions, receptive fields, U-Nets

The message-passing story has an architectural mirror, and it supplies the vocabulary for the locality questions of the research chapters. A convolutional network processes a signal by applying the same local filter at every position, stacking such layers so that depth enlarges the *receptive field*: the window of input coordinates that can influence a given output coordinate (LeCun et al., 2015). Convolutional architectures powered the breakthrough of deep learning in vision (Krizhevsky et al., 2012), and their central inductive bias is precisely locality plus translation invariance: the value of an output pixel is computed from a bounded neighbourhood, identically across the image. The standard denoising backbone of diffusion models, the U-Net (Ronneberger et al., 2015; Ho et al., 2020), is a convolutional encoder–decoder with skip connections, whose receptive field grows along the contracting path and whose skip connections reinject fine-scale local information along the expanding path.

Why local architectures should be *sufficient* for score estimation is a theoretical question, and message passing has begun to answer it. Mei (2024) shows that for generative hierarchical models (tree-structured graphical models over image-like data), the denoising function computed by belief propagation can be implemented efficiently by a U-Net, with the network’s layers mirroring the sweeps of the message-passing schedule; convolutional layers are efficient implementations of the local BP updates. This gives the U-Net an inference-algorithm reading: the architecture is good at denoising because denoising, on structured data, *is* message passing, and message passing is local.

The present thesis works one step below this correspondence, on a model small enough that the locality of the score is not an architectural hypothesis but a computable quantity. For the Gaussian chain, the exact score at frame k depends on the observation at frame $k \pm d$ with a weight that decays geometrically in d , at a rate $q(\alpha, t)$ computed in closed form in Chapter 7; the error of the best radius- r local approximation decays as q^r , and the comparison between truncating the inference and truncating the score matrix quantifies,

in one solvable case, how a local computation should be organised. We state the obvious caveat once: no network is trained in this thesis, so these results describe the function a network would have to learn, not the network that learns it.

Chapter 6

Preliminary Work: From Toy Models to a Minimal Model

This chapter records how the questions of Chapter 1 took their final form: the initial framing, the exploratory models built in the first months of the project, the conceptual error those models exposed, and the corrections from supervision that fixed the object of study. We record this material because each failed formulation constrained the next one, and the sequence explains why the final model looks the way it does. All preliminary code, interactive laboratories, and notes are archived in a dedicated preliminary-work repository (`Score_Diffusion`, directory `Toy-models`), kept separate from the reproducibility package of Appendix C and available together with it.

6.1 The initial framing: dynamic objects

The project began from a simple observation about the diffusion literature. Standard diffusion models generate *static* objects; the data we wanted to understand are *dynamic*: a data point is a sequence of frames indexed by an internal time $u = 0, 1, \dots, K - 1$ (a video, a rotating shape, a stochastic trajectory). The naive treatment embeds the whole trajectory in $\mathbb{R}^{dK}$ and learns a single monolithic score, which ignores the fact that consecutive frames are strongly correlated. The initial research hypothesis was correspondingly

ambitious: that the joint score field of a dynamic object has an internal structure of its own, concretely that consecutive frame-blocks of the score might be related by a fixed map (a “propagator”),

$$S_{:,u}(x, t) \approx \mathcal{P}_t(S_{:,u-1}(x, t)), \quad (6.1)$$

which a learning system could exploit by learning one block and one map instead of K blocks. Two distinct times appear already in this formulation, and keeping them apart became a permanent discipline of the project: the *internal* time u of the sequence, and the *diffusion* time t of the noising channel, which acts independently on every frame.

6.2 First models from supervision: structured data on a circle

The first concrete models were proposed in early supervision meetings, and they were geometric. Take planar data on a circle, $x = r \cos \theta$, $y = r \sin \theta$, and give the two polar coordinates simple independent dynamics in the internal time: the radius fluctuates around the unit circle under a restoring potential $V(r) = \frac{1}{2}(r - 1)^2$, i.e. an Ornstein–Uhlenbeck process $dr = (1 - r) du + \Gamma dB_u$, while the phase advances with a drift. A data point is the whole trajectory of T frames (the working value on the board was $T = 40$), embedded in $\mathbb{R}^{2T}$; generative diffusion then corrupts every coordinate as $x^\mu(t) = a^\mu e^{-t} + \sqrt{1 - e^{-2t}} \xi^\mu$, which is exactly the channel of Section 3.6. A second variant from the same discussions kept the circle but made the dynamics deterministic: points on a circle rotated by a fixed angle per frame, observed through noise. These models have precisely the structure the thesis is about (an internal time with a simple dynamical rule, coupled to a diffusion time), but they are nonlinear in the coordinates in which the diffusion acts, so no closed form for their joint score was available at that stage. They set the target, and something simpler had to supply the intuition first. Chapter 9 returns to the first of them once the solvable case is in place, and shows that a Cartesian surrogate of it is exactly solvable while the model as drawn is solvable numerically to a measured

accuracy.

6.3 The toy models

The exploratory phase produced seven small, self-contained interactive models (browser-based laboratories with exact formulas inside, no neural networks), built to make score fields visible before attempting theorems. In the numbering of the archive:

- *TM1: the OU channel on one frame.* A single univariate distribution transported by the forward OU flow, with the score and the reverse flow animated. The purpose was calibration: seeing (4.4) act on a multimodal density, and the reverse drift reassemble it.
- *TM2: two frames, additive dynamics.* The first genuinely dynamic object: $x_1 = x_0 + c + \eta$ with a trimodal prior on x_0 , constant drift c , and Gaussian innovation η . The joint density $p(x_0, x_1)$ is a product of the prior and the transition kernel, and the two-dimensional joint score is computable in closed form. Figure 6.1 shows the object this toy model was built around: the joint density and its score field, whose diagonal orientation is the visible fingerprint of the frame coupling.
- *TM3: rotation observed through OU noise.* The deterministic-dynamics variant of the board models: a planar rotation per frame, with the diffusion channel acting on the coordinates.
- *TM4: bounded deterministic dynamics.* A bouncing cube: a deterministic bounded trajectory, chosen to probe score fields for dynamics with reflections, where the stationary law is far from Gaussian.
- *TM5: two clocks.* Two coupled periodic phases advancing at different rates, the minimal model of a system with two internal frequencies.
- *TM6: the circle prior.* The static circle distribution itself, isolating the geometry of the score of a curved data manifold from any internal dynamics.

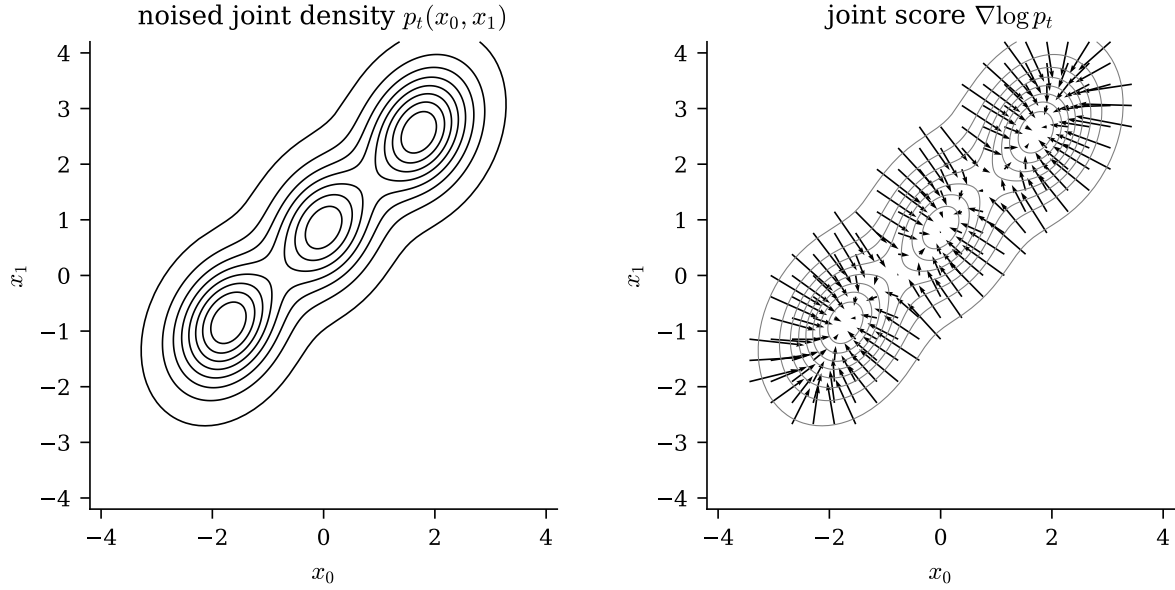


Figure 6.1: The two-frame toy model (TM2/TM7): $x_1 = x_0 + c + \eta$ with a trimodal Gaussian-mixture prior on x_0 , drift $c = 1$, innovation standard deviation $\sigma = 0.45$, after diffusion time $t = 0.15$. Left: the noised joint density $p_t(x_0, x_1)$. Right: the joint score field $\nabla \log p_t$ (arrows) over the same density (contours). Both panels are computed from the closed-form mixture expressions, on a grid; the arrows point towards the modes and tilt along the diagonal, the visible effect of the coupling between the two frames.

- *TM7: the two-frame joint score laboratory.* The synthesis of TM1–TM6 built after the correction described next: an interactive computation of the *joint* two-frame score, with the coupling between frames on explicit display.

The toy models delivered the geometric intuition they were built for, and one lasting technical observation: everywhere a closed form existed, the score’s cross-frame components were controlled by the *conditional* structure of the joint law, not by the marginal law of single frames. The toy models also exposed a conceptual error in how the project was computing scores.

6.4 The joint-score correction

In the early experiments, scores were computed *per frame*: each frame’s noisy marginal was treated as a one-dimensional distribution, its score computed independently, and the

trajectory treated in effect as K separate samples. Supervision corrected the object of study: a sequence is *one* sample from a joint law, and the quantity that drives the reverse dynamics of the sequence is the joint score of the whole noisy trajectory, i.e. the gradient of $\log P_t(x_0, \dots, x_{K-1})$, whose k -th component is a conditional expectation over the *full* posterior of the clean frames, as in (4.8).

For the stationary models of this thesis the correction changes the object of study completely. Under the variance-preserving channel every noisy frame is marginally $\mathcal{N}(0, 1)$ at every diffusion time, so the per-frame marginal score is $-x_k$: independent of t , independent of the dynamics, identical for every model in this thesis. All information about the temporal structure lives in the joint object. Chapter 7 exhibits the contrast in closed form at $K = 2$. The correction also reframed the propagator hypothesis (6.1): the structure worth studying is not a map between per-frame scores but the coupling pattern of the joint score, which is a statement about a matrix (in the Gaussian case) or about a posterior (in general).

6.5 The minimal model

The joint-score reframing selected the model of the research chapters. Among the candidate dynamic objects (circles, rotations, clocks), the stationary Gaussian AR(1) chain under coordinatewise OU noise is the minimal one in which the joint score is nontrivial yet exactly computable: one coupling parameter α for the internal dynamics against one noise parameter t for the corruption, linear algebra available at every step, and every structural question (coupling range, locality, message passing) posable in closed form. The nonlinear board models were set aside at that point, deliberately and explicitly, as targets for after the solvable case was understood. Chapter 9 takes up the first of them.

6.6 The message-passing programme and its bottleneck

Since the joint score is a smoothing posterior mean, supervision proposed computing it by belief propagation on the chain’s factor graph and asking three questions: whether the recursion reproduces the exact Gaussian score, what happens for non-Gaussian innovations, and how accurate local truncations are. Working out the recursion identified the central obstruction, already stated abstractly in Section 5.5: for continuous variables the messages are functions, and a forward–backward sweep is computable only if the messages are representable. This bottleneck fixed the final protocol of the thesis: prove the Gaussian closure where it is exact (Chapter 7), and for a non-Gaussian chain compare the Gaussian-projected recursion against a numerically validated grid representation of the true functional messages (Chapter 8).

6.7 A discarded conjecture

One intermediate conjecture is recorded because its failure shaped the final scope. For the Laplace chain at $K = 2$, numerical explorations suggested that the posterior log-density Hessian might be approximately rank-two, which would have opened a cheap approximate inference route. Systematic tests falsified the conjecture outside a narrow parameter regime, and it was dropped; the negative result, with the regime in which the approximation does hold, is stated in Section 8.6. The episode reinforced the working rule adopted throughout the thesis: a pattern observed in one corner of parameter space is a hypothesis, not a result, until an audit sweeps the space.

Chapter 7

The Gaussian Chain: Exact Score, Belief Propagation, and Locality

This chapter answers RQ1, RQ2, and the Gaussian half of RQ4. For the stationary AR(1) chain under the OU channel we compute the joint score exactly by two independent routes (linear algebra and Bayesian smoothing), characterise how its coupling structure evolves with diffusion time, derive the belief-propagation computation of the same score and identify it with the Kalman filter and smoother, and quantify how rapidly the influence of remote observations decays. Everything in this chapter is exact unless explicitly marked as a numerical check; each boxed formula is covered by a named check of the audit suite (Appendix C).

7.1 The joint law and the object of study

A data point is a trajectory $a = (a_0, \dots, a_{K-1})$ of the stationary AR(1) chain (3.5) with $\sigma_\eta^2 = 1 - \alpha^2$. Each frame passes independently through the OU channel (3.11) for a time t ,

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (7.1)$$

and the object of study is the joint score of the noisy sequence,

$$S(x, t) = \nabla_x \log P_t(x) \in \mathbb{R}^K, \quad P_t(x) = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}(x_k; e^{-t}a_k, \Delta_t) da. \quad (7.2)$$

The gradient is taken with respect to all K noisy frames simultaneously, under the law of the whole noisy trajectory (Section 4.6).

7.2 The clean law: dense covariance, sparse precision

The chain factorisation (3.2) gives the clean joint density

$$P_0(a) = \mathcal{N}(a_0; 0, 1) \prod_{k=0}^{K-2} \mathcal{N}(a_{k+1}; \alpha a_k, \sigma_\eta^2) = \mathcal{N}(a; 0, \Sigma_0). \quad (7.3)$$

Proposition 7.1 (Stationary covariance). $(\Sigma_0)_{ij} = \alpha^{|i-j|}$: a dense, symmetric Toeplitz matrix with geometrically decaying off-diagonals (Toolbox 3.1).

Proposition 7.2 (Tridiagonal precision). $Q_0 := \Sigma_0^{-1}$ is exactly tridiagonal:

$$(Q_0)_{kk} = \begin{cases} \frac{1}{\sigma_\eta^2}, & k \in \{0, K-1\}, \\ \frac{1+\alpha^2}{\sigma_\eta^2}, & 0 < k < K-1, \end{cases} \quad (Q_0)_{k,k\pm 1} = -\frac{\alpha}{\sigma_\eta^2}, \quad (7.4)$$

and every entry at distance ≥ 2 vanishes identically.

Toolbox 7.1 — Deriving the tridiagonal precision from the log-density

For a centred Gaussian, the precision is the Hessian of the negative log-density. From (7.3),

$$-2 \log P_0(a) = \frac{a_0^2}{\sigma_0^2} + \sum_{k=0}^{K-2} \frac{(a_{k+1} - \alpha a_k)^2}{\sigma_\eta^2} + \text{const}, \quad \sigma_0^2 = 1.$$

Read off the quadratic form term by term. The only term coupling a_k and a_{k+1} is the transition $(a_{k+1} - \alpha a_k)^2 / \sigma_\eta^2$, with cross coefficient $-2\alpha / \sigma_\eta^2$, hence precision entry $-\alpha / \sigma_\eta^2$; no term couples variables at distance ≥ 2 , so those entries are exactly zero. An

interior a_k appears in its left transition with coefficient $1/\sigma_\eta^2$ and in its right transition with α^2/σ_η^2 , summing to $(1 + \alpha^2)/\sigma_\eta^2$. At the left boundary the prior term $1/\sigma_0^2 = (1 - \alpha^2)/\sigma_\eta^2$ plus the right transition α^2/σ_η^2 give $1/\sigma_\eta^2$; the right boundary is symmetric.

□

Remark 7.3 (Tridiagonality is Markovianity, not Gaussianity). A zero in the precision matrix is a conditional independence given all other variables; by the Hammersley–Clifford theorem (Section 5.3), factorisation over the cliques of a graph and conditional independence with respect to that graph are the same property. Gaussianity entered the derivation only to identify the precision with the Hessian of $-\log P_0$; the support of that Hessian, nearest neighbours only, comes from the chain factorisation alone. Any Markov chain with smooth transitions has this sparsity pattern; the Gaussian case makes the entries constants.

The covariance is dense (every frame correlates with every other) while the precision is tridiagonal: correlation is global, conditional dependence is local. The score is governed by the second object.

7.3 The score, two ways

7.3.1 Linear algebra

Proposition 7.4 (Noised law). $P_t(x) = \mathcal{N}(x; 0, \Sigma_t)$ with

$$\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t I_K \quad (7.5)$$

This is (3.12). The diagonal is $e^{-2t} + \Delta_t = 1$ at every t (variance preservation); only the off-diagonal entries, $\alpha^{|i-j|} e^{-2t}$, depend on the diffusion time, because the noise of different frames is independent. Since $\log P_t(x) = -\frac{1}{2} x^\top \Sigma_t^{-1} x + \text{const}$,

$$S(x, t) = -Q_t x, \quad Q_t := \Sigma_t^{-1}. \quad (7.6)$$

The joint score is linear in x , the hallmark of Gaussianity, and its entire structure is the noisy precision matrix Q_t : which frames couple, how strongly, and at which t , are questions about this one matrix, answered in Sections 7.4 and 7.7.

7.3.2 Bayesian inference

The score–posterior identity (4.8) expresses the same score through the posterior mean of the clean chain,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x] - x_k}{\Delta_t}, \quad (7.7)$$

and the posterior is Gaussian with parameters used throughout:

Proposition 7.5 (Posterior of the clean chain). $P(a | x) = \mathcal{N}(a; m, J^{-1})$ with

$$\boxed{J = \frac{e^{-2t}}{\Delta_t} I + Q_0, \quad J m = h := \frac{e^{-t}}{\Delta_t} x.} \quad (7.8)$$

Toolbox 7.2 — Posterior precision and field, by completing the square

Bayes' rule with the likelihood (7.1):

$$\begin{aligned} -2 \log P(a|x) &= a^\top Q_0 a + \sum_k \frac{(x_k - e^{-t} a_k)^2}{\Delta_t} + \text{const} \\ &= a^\top \left(Q_0 + \frac{e^{-2t}}{\Delta_t} I \right) a - 2 \frac{e^{-t}}{\Delta_t} x^\top a + \text{const}. \end{aligned}$$

Matching against the generic form $a^\top J a - 2h^\top a$ gives (7.8). Two structural observations. (i) The likelihood is diagonal in a : observing each frame through its own channel adds the channel precision e^{-2t}/Δ_t to every diagonal entry of the prior precision and touches nothing else. (ii) Consequently J is tridiagonal at every t : the *posterior* of the clean chain remains a nearest-neighbour Markov field; what fills up with t is the precision Q_t of the noisy marginal law. Keeping J (tridiagonal, posterior) and Q_t (dense, noisy marginal) distinct prevents much confusion. $\square$

Substituting $m = J^{-1}h$ into (7.7) must return $-Q_t x$; the operator identity behind this

is Woodbury's (Appendix A), and the audit suite checks the two routes agree to 2.7×10^{-15} across sweeps of (K, α, t) . The agreement is an end-to-end check of two computations built from different formulas.

7.3.3 The smallest example, $K = 2$

For $K = 2$, with $r := \alpha e^{-2t}$,

$$\Sigma_t = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix} \implies S_0(x, t) = -\frac{x_0 - r x_1}{1 - r^2}, \quad S_1(x, t) = -\frac{x_1 - r x_0}{1 - r^2}. \quad (7.9)$$

By contrast, each x_k is marginally $\mathcal{N}(0, 1)$ at every t (variance preservation), so the per-frame marginal score is $-x_k$: independent of t and of the other frame, it carries no information about the dynamics. The joint score contains the cross term $-rx_1/(1 - r^2)$: the denoiser of frame 0 uses evidence from frame 1 with a weight that equals α at $t = 0$ and decays as e^{-2t} . This is the coupling structure whose full lifecycle the next section derives, in miniature.

7.4 The spectral form and the lifecycle of the precision

How does Q_t interpolate between the tridiagonal Q_0 at $t = 0$ and the identity at $t = \infty$? The analysis rests on the spectral theorem and its affine-family corollary (Section 5.2): affine images of a symmetric matrix share its eigenbasis, $(cA + dI)u_i = (c\omega_i + d)u_i$. Applying this to the forward map (7.5), which is affine in Σ_0 , gives the representation from which everything in this section follows.

Theorem 7.6 (Spectral form). *Let $\Sigma_0 = U \operatorname{diag}(\omega_i) U^\top$. Then for every $t \geq 0$,*

$$\Sigma_t = U \operatorname{diag}(\lambda_i(t)) U^\top, \quad Q_t = U \operatorname{diag}\left(\frac{1}{\lambda_i(t)}\right) U^\top, \quad \lambda_i(t) = e^{-2t}\omega_i + \Delta_t. \quad (7.10)$$

Diffusion never rotates the problem: the eigenvectors of Σ_t and Q_t are those of Σ_0 , frozen for all t , while each eigenvalue flows along the affine path $\omega_i \mapsto e^{-2t}\omega_i + \Delta_t$, monotonically towards 1. Because Σ_0 is symmetric Toeplitz, its eigenvectors are asymptotically the Fourier modes of the sequence (Section 5.2): in frequency space the denoising problem is K independent scalar problems at every diffusion time. Two computational corollaries are recorded for later use: factoring Δ_t out of (7.5) gives the SNR form

$$Q_t = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}, \quad \gamma_t := e^{-2t}/\Delta_t, \quad (7.11)$$

and multiplying (7.5) by Q_0 and inverting gives the resolvent form

$$Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}, \quad c_t := e^{2t} - 1, \quad (7.12)$$

whose Neumann series converges for $\|c_t Q_0\| < 1$, i.e. at small t .

7.4.1 Losing tridiagonality: when, how, and how much

Sparsity is a property of a matrix in a basis. In the eigenbasis the problem is diagonal at all times; in the frame basis, the operationally meaningful one, the zeros of Q_0 are a cancellation that diffusion destroys. The destruction is quantifiable, and it answers three questions: when tridiagonality is lost, how, and how much structure survives at each t .

When: immediately. By (7.10) every entry of Q_t is analytic in t . The distance- d entries, identically zero at $t = 0$ for $d \geq 2$, are non-zero for every $t > 0$: no exact zero survives any positive diffusion time.

How: one diagonal per order in t . The loss is nevertheless graded, and the resolvent form gives the exact rate at which each diagonal switches on.

Theorem 7.7 (Band-fill leading order). *For small t and frame distance $d \geq 1$,*

$$(Q_t)_{i,i+d} = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i,i+d} + O(t^d). \quad (7.13)$$

Toolbox 7.3 — Band fill from the Neumann series

In the resolvent form (7.12), substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$:

$$Q_t = (1 + O(t)) \left[Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \cdots \right].$$

Q_0 is tridiagonal, so Q_0^m has bandwidth m : the entry at distance d receives its first contribution from the single term $m = d$, with prefactor $(-c_t)^{d-1} = (-1)^{d-1} (2t)^{d-1} (1 + O(t))$; every later term is $O(t^d)$. Each additional frame of coupling range costs one power of t : the band fills outward, one diagonal per order. $\square$

Remark 7.8 (Correction of an earlier formula). An earlier working note of this project stated (7.13) with an extraneous factor $1/(d-1)!$. The factor does not follow from the expansion (the leading distance- d term is a single Neumann term, with no combinatorial sum behind it), and the numerical audit rejects it: the no-factorial coefficient matches the measured $(Q_t)_{i,i+d}/t^{d-1}$ to better than 0.9% for every $d \in \{1, \dots, 5\}$ (at $\alpha = 0.9$, $K = 25$), while the factorial version fails by 86% already at $d = 3$. The error had survived because slope fits in $\log t$ are blind to constant prefactors; every prefactor in this thesis has since been audited, not only every exponent.

How much: the off-tridiagonal mass. Define the off-tridiagonal Frobenius mass of Q_t : the norm of everything outside the band that Markov structure would allow. At small t the band-fill law makes it rise from zero, dominated by the $d = 2$ diagonal, linearly in t . At large t , expanding the inverse of $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ gives

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}), \quad (7.14)$$

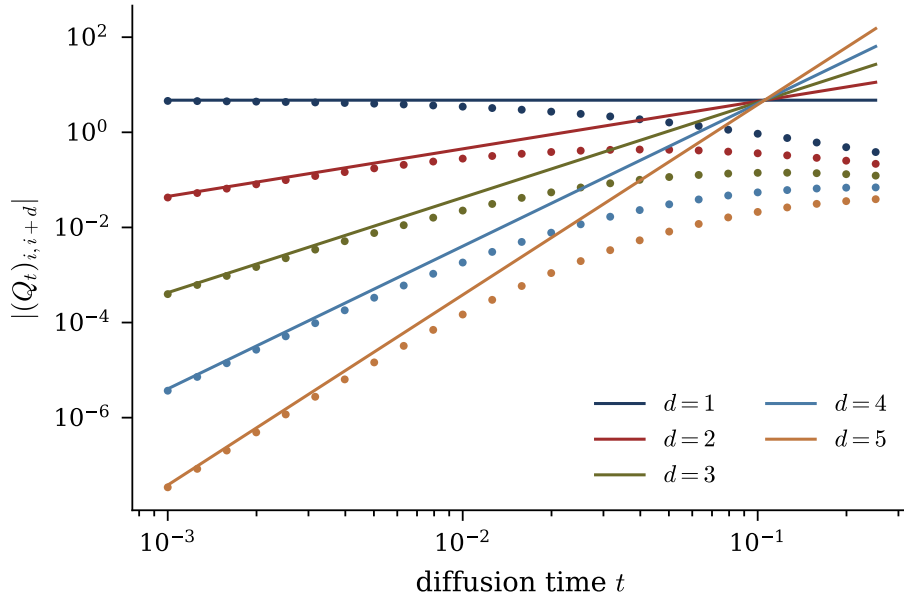


Figure 7.1: Band-fill law. Measured $|(Q_t)_{i,i+d}|$ against t (dots) and the leading order (7.13) (solid lines, no fitted parameters), for $\alpha = 0.9$, $K = 25$, $d = 1, \dots, 5$. The slopes are $d - 1$; agreement at small t is at the percent level for every d .

so $\|Q_t - I\|_F \simeq e^{-2t} \|\Sigma_0 - I\|_F$: all structure, off-tridiagonal and tridiagonal alike, decays at the rate e^{-2t} , and the score approaches the isotropic $-x$. In between, the off-tridiagonal mass peaks (Figure 7.2): the noisy sequence is most non-Markov at intermediate diffusion times. The lifecycle in one line: tridiagonal at $t = 0$; densifying at rate t^{d-1} per diagonal d ; maximally coupled at intermediate t ; isotropic as $t \rightarrow \infty$ at rate e^{-2t} .

7.5 Belief propagation on the chain

The matrix route inverts a $K \times K$ matrix, a global computation over the whole sequence. The score-posterior identity says the same object is a posterior expectation, and the posterior has a structure that supports local computation. This section develops that computation once, in full.

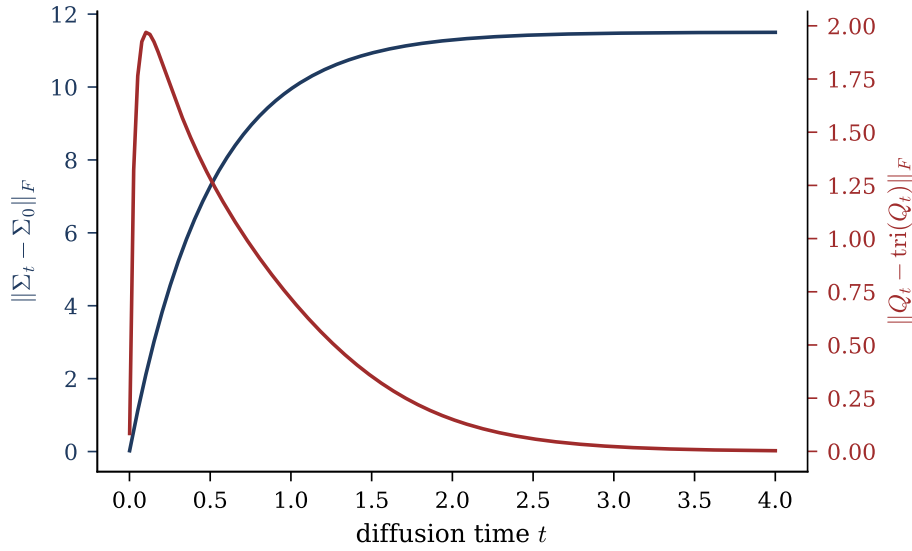


Figure 7.2: Structure across diffusion time ($\alpha = 0.8$, $K = 40$). $\|\Sigma_t - \Sigma_0\|_F$ (navy) grows monotonically as the covariance is driven to the identity; the off-tridiagonal Frobenius mass of Q_t (red) rises from zero, peaks at intermediate t , and decays to zero: the band fills, then empties.

7.5.1 The factor graph of the posterior

For any Markov-chain prior and per-frame OU evidence, Bayes' rule gives

$$p(a | x) \propto \psi_{\text{pr}}(a_0) \prod_{k=0}^{K-2} \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \prod_{k=0}^{K-1} \psi_{\text{ob}}^{(k)}(a_k), \quad \psi_{\text{ob}}^{(k)}(a_k) = \mathcal{N}(x_k; e^{-t} a_k, \Delta_t), \quad (7.15)$$

a hidden Markov model (Rabiner, 1989): a latent chain observed through independent per-frame noise. By (4.8), the k -th score component requires exactly the smoothing marginal $\mathbb{E}[a_k | x]$ of (7.15).

The factor graph of this posterior (Section 5.4) has K variable nodes, $2K$ factor nodes (one prior, $K-1$ transitions, K observations; the data x_k are constants clamped inside their factors) and $3K - 1$ edges. A connected graph with one fewer edge than nodes is a tree: here, a spine of variables and transition factors with one evidence leaf per variable.

Proposition 7.9. *For any Markov-chain prior and any per-frame observation model, the posterior $p(a | x)$ factorises over a tree-shaped factor graph, and all K smoothing marginals*

are computed exactly by one forward and one backward sweep of sum-product messages.

Proposition 7.9 counts sweeps, not work: it says nothing about how the messages are represented, and for continuous variables each message is a function (Section 5.5). Keeping tree-exactness of the algorithm (a graph property) separate from representability of the messages (a property of the model's function class) is the conceptual hinge of this thesis; the Gaussian case closes the gap, and Chapter 8 measures what happens when it no longer does.

7.5.2 The sum-product recursion, with the bookkeeping explicit

Specialising the sum-product rules (5.6) to this tree, the schedule collapses into one forward and one backward sweep (Theorem 5.5), and the bookkeeping deserves care.

Proposition 7.10 (Chain sweeps, Convention A). *Define messages*

$$\mu_{\rightarrow 0}(a_0) := \psi_{\text{pr}}(a_0), \quad \mu_{\rightarrow k+1}(a_{k+1}) := \int \psi_{\text{tr}}^{(k)}(a_k, a_{k+1}) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\rightarrow k}(a_k) \, da_k, \quad (7.16)$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \quad \mu_{\leftarrow k-1}(a_{k-1}) := \int \psi_{\text{tr}}^{(k-1)}(a_{k-1}, a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) \, da_k. \quad (7.17)$$

Then for every k ,

$$p(a_k | x) \propto \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) : \quad (7.18)$$

the forward message into a_k carries the evidence $x_0, \dots, x_{k-1}$ only, the backward message carries $x_{k+1}, \dots, x_{K-1}$ only, and the local evidence enters once, at combination time.

The proof is direct marginalisation: integrating (7.15) over all a_j , $j \neq k$, splits at a_k by the Markov property into independent left and right halves, and peeling the innermost variable of each half yields the recursions.

Remark 7.11 (The double-counting trap). An alternative convention absorbs the local evidence into the forward message, $\mu_{\rightarrow k}^{\text{B}} = \mu_{\rightarrow k} \cdot \psi_{\text{ob}}^{(k)}$; the product (7.18) then counts x_k twice and must be corrected by dividing one factor back out. An early version of this

project's notes fell into this trap, caught by the numerical audit: under Convention A the BP score matches the matrix score to 10^{-14} , under the naive alternative the disagreement is of order one. Every message in this thesis is therefore stated with its evidence content explicit.

7.6 The Gaussian closure and the Kalman smoother

For the Gaussian chain, the two closure lemmas of Toolbox 5.1 are the only facts needed: the recursions (7.16)–(7.17) perform exactly one product and one affine marginalisation per step. The local evidence, read as a function of a_k , is itself Gaussian, $\psi_{\text{ob}}^{(k)}(a_k) \propto \mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$: a pseudo-observation at the deconvolved location $e^t x_k$ with precision e^{-2t}/Δ_t , the pinning strength that appears throughout this chapter.

Theorem 7.12 (Gaussian closure of BP on the chain). *For the Gaussian AR(1) chain under the OU channel, every BP message and every belief is exactly Gaussian, at every node, for every (K, α, t) . Each message is carried by two numbers, each update is a few arithmetic operations, and the two sweeps compute the exact posterior marginals, hence the exact joint score. Here, and only here, the sweep count of Proposition 7.9 translates into an actual computation of fixed, small cost per frame.*

Derivation. Induction along each sweep. The initial messages are Gaussian (the prior; the flat message is the zero-precision limit of the family). Each step of (7.16)–(7.17) is one application of Lemma 1 of Toolbox 5.1 (absorb the local evidence) followed by one of Lemma 2 (push through the transition); the combination (7.18) is Lemma 1 twice. Exactness follows from tree exactness of sum-product (Theorem 5.5). $\square$

Remark 7.13 (Closure is algebra, not a central limit theorem). The closure just proved holds at node degree two exactly as it would at any degree: it is algebra, not an aggregation effect, as discussed in Section 5.6. The central-limit argument matters only for the per-node closures of statistical physics (Section 5.7), and on the chain those fail in a way that

can be made exact, with a breakdown time $t_c(\alpha)$ and a critical coupling $\alpha_c = \sqrt{2} - 1$ (Appendix B).

Executing the Gaussian sweeps in mean–variance arithmetic reproduces, update by update, the classical algorithms of linear estimation.

Toolbox 7.4 — The Kalman filter, derived from the BP sweep

Write the forward message in Convention A as $\mu_{\rightarrow k} = \mathcal{N}(a_k; m_{k|k-1}, P_{k|k-1})$, the predicted state given past observations. One BP step does two things.

Update (absorb the evidence; Lemma 1). Multiplying by the pseudo-observation $\mathcal{N}(a_k; e^t x_k, \Delta_t e^{2t})$ gives the filtered Gaussian $\mathcal{N}(a_k; m_{k|k}, P_{k|k})$ with

$$\begin{aligned} \frac{1}{P_{k|k}} &= \frac{1}{P_{k|k-1}} + \frac{e^{-2t}}{\Delta_t}, \\ m_{k|k} &= P_{k|k} \left(\frac{m_{k|k-1}}{P_{k|k-1}} + \frac{e^{-t} x_k}{\Delta_t} \right) = m_{k|k-1} + G_k (e^t x_k - m_{k|k-1}), \end{aligned}$$

where $G_k = P_{k|k} e^{-2t} / \Delta_t \in (0, 1)$ is the Kalman gain: the filtered mean is the prediction corrected by a gain-weighted innovation (Kalman, 1960).

Predict (push through the dynamics; Lemma 2).

$$m_{k+1|k} = \alpha m_{k|k}, \quad P_{k+1|k} = \alpha^2 P_{k|k} + \sigma_\eta^2.$$

The forward BP sweep is the Kalman filter, message by message.

Smoothing (combine both directions). The belief (7.18) multiplies the filtered forward Gaussian by the backward message (Lemma 1 once more), producing the posterior marginal given all observations, $\mathcal{N}(a_k; m_{k|K-1}, P_{k|K-1})$; carrying out the same algebra for the backward recursion yields the fixed-interval smoother of Rauch et al. (1965). The two-sweep BP schedule on the chain is the Kalman filter followed by the RTS smoother, exactly as the generic correspondence of Section 5.6 predicts. $\square$

Combining with the score–posterior identity answers RQ2: the k -th component of the

joint diffusion score is an affine function of the Kalman-smoothed estimate of frame k ,

$$S_k(x, t) = \frac{e^{-t} m_{k|K-1} - x_k}{\Delta_t}. \quad (7.19)$$

For Markov sequences, score computation is a smoothing problem, and the classical theory of linear smoothing applies directly.

Numerical agreement of the two routes. The audit suite runs BP against the matrix route over 80 configurations ($K \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1, 3\}$, random x): maximum disagreement 1.3×10^{-15} on posterior means, 3.9×10^{-15} on variances, 1.2×10^{-14} on scores, i.e. double-precision rounding accumulated along the recursion. An independent implementation, written from scratch against the recursion specification alone, reproduces the same agreement (Appendix C). One qualitative detail from a worked $K = 3$ instance: at $x = (1.2, -0.4, 0.8)$ the score component S_1 is positive although $x_1 < 0$, because the neighbours pull the posterior mean of the middle frame positive. No per-frame marginal score can produce this; it is inter-frame evidence sharing in numbers.

7.7 The bulk in closed form, and the price of locality

For a long chain the interior (bulk) is translation invariant and can be solved exactly. By Proposition 7.5 the posterior precision in the bulk is tridiagonal Toeplitz with

$$\boxed{J_d = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2}{\sigma_\eta^2}, \quad \beta = -\frac{\alpha}{\sigma_\eta^2}, \quad \sigma_\eta^2 = 1 - \alpha^2.} \quad (7.20)$$

The diagonal J_d is the sum of the evidence precision e^{-2t}/Δ_t (divergent as $1/2t$ at small t , vanishing at large t) and the constant prior precision; the coupling β is pure prior. The dimensionless ratio $|\beta|/J_d \in (0, \frac{1}{2})$ controls everything below.

Theorem 7.14 (Exact bulk variance and geometric covariance). *In the bulk of a long*

chain, the posterior variance and covariance are

$$V = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (J^{-1})_{i,i+d} = q^d V, \quad q = \frac{J_d - \sqrt{J_d^2 - 4\beta^2}}{2|\beta|} \in (0, 1). \quad (7.21)$$

Toolbox 7.5 — Transfer-matrix inversion of a tridiagonal Toeplitz matrix

Seek the inverse row in geometric form, $(J^{-1})_{i,i+d} = Cq^d$ for $d \geq 0$, symmetric in $\pm d$. The defining relation $\sum_j J_{ij}(J^{-1})_{j,i+d} = \delta_{0d}$ reads, for $d \geq 1$,

$$\beta Cq^{d-1} + J_d Cq^d + \beta Cq^{d+1} = 0 \iff \beta q^2 + J_d q + \beta = 0,$$

a quadratic whose root inside the unit disc is $q = (J_d - \sqrt{J_d^2 - 4\beta^2})/(2|\beta|)$ (for $\alpha > 0$ all inverse entries are positive; for $\alpha < 0$ they alternate in sign with the same modulus). The $d = 0$ relation fixes the prefactor: $J_d C + 2\beta Cq = 1$, and since $2\beta q = -(J_d - \sqrt{J_d^2 - 4\beta^2})$, $C = 1/\sqrt{J_d^2 - 4\beta^2} = V$. Existence of the real square root requires $J_d > 2|\beta|$, which always holds here:

$$J_d - 2|\beta| = \frac{e^{-2t}}{\Delta_t} + \frac{1 + \alpha^2 - 2|\alpha|}{\sigma_\eta^2} = \frac{e^{-2t}}{\Delta_t} + \frac{(1 - |\alpha|)^2}{\sigma_\eta^2} > 0$$

for every $|\alpha| < 1$ and $t > 0$. This positivity margin is also what guarantees the belief-propagation fixed point exists for every coupling and diffusion time (Appendix B).

□

The number $q(\alpha, t)$ measures how far evidence travels along the chain: the posterior correlation length is $\xi = 1/\log(1/q)$. Its limits are

$$\begin{aligned} t \rightarrow 0 : \quad q &\approx \frac{|\beta|}{J_d} \rightarrow 0 \quad (\text{each frame pinned by its own observation}); \\ t \rightarrow \infty : \quad q &\rightarrow |\alpha| \quad (\text{posterior} \rightarrow \text{prior}), \end{aligned} \quad (7.22)$$

the second because $q \rightarrow ((1+\alpha^2) - (1-\alpha^2))/(2|\alpha|) = |\alpha|$ when the evidence term vanishes. As diffusion time runs, the posterior correlation length climbs monotonically from 0 to

the prior's own $1/\log(1/|\alpha|)$.

7.7.1 The locality law

RQ4 asks how rapidly the influence of remote observations decays. Define the local estimator of radius r : the posterior mean of a_k given only the observations within distance r , $m_k^{(r)} = \mathbb{E}[a_k | x_{k-r}, \dots, x_{k+r}]$. Two facts make the comparison clean. Any contiguous window of a stationary AR(1) chain is itself a stationary AR(1) chain, so $m_k^{(r)}$ is exact on its window; the only approximation is the truncation of range. And both estimators are linear in x , so the error $m_k - m_k^{(r)} = e^\top x$ has a closed-form RMS over the data law, $\text{RMS}_r = \sqrt{e^\top \Sigma_t e}$, a deterministic number with no sampling involved.

Theorem 7.15 (Locality error decays as q^r). *In the bulk, $\log \text{RMS}_r$ decreases linearly in r with slope $\log q(\alpha, t)$: truncating the evidence range costs a factor q per discarded frame.*

The mechanism is Theorem 7.14: the influence of evidence at distance d on the posterior mean is proportional to $(J^{-1})_{k, k \pm d} \propto q^d$, and the leading discarded evidence sits at distance $r + 1$. The audit verifies the rate: fitted slopes of $\log \text{RMS}_r$ over $r = 2, \dots, 13$ at the centre of a $K = 121$ chain match $\log q$ within 0.2% at $t \in \{0.1, 0.5, 2.0\}$ ($\alpha = 0.8$); see Figure 7.3.

7.7.2 Truncating the matrix versus truncating the inference

A truncated score can be built in two inequivalent ways. *Truncate the answer*: band the matrix, $\hat{S} = -Q_t^{(b)} x$ with entries beyond $|i - j| \leq b$ zeroed. *Truncate the algorithm*: let messages travel at most r hops, i.e. run exact inference on the window $x_{k-r}, \dots, x_{k+r}$ (the local estimator above). Both are linear in x , so their RMS errors are again deterministic. At equal locality budget ($b = r = 1$, $K = 40$, $\alpha = 0.8$):

relative RMS score error	$t = 0.05$	$t = 0.3$	$t = 1$	$t = 3$
banded matrix ($b = 1$)	0.210	0.302	0.135	0.0035
truncated inference ($r = 1$)	0.123	0.235	0.130	0.0035

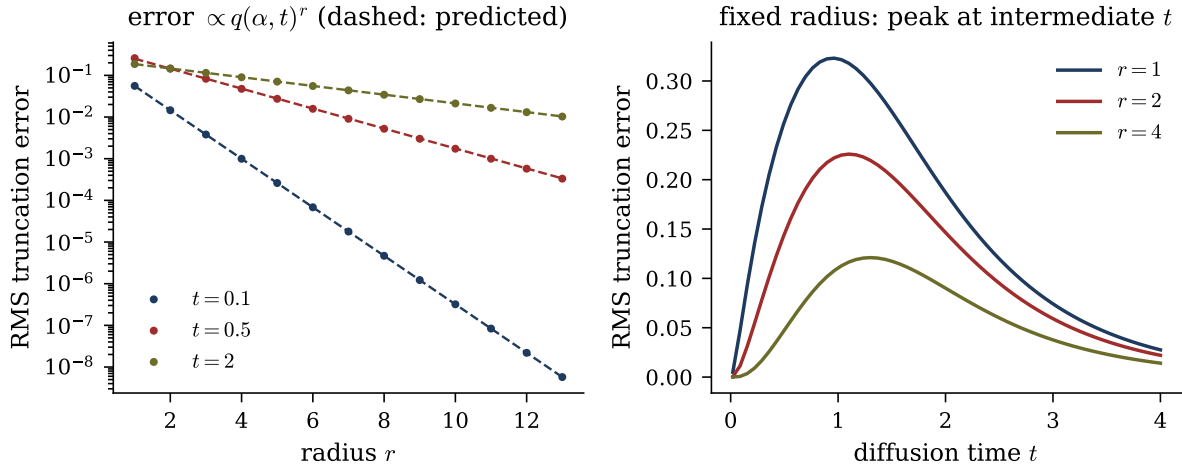


Figure 7.3: Locality of the joint score ($\alpha = 0.8$, $K = 121$, deterministic RMS over the data law). Left: exact truncation error of the radius- r local estimator (dots) at three diffusion times; the dashed lines have the predicted slopes $q(\alpha, t)^r$ from (7.21), with no fitting. Right: the same error at fixed radius as a function of t : it vanishes at both ends and peaks at intermediate t , where the band of Q_t is fullest.

Two observations. The cost of locality is non-monotone in t and peaks at intermediate times, consistent with Section 7.4.1. And at small t truncating the algorithm beats truncating the answer (12.3% against 21.0% at $t = 0.05$): the windowed inference solves a correctly normalised problem on its window, while zeroing entries of an inverse produces a matrix that is the inverse of nothing and misstates the diagonal. The two coincide at large t , where the off-diagonal content vanishes. Either error decays geometrically in the allowed range at the rate $q(\alpha, t)$ of Theorem 7.15.

7.8 Status of the results

Exact, by derivation: the score formulas (7.6) and (7.7), the spectral form (7.10), the band-fill law (7.13) as a small- t asymptotic, the large- t law (7.14), tree exactness and Gaussian closure of BP, the Kalman identification, and the bulk formulas (7.20)–(7.21) with the locality rate. Numerical, within stated grids and tolerances: the agreement of the two score routes, the audited band-fill coefficients, the locality slopes, and the truncation table. Interpretation, limited to this model: within the Gaussian benchmark, a local score

computation with receptive field $r \gtrsim \xi \log(1/\varepsilon)$, $\xi = 1/\log(1/q)$, achieves relative error ε , and the requirement is most demanding at intermediate diffusion times; whether trained score networks on real sequence data behave accordingly is not tested here.

Chapter 8

Beyond Gaussianity: Laplace Innovations and the Gaussian-Message Approximation

Every exact result of Chapter 7 rests on the closure of the Gaussian family under the two operations the model performs, multiplying densities and marginalising. This chapter removes that property in the most controlled way available and answers RQ3. We keep the AR(1) recursion, the OU channel, and the factor-graph structure, and change only the innovation law from Gaussian to Laplace: heavy-tailed and kinked at the origin, yet log-concave and analytically tractable, it is a minimal step outside the Gaussian family, so every deviation we find is attributable to non-Gaussianity alone. The chapter first establishes what remains exact (the score-posterior identity, the tree structure, one boundary integral, and a differential characterisation of the $K = 1$ marginal), then locates precisely where closed forms stop, and finally measures, against a validated numerical reference, what a Gaussian message family costs.

8.1 What changes, and what survives

The Laplace AR(1) chain replaces the Gaussian innovations of (3.5) with Laplace ones:

$$P_0(a) = \frac{1}{(2b)^K} \exp\left(-\frac{1}{b} \sum_{k=0}^{K-1} |a_k - \alpha a_{k-1}|\right), \quad a_{-1} := 0, \quad (8.1)$$

observed through the same per-frame OU channel. Three structural facts survive unchanged, because none of them used Gaussianity of the prior. The score-posterior identity (4.8) holds for any p_0 . The posterior factorises over the same tree-shaped factor graph (Proposition 7.9), so the sum-product recursion (7.16)–(7.17) still computes exact smoothing marginals in one forward and one backward sweep. And the OU likelihood factors are still Gaussian in a_k . What is lost is representability: the messages are no longer Gaussian, no finite-dimensional family we know of is closed under the updates, and the cost of an update is no longer bounded. The score also ceases to be linear in x , which is the first thing to make precise.

8.2 One frame: the score becomes nonlinear

For $K = 1$ there is no chain; the model isolates the ingredient “non-Gaussian prior meets Gaussian channel”. Let $a \sim \text{Laplace}(0, b)$, $p_0(a) = e^{-|a|/b}/2b$, and write $\mu := e^{-t}$. The noised marginal is the convolution

$$p_t^{(1)}(x) = \int p_0(a) \mathcal{N}(x; \mu a, \Delta_t) da. \quad (8.2)$$

As $t \rightarrow 0$ the Gaussian kernel concentrates and $p_t^{(1)} \rightarrow p_0$; as $t \rightarrow \infty$, $p_t^{(1)} \rightarrow \mathcal{N}(0, 1)$, the OU attractor. By the score-posterior identity, $S(x, t) = (\mu \mathbb{E}[a|x] - x)/\Delta_t$; for a Laplace prior the conditional mean is a nonlinear function of x , and the score inherits the

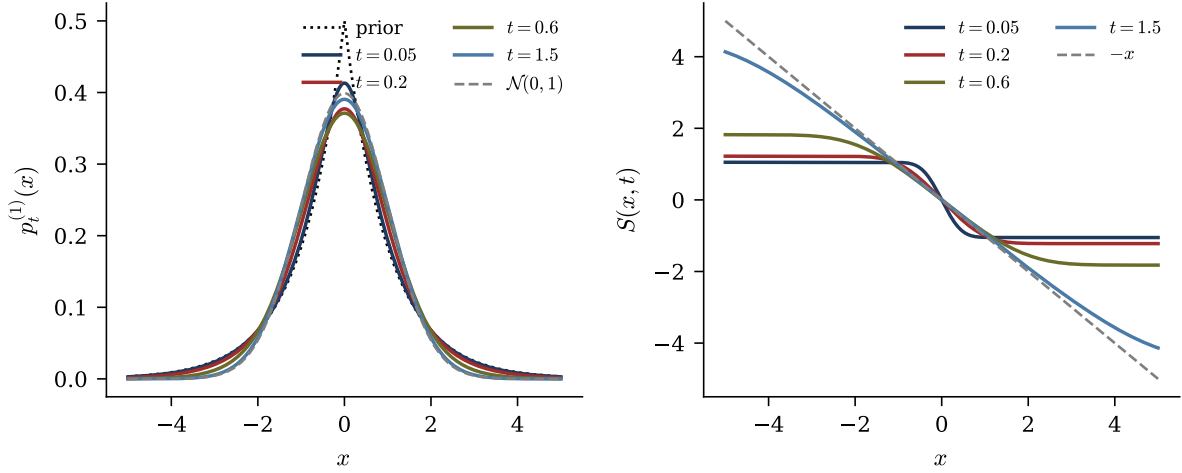


Figure 8.1: The Laplace $K = 1$ problem ($b = 1$). Left: noised marginal $p_t^{(1)}$ at increasing diffusion times, from the kinked Laplace prior (dotted) to the Gaussian attractor (dashed). Right: the score at the same times; the dashed line is the attractor $-x$. The nonlinearity lives near $x = 0$ and flattens as t grows; the limits are the exact statements (8.3). Curves computed by deterministic quadrature.

nonlinearity. Its limits are exact: for every fixed $x \neq 0$,

$$t \rightarrow 0 : \quad S(x, t) \longrightarrow \partial_x \log p_0(x) = -\frac{\text{sign}(x)}{b}, \quad t \rightarrow \infty : \quad S(x, t) \longrightarrow -x. \quad (8.3)$$

At small t the score converges pointwise to the sign-shaped score of the prior, with the nonlinearity concentrated in the kink near $x = 0$; at large t it flattens into the linear attractor score $-x$ (Figure 8.1). The audit verifies the posterior-expectation route against central finite differences of $\log p_t^{(1)}$ computed by quadrature: maximum disagreement 1.19×10^{-7} over a grid of (b, t, x) , at the truncation floor of the finite difference.

8.2.1 A differential characterisation of the noised marginal

The channel writes x as a sum of independent variables, so (8.2) is a convolution, which factorises in Fourier space:

$$\hat{p}_t^{(1)}(k) = \frac{1}{1 + b^2 \mu^2 k^2} e^{-\Delta_t k^2 / 2}, \quad (8.4)$$

the product of the (rescaled) Laplace and Gaussian characteristic functions.

Theorem 8.1 (Screened Poisson equation). *The Laplace noised marginal solves the linear, constant-coefficient ODE*

$$\boxed{(1 - b^2 \mu^2 \partial_x^2) p_t^{(1)}(x) = G_{\Delta_t}(x)}, \quad G_{\Delta_t}(x) = \frac{e^{-x^2/2\Delta_t}}{\sqrt{2\pi\Delta_t}}. \quad (8.5)$$

Toolbox 8.1 — From the characteristic function to the screened Poisson equation

Multiply (8.4) by $(1 + b^2 \mu^2 k^2)$:

$$(1 + b^2 \mu^2 k^2) \hat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2},$$

whose right-hand side is the characteristic function of G_{Δ_t} . Multiplication by k^2 in Fourier space is $-\partial_x^2$ in real space, so the polynomial $(1 + b^2 \mu^2 k^2)$ becomes the operator $(1 - b^2 \mu^2 \partial_x^2)$, and (8.5) follows by inverse transformation. The audit checks (8.4) against direct quadrature of the characteristic function to 2.9×10^{-6} , the floor of oscillatory quadrature. $\square$

Equation (8.5) states that the noised marginal is the solution of a screened Poisson equation with a Gaussian source, with screening length $b\mu = be^{-t}$: the memory of the prior's tails shrinks as e^{-t} . The deviation from Gaussianity is governed by a single local differential operator whose strength dies with diffusion time.

One further exact consequence identifies what a Gaussian approximation must get wrong. Differentiating the score-posterior identity once more,

$$H_t(x) := -\partial_x^2 \log p_t^{(1)}(x) = \frac{1}{\Delta_t} \left(1 - \frac{\mu^2}{\Delta_t} \text{Var}[a | x] \right). \quad (8.6)$$

For a Gaussian prior the posterior variance is independent of x and H_t is constant. For the Laplace prior $\text{Var}[a|x]$ depends on x and H_t is a nontrivial function: the breakdown of a constant curvature is the second-order form of the breakdown of score linearity. A Gaussian message is exactly the object that cannot represent an x -dependent curvature,

so (8.6) states what the Gaussian closure of Section 8.5 will miss, before it is measured.

8.3 The chain: where closed forms stop

Theorem 8.2 (Joint score from chain messages). *For any chain prior, in particular (8.1), and the OU likelihood, the joint score is, componentwise,*

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k|x] - x_k}{\Delta_t}, \quad \mathbb{E}[a_k|x] = \frac{\int a_k \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}{\int \mu_{\rightarrow k}(a_k) \psi_{\text{ob}}^{(k)}(a_k) \mu_{\leftarrow k}(a_k) da_k}, \quad (8.7)$$

with the messages of Proposition 7.10.

This is a complexity statement: the K -dimensional posterior integral is replaced by two one-dimensional message recursions plus one one-dimensional integral per frame. Computational locality survives non-Gaussianity; what does not survive is the collapse of those one-dimensional integrals into closed-form arithmetic. Exactly one such integral has a closed form.

Proposition 8.3 (One-step boundary integral). *For the Laplace transition $M(a'|a) = \frac{1}{2b} e^{-|a' - \alpha a|/b}$ and OU evidence $\mathcal{N}(x'; \mu a', \Delta_t)$,*

$$\int_{\mathbb{R}} M(a'|a) \mathcal{N}(x'; \mu a', \Delta_t) da' = p_t^{(1)}(x' - \mu \alpha a) : \quad (8.8)$$

the first backward message of the chain is the $K = 1$ noised marginal evaluated at a shifted argument.

The proof is the innovation substitution $r := a' - \alpha a$, which turns the transition into the bare Laplace density at r and the evidence into $\mathcal{N}(x' - \mu \alpha a; \mu r, \Delta_t)$; the integral is then the definition (8.2). The shift $\mu \alpha a$ is what frame a predicts for the next noisy observation, and the whole $K = 1$ analysis, screened Poisson equation included, recycles as the chain's boundary building block.

Remark 8.4 (Why iteration destroys the closed form). Equation (8.8) works because the integrand contains exactly one Laplace kink times exactly one Gaussian. Every subsequent BP step violates this: the incoming message already carries the propagated structure of several frames, so the next integrand contains two or more absolute-value kinks. On each sign-quadrant the integral reduces to half-line Gaussians, but globally it is a sum of error functions whose arguments depend on the conditioning variable non-affinely, and the clean form does not survive one iteration. In the Gaussian chain every step is Gaussian-times-Gaussian and closure propagates forever; this is the precise structural location of the difference between the two models, and the reason the Gaussian projection below is an approximation with an error to be measured, not an identity.

A change of coordinates shows the obstruction is intrinsic, not an artefact of the frame basis. In innovation coordinates $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ (unit Jacobian), the prior factorises into i.i.d. Laplace variables, but the likelihood becomes $p(x|r) \propto \exp(-\frac{\mu^2}{2\Delta_t} r^\top L^\top L r + \frac{\mu}{\Delta_t} x^\top L r)$ with L the unit lower-triangular map $a = Lr$, and $L^\top L$ is dense. The coupling does not disappear; it migrates from the prior to the likelihood. There is no basis in which both are diagonal at once.

8.4 A validated numerical reference: grid BP

Since functional messages admit no closed form, the exact recursion is approximated numerically: messages are represented by their values on M equispaced points in $[-A, A]$ and the update integrals are carried out by trapezoidal quadrature, each update touching every pair of grid points. This *grid BP* is a numerical approximation of the functional-message recursion, and it can serve as a reference only inside a validated regime, which we establish on the exactly solvable Gaussian chain. There the trapezoid rule on analytic, rapidly decaying integrands converges spectrally, so failure is abrupt rather than gradual: at $M = 101$, $A = 8$ the score error sits at the 10^{-15} floor down to $t = 0.02$ and fails only when the likelihood width $\sqrt{2t}$ approaches the grid step. The working configuration for all non-Gaussian experiments is $M = 401$, $A = 8$, with a reference solution at $M = 801$ and

the resolution constraint $\sqrt{2t} \gtrsim 3 dx$ checked automatically; grid self-convergence error remains 10^4 to 10^7 times smaller than every effect measured below. Within this regime we treat grid BP as the reference score; outside it no claim is made.

All score errors are amplified posterior-mean errors: for any two solvers with means $\widehat{m}$ and m_{ref} , the score-posterior identity gives, exactly,

$$\widehat{s} - s_{\text{ref}} = \frac{e^{-t}}{\Delta_t} (\widehat{m} - m_{\text{ref}}), \quad \frac{e^{-t}}{\Delta_t} \sim \frac{1}{2t} \quad (t \downarrow 0). \quad (8.9)$$

The numerical residual of this identity is logged in every experiment and stays at 10^{-15} to 10^{-12} . Equation (8.9) explains in advance why small diffusion times are the difficult regime: whatever an approximation does to the posterior mean is amplified by $1/2t$ in the score.

8.5 The Gaussian-message approximation, measured

The object under test is Gaussian-projected BP: the same sweeps, with every message forced into the Gaussian family by moment matching. This is the algorithm one would deploy when quadrature is unaffordable, and the chain analogue of the per-node closures of statistical physics (Appendix B).

On the Laplace chain (chain correlation $\alpha = 0.85$, $K = 40$ frames, 60 trials per t , reference $M = 801$), the median relative score error of the Gaussian closure is

t	0.02	0.08	0.4	2.4
median relative score error	0.39	0.17	3.5×10^{-2}	7.6×10^{-5}

with failure probability $\mathbb{P}(\text{err} > 0.1)$ equal to 1 for $t \leq 0.08$ and 0 for $t \geq 0.9$ (Figure 8.2). The closure error is of order tens of percent at small diffusion time and decays below 10^{-3} once diffusion noise Gaussianises the tilted posteriors. Inspection of the worst trials shows the mechanism: the reference belief is sharply non-Gaussian, occasionally bimodal; the single Gaussian commits to a compromise mean; and (8.9) amplifies the shift. The score *direction* is much more robust than its magnitude: the median cosine similarity between

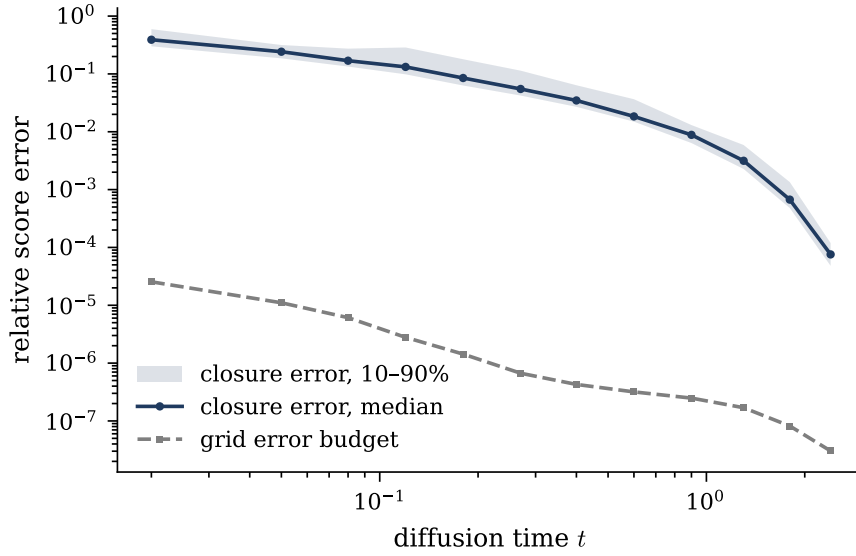


Figure 8.2: Gaussian-closure score error on the Laplace chain ($\alpha = 0.85$, $K = 40$, 60 trials per t ; median and 10–90% band) against the grid error budget measured on identical data. The closure error exceeds the reference error by 10^4 to 10^7 at every t , so the measurement is trustworthy where it is used.

projected and reference scores is 0.92 at $t = 0.02$ (where the magnitude error is 39%), exceeds 0.99 for all $t \geq 0.12$, and the worst single trial of the sweep stays above 0.78. These are numerical observations for the stated parameter ranges, not theorems.

8.5.1 Varying the innovation law

Sweeping six innovation families with identical covariance $\alpha^{|i-j|}$ (Laplace, Student- t with $\nu = 5, 8$, and symmetric Gaussian mixtures of increasing separation) isolates the shape of the innovation law as the only moving part. Three observations, all within the tested grids. First, the noise level dominates: every family’s error decays to about 10^{-3} or below by $t = 2.4$; all substantial closure error lives at $t \lesssim 0.4$. Second, bimodality is the worst case: at $t = 0.05$, $\alpha = 0.5$, the median error is 0.94 for a strongly bimodal mixture, 0.52 for Laplace, 0.32 for Student- $t_{\nu=5}$, and 0.07 for a mildly bimodal mixture; a single Gaussian cannot represent two posterior modes and commits to the gap between them. Third, stronger chain correlation helps rather than hurts: for Laplace at $t = 0.05$ the error is 0.43, 0.27, 0.13 at $\alpha = 0.5, 0.85, 0.95$; informative neighbours act like additional

Gaussian observations and Gaussianise the local tilted posteriors.

8.6 What can and cannot be concluded

Exact: the surviving structure of Section 8.1, the limits (8.3), the factorisation (8.4), the screened Poisson equation (8.5), the curvature identity (8.6), the message form (8.7), the boundary integral (8.8), the breakdown argument of Remark 8.4, and the error identity (8.9). Numerical, within validated regimes: the quadrature audits of Section 8.2, the grid-BP calibration, and the closure-error measurements, which are specific to the stated innovation laws, correlation levels, chain lengths, and diffusion times. Not concluded: closed forms for general (non-boundary) messages at $K \geq 2$; any locality law for the non-Gaussian score beyond the structural delimitation above; and any statement about learned scores. An earlier conjecture of this project, that the Laplace $K = 2$ posterior Hessian is approximately rank-two, proved regime-dependent under numerical tests and is not asserted.

Chapter 9

The rotating ring: a two-dimensional dynamic object

Chapters 7 and 8 studied a chain of scalar frames, where the joint score is either exactly Gaussian or a controlled departure from it. Section 6.2 recorded a second family of dynamic objects proposed in supervision, planar trajectories near a circle, and set them aside because they are nonlinear in the coordinates in which the diffusion acts. This chapter returns to the first of them. It shows that a Cartesian surrogate of the model is exactly solvable, that the model as drawn is solvable numerically to a measured accuracy, and that both admit the same structural description of the joint score.

The question the chapter answers is narrower than the models it uses.

How can the Markov structure of the clean rotating trajectory be exploited to compute and interpret the joint diffusion score?

Four measurable subquestions make this operational: how far the joint score differs from a collection of one-frame marginal scores; how strongly the observation at frame j influences the score block at frame k ; how that influence decays with the temporal lag $|j - k|$; and how many neighbouring frames are needed to reproduce the score block to a stated accuracy. Every answer below is a statement about one model at one parameter set, in the sense fixed in Section 1.3.

9.1 Model and notation

9.1.1 The clean process

The clean state is a radius and a phase, evolving in the internal time u as two independent diffusions,

$$dR_u = -\kappa(R_u - 1) du + \sqrt{2D_r} dB_u^{(r)}, \quad (9.1)$$

$$d\Theta_u = \omega du + \sqrt{2D_\theta} dB_u^{(\theta)}, \quad (9.2)$$

observed through the polar embedding

$$a_u = R_u (\cos \Theta_u, \sin \Theta_u) \in \mathbb{R}^2. \quad (9.3)$$

We use the simplest constant-angular-drift reading of the model; Section 9.10 records the alternatives.

A point of $\mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis, and at each point the plane carries the orthonormal moving frame

$$e_r = (\cos \theta, \sin \theta), \quad e_\theta = (-\sin \theta, \cos \theta), \quad (9.4)$$

in which e_θ is e_r turned by a quarter revolution. Moving along e_r changes the radius at fixed angle, moving along e_θ changes the angle at fixed radius, and the area element is $dx dy = r dr d\theta$. The decomposition matters because the two directions are treated very differently by (9.1) and (9.2): the radius is pulled back towards one, the phase drifts and diffuses freely. Measuring the consequence of that asymmetry for the score is one purpose of the chapter.

9.1.2 The radial drift is a confining force

Equation (9.1) is the overdamped Langevin equation (3.14) for the quadratic potential

$$V(r) = \frac{\kappa}{2} (r - 1)^2. \quad (9.5)$$

A potential stores energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa (r - 1). \quad (9.6)$$

The sign carries the content: a system driven by $-V'$ moves downhill in V , since $\frac{d}{du}V(r_u) = V'(r)\dot{r} = -V'(r)^2 \leq 0$. Reading the two cases of (9.6), if $r > 1$ then $F < 0$ and the force points inward, and if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards one from either side at a rate proportional to the displacement, with κ setting the stiffness. By (3.15) the stationary radial law is $\mathcal{N}(1, D_r/\kappa)$.

9.1.3 One trajectory is one data point

The trajectory is sampled at internal times $u_k = kh$ for $k = 0, \dots, K-1$, and the complete datum is the stacked vector

$$a = (a_0, a_1, \dots, a_{K-1}) \in \mathbb{R}^{2K}. \quad (9.7)$$

The K frames are correlated coordinates of one data point, not K samples. Relative to Chapter 7 the only change of setting is that each frame is now two-dimensional, so the score has K blocks of size two rather than K scalar components; the forward channel is unchanged,

$$x = e^{-t}a + \sqrt{\Delta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_{2K}), \quad (9.8)$$

which is the channel of Section 3.6 acting on all $2K$ coordinates at once. Figure 9.1 shows the model and the two clocks: internal time orders the frames along the path, diffusion time corrupts them simultaneously.

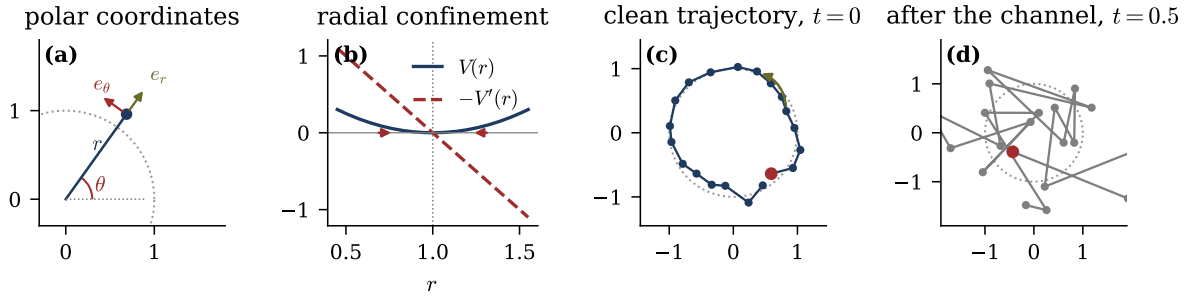


Figure 9.1: The rotating-ring model, at the parameters used throughout this chapter ($K = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, $\omega = 1$, $h = 2\pi/20$). (a) Polar coordinates and the moving frame (9.4). (b) The potential (9.5) and the force (9.6) it generates; the force changes sign at $r = 1$, which is therefore the stable radius. (c) One clean trajectory, a noisy coherently turning path near the unit circle; the arrow marks the sense of rotation. (d) The same trajectory after the channel (9.8) at $t = 0.5$, from a single standardised noise realisation.

9.2 Two identities for the joint score

The score-posterior identity of Section 4.6 applies unchanged. Written per block, with the expectation taken under the posterior of the clean trajectory given the whole noisy one,

$$S_k(x, t) = \frac{e^{-t} \mathbb{E}[a_k | x_0, \dots, x_{K-1}] - x_k}{\Delta_t} \in \mathbb{R}^2. \quad (9.9)$$

The score block at frame k is a *smoothing* estimate: it uses every observation, before and after k , to estimate one clean frame. Two consequences of (9.9) are used repeatedly and are derived here.

9.2.1 The joint and marginal objects differ by a conditioning set

The one-frame marginal score denoises frame k from frame k alone,

$$s_k^{\text{marg}}(x_k, t) = \frac{e^{-t} \mathbb{E}[a_k | x_k] - x_k}{\Delta_t}, \quad (9.10)$$

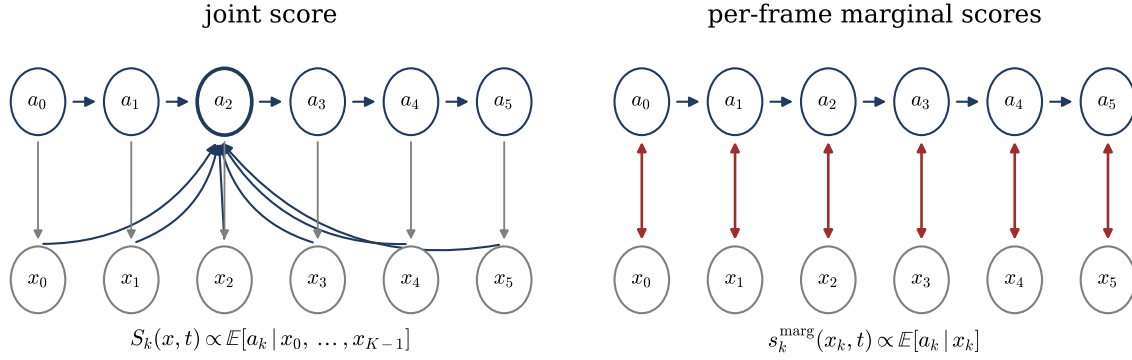


Figure 9.2: The two objects compared in (9.11). Clean frames a_k form a Markov chain in internal time and each emits one noisy observation x_k through the channel (9.8). Left: computing S_2 requires the posterior of a_2 given all observations, so every x_j has a path to a_2 . Right: the per-frame marginal score conditions on x_k alone. The magnitude of the difference is measured in Sections 9.4.5 and 9.7.

and subtracting it from (9.9) isolates the difference exactly:

$$S_k(x, t) - s_k^{\text{marg}}(x_k, t) = \frac{e^{-t}}{\Delta_t} \left(\mathbb{E}[a_k | x_{0:K-1}] - \mathbb{E}[a_k | x_k] \right) \quad (9.11)$$

The gap is the extra denoising information supplied by the rest of the trajectory, and nothing else. Figure 9.2 states the same thing as a graphical model, in the form used in Chapter 5: the joint score is a smoother, in which every observation has a path to every clean frame, whereas the product-of-marginals score severs all cross-frame paths before denoising begins.

9.2.2 The response of the score is a posterior covariance

The second subquestion asks how strongly the observation at frame j influences the score block at frame k . The answer is an exact identity, and every quantity measured later in the chapter is an instance of it.

Toolbox 9.1 — The observation as an external field

Step 1: write the posterior as a Gibbs measure with a linear field. By Bayes and (9.8),

$$\begin{aligned}
 p(a | x) &\propto P_0(a) \exp \left[- \frac{\|x - e^{-t}a\|^2}{2\Delta_t} \right] \\
 &= P_0(a) \underbrace{\exp \left[- \frac{\|x\|^2}{2\Delta_t} \right]}_{\text{no } a} \exp \left[- \frac{e^{-2t}}{2\Delta_t} \|a\|^2 + \frac{e^{-t}}{\Delta_t} x^\top a \right] \\
 &\propto \underbrace{P_0(a) \exp \left[- \frac{e^{-2t}}{2\Delta_t} \|a\|^2 \right]}_{\nu(a), \text{ independent of } x} e^{h^\top a}, \quad h = \frac{e^{-t}}{\Delta_t} x.
 \end{aligned}$$

Expanding $\|x - e^{-t}a\|^2$ produces exactly three kinds of term: one free of a , absorbed by the normalisation; one quadratic in a and free of x ; and one bilinear term. The whole dependence on the data therefore enters through a single vector h coupled *linearly* to a . In the language of Section 2.4, the observation at frame j acts as an external field $h_j = (e^{-t}/\Delta_t) x_j$ conjugate to the clean state a_j .

Step 2: differentiate the log-normalisation once. With $Z(h) = \int \nu(a) e^{h^\top a} da$, so that $p(a | x) = \nu(a) e^{h^\top a} / Z(h)$,

$$\frac{\partial \log Z}{\partial h_j} = \frac{1}{Z} \int a_j \nu(a) e^{h^\top a} da = \mathbb{E}[a_j | x]. \quad (9.12)$$

$\log Z$ is the cumulant generating function of the posterior.

Step 3: differentiate twice. Differentiating (9.12) again, using $\partial_{h_k} e^{h^\top a} = a_k e^{h^\top a}$ and $\partial_{h_k} Z^{-1} = -Z^{-2} \partial_{h_k} Z$,

$$\frac{\partial^2 \log Z}{\partial h_k \partial h_j} = \mathbb{E}[a_k a_j | x] - \mathbb{E}[a_k | x] \mathbb{E}[a_j | x] = \text{Cov}(a_k, a_j | x), \quad (9.13)$$

so that the derivative of a posterior mean with respect to the field is a posterior covariance, $\partial_{h_j} \mathbb{E}[a_k | x] = \text{Cov}(a_k, a_j | x)$.

Step 4: change variables back to the data. Since $h = (e^{-t}/\Delta_t)x$, the chain rule gives $\partial/\partial x_j = (e^{-t}/\Delta_t)\partial/\partial h_j$ and

$$\frac{\partial}{\partial x_j} \mathbb{E}[a_k | x] = \frac{e^{-t}}{\Delta_t} \text{Cov}(a_k, a_j | x). \quad (9.14)$$

Step 5: differentiate the score. Apply $\partial/\partial x_j$ to (9.9). The posterior mean contributes (e^{-t}/Δ_t) times (9.14), the subtracted x_k contributes $-\delta_{kj}I_2/\Delta_t$, which gives (9.15). $\square$

Proposition 9.1 (Posterior response identity). *For every pair of frames k, j and every $t > 0$,*

$$J_{kj}(x, t) := \frac{\partial S_k}{\partial x_j} = -\frac{\delta_{kj}}{\Delta_t} I_2 + \frac{e^{-2t}}{\Delta_t^2} \text{Cov}(a_k, a_j | x) \quad (9.15)$$

where $\text{Cov}(a_k, a_j | x) \in \mathbb{R}^{2 \times 2}$ is the cross-covariance block of the two clean frames under the posterior given the whole noisy trajectory.

Two remarks fix the status of (9.15). First, it is a consistency check on itself: J is the Hessian $\nabla^2 \log P_t$, so it must be symmetric under exchanging the pairs (k, α) and (j, β) , and it is, because transposing the block of (9.15) exchanges a_k and a_j . Second, it is the static linear-response relation of Section 2.4: a derivative with respect to a field equals a covariance in one fixed measure. It is not a two-time dynamical fluctuation–dissipation theorem, and none is claimed.

For $j \neq k$ the response is proportional to a posterior cross-covariance. Perturbing a distant observed frame changes S_k if and only if the two clean states remain statistically coupled *after* conditioning on the whole noisy trajectory. The pattern of the score Jacobian is therefore a map of posterior information flow, and the prefactor e^{-2t}/Δ_t^2 , which tends to zero as t grows, is separate from that pattern. Keeping the two apart is the purpose of the next section.

9.3 Diagnostics

Subquestions two to four concern strength, reach, and sufficiency. These are three properties of one profile, and in the measurements below they move in different directions with diffusion time. Fix a central frame k and let

$$C_t(\ell) = \mathbb{E}_x \left[\|J_{k,k+\ell}(x, t)\|_F \right], \quad \ell = 0, 1, \dots, L_{\max}, \quad (9.16)$$

be the *response profile*: the mean Frobenius norm of the response block at lag ℓ , averaged over noisy trajectories and over the two sides $k \pm \ell$. By (9.15) the self-response $C_t(0)$ is dominated by $1/\Delta_t$.

Strength. The total cross-frame response is

$$I_{\text{off}}(t) = \sum_{\ell \geq 1} C_t(\ell). \quad (9.17)$$

It answers how much the other frames matter in total, and says nothing about where they sit.

Reach. The natural first choice is an e-folding length obtained by fitting $C_t(\ell) \approx C_t(1) \exp[-(\ell - 1)/\xi^{\text{exp}}]$. It is the finite-chain estimate of a correlation length, and in the low-noise regime it is informative. It is also unbounded: once the measured profile is nearly flat the fit is ill-conditioned, and in our measurements the fitted $\xi_{\theta}^{\text{exp}}$ exceeds 290 frames in a chain of 20 at the largest diffusion times. We therefore report it only where it is meaningful, and use as the working measure of reach the first moment of the response mass,

$$\bar{\ell}(t) = \frac{\sum_{\ell \geq 1} \ell C_t(\ell)}{\sum_{\ell \geq 1} C_t(\ell)}, \quad (9.18)$$

the centre of mass of the cross-frame response. It is measured in frames, is bounded by $L_{\max}$, and assumes no particular shape of profile. It has a known *structureless ceiling*: a flat profile gives $\bar{\ell} = (1 + L_{\max})/2$, which is 5.5 for the $L_{\max} = 10$ of this chapter. A

measured $\bar{\ell}$ near 5.5 therefore indicates that the temporal structure has been erased, not that the correlation is long. The same caveat applies to the shape-only normalised range $\xi^{\text{norm}}(t) = \sum_{\ell \geq 1} C_t(\ell)/C_t(1)$, whose ceiling is L_{max} , and which by construction cannot distinguish a strong short-range coupling from a negligible long-range one.

Strength and reach together. Intensity and mean lag answer complementary questions, and their product is a single number that responds to either:

$$\Xi(t) = I_{\text{off}}(t) \bar{\ell}(t) = \sum_{\ell \geq 1} \ell C_t(\ell), \quad \boxed{\tilde{\Xi}(t) = \frac{\Xi(t)}{C_t(0)}} \quad (9.19)$$

Ξ is the unnormalised first moment of the profile: it increases if the coupling becomes stronger or if it arrives from farther away, and it factorises exactly into how much times how far, so the two ingredients can always be read off separately. Dividing by the self-response makes it comparable across diffusion times, since $C_t(0) \simeq 1/\Delta_t$ removes the overall scale of the Jacobian, and leaves a quantity measured in frames that expresses the reach of the context relative to the local term competing with it. Like $\bar{\ell}$ it inherits a ceiling and must be read against it.

Sufficiency. Every diagnostic so far is infinitesimal. A direct test of the fourth subquestion is to recompute the score block from a truncated window $x_{k-L:k+L}$, ignoring the rest of the trajectory. Writing $S_k^{(L)}$ for that windowed score,

$$\varepsilon_L(t) = \left[\frac{\mathbb{E} \|S_k^{(L)} - S_k^{\text{full}}\|^2}{\mathbb{E} \|S_k^{\text{full}}\|^2} \right]^{1/2}, \quad L_\epsilon(t) = \min\{L : \varepsilon_L(t) \leq \epsilon\}. \quad (9.20)$$

L_ϵ carries the most direct modelling reading of the four, in the sense of Section 5.8: it is the temporal window a denoiser would need in order to reproduce the score to accuracy ϵ at that noise level. It is also the most expensive to compute, which is why the infinitesimal diagnostics are worth having.

Remark 9.2 (What a large reach does not imply). If every off-diagonal response is small

but nearly equal, then ξ^{norm} and $\bar{\ell}$ both saturate at their structureless ceilings while the total correction to the score is negligible. That is what occurs at large diffusion time, where the score approaches the field $-x$. Conversely I_{off} alone cannot separate a tightly local coupling from a diffuse one. Only the pair, or the weighted reach that combines them, is safe to quote, and no diagnostic here should be read as an infinite-volume correlation length: $K = 20$ is finite and no thermodynamic limit is taken.

9.4 An exactly solvable surrogate

9.4.1 The surrogate, and what it keeps

The surrogate retains a ring, a coherent rotation, and dependence along the trajectory, and drops the rest:

$$z_0 \sim p_\lambda(z) \propto \exp\left[-\frac{(\|z\| - 1)^2}{2\lambda}\right], \quad (9.21)$$

$$z_{k+1} = R_\psi z_k + \sigma \eta_{k+1}, \quad \eta_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad (9.22)$$

with R_ψ the planar rotation through the one-frame angle $\psi = \omega h$.

Remark 9.3 (The surrogate is not a discretisation of the polar model). In (9.22) the ring acts only on z_0 . Later frames form a rotating Cartesian random walk and their radial variance grows without bound, whereas in (9.1) a confining force acts at every internal time. The surrogate is a solvable companion whose role is to exhibit the mechanism in closed form, not to approximate the model of Section 9.1.

9.4.2 The known rotation is an orthogonal change of frame

The rotation is the object to be removed, so we write it out:

$$R_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}. \quad (9.23)$$

Its columns are the standard basis vectors, each turned through ψ . They are unit vectors, since $\cos^2 \psi + \sin^2 \psi = 1$, and orthogonal, since $\cos \psi(-\sin \psi) + \sin \psi \cos \psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \quad \det R_\psi = 1, \quad R_\psi^\top = R_{-\psi}, \quad R_\psi R_\varphi = R_{\psi+\varphi}, \quad (9.24)$$

and orthogonality gives $\|R_\psi z\|^2 = z^\top R_\psi^\top R_\psi z = \|z\|^2$: lengths and inner products are preserved.

Lemma 9.4 (Rotations preserve the standard isotropic Gaussian). *If $\eta \sim \mathcal{N}(0, I_2)$ and R is orthogonal, then $R\eta \sim \mathcal{N}(0, I_2)$.*

Proof. A linear map of a Gaussian is Gaussian, so $R\eta$ is fixed by its first two moments: $\mathbb{E}[R\eta] = R\mathbb{E}[\eta] = 0$ and $\text{Cov}(R\eta) = R I_2 R^\top = R R^\top = I_2$. Equivalently, in terms of densities, $(2\pi)^{-1} e^{-\|z\|^2/2}$ depends on z only through $\|z\|$, and the change of variables $z \mapsto Rz$ changes neither $\|R^{-1}z\| = \|z\|$ nor $|\det R^{-1}| = 1$: the level sets are circles centred at the origin and a rotation maps each circle to itself. $\square$

The isotropy is essential. A rotation would not preserve $\mathcal{N}(0, \Sigma)$ for general Σ , and the reduction below is exact only because both the innovation in (9.22) and the channel (9.8) are isotropic.

Lemma 9.5 (Gauge reduction). *Define the de-rotated frame $y_k = R_{-k\psi} z_k$ and stack the per-frame rotations into $U_\psi = \text{diag}(I_2, R_{-\psi}, \dots, R_{-(K-1)\psi})$. Then*

$$y_{k+1} = y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} \stackrel{iid}{\sim} \mathcal{N}(0, I_2), \quad (9.25)$$

U_ψ is orthogonal, and the noised density and score satisfy

$$P_t(z | \psi) = P_t^{(0)}(U_\psi z), \quad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t), \quad (9.26)$$

where the superscript zero denotes the quantities of the non-rotating model.

Proof. Substituting (9.22) and using the composition rule of (9.24),

$$y_{k+1} = R_{-(k+1)\psi} \left(R_\psi z_k + \sigma \eta_{k+1} \right) = \underbrace{R_{-(k+1)\psi} R_\psi}_{=R_{-k\psi}} z_k + \sigma R_{-(k+1)\psi} \eta_{k+1} = y_k + \sigma \tilde{\eta}_{k+1},$$

and by Lemma 9.4 each $\tilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$; they stay independent because each is a fixed linear map of a different independent η . Next, U_ψ is block diagonal with orthogonal blocks, so $U_\psi^\top U_\psi = \text{diag}(I_2, \dots, I_2) = I_{2K}$ and $\det U_\psi = \prod_k \det R_{-k\psi} = 1$. Applying U_ψ to (9.8) gives $U_\psi x = e^{-t} U_\psi a + \sqrt{\Delta_t} U_\psi \xi$ with $U_\psi \xi \sim \mathcal{N}(0, I_{2K})$, so $U_\psi x$ has exactly the law of the noised non-rotating trajectory: the de-rotation commutes with the channel. For the densities, $z = U_\psi^\top y$ and $P_t(z | \psi) = P_t^{(0)}(U_\psi z) |\det U_\psi|$, the Jacobian factor being one. Differentiating that identity componentwise, with $\partial y_b / \partial z_c = (U_\psi)_{bc}$,

$$\frac{\partial}{\partial z_c} \log P_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{bc} \left(\nabla \log P_t^{(0)} \right)_b (U_\psi z) = \left[U_\psi^\top S^{(0)}(U_\psi z, t) \right]_c.$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$ while a point transforms with U_ψ . $\square$

The term *gauge* is used in the sense standard in physics, of a change of description that leaves the content untouched and may be chosen to simplify the equations. Equation (9.26) says that a known rotation adds no statistical difficulty: de-rotate the data, solve the non-rotating problem, rotate the score back. Since $U_\psi^\top = U_\psi^{-1}$, the operation is exactly invertible and destroys no information. It would not hold for an *unknown* ψ , which is a genuine latent variable (Section 9.10). Figure 9.3 shows the two frames.

9.4.3 The clean law after the gauge

Iterating (9.25) and letting the intermediate terms cancel,

$$y_k - y_0 = \sum_{r=1}^k (y_r - y_{r-1}) = \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad \text{so} \quad y_k = A + \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad (9.27)$$

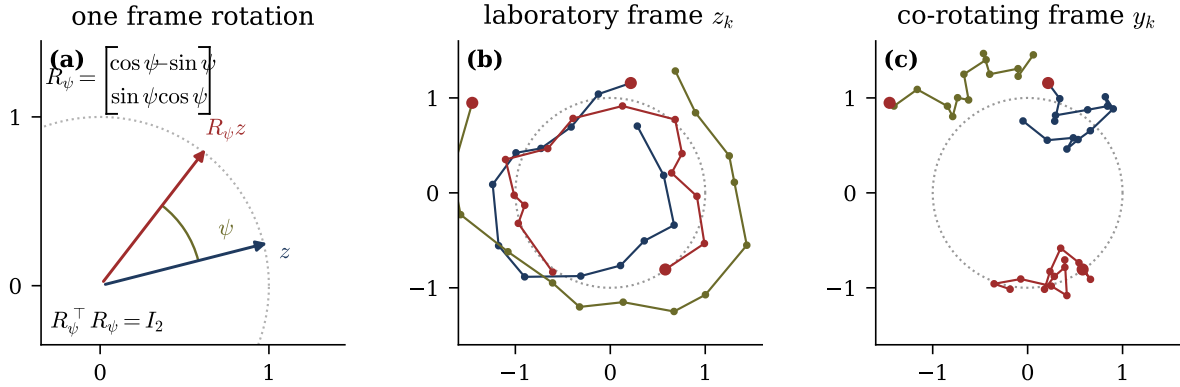


Figure 9.3: The rotation and the gauge. (a) R_ψ turns every vector through the same angle without changing its length, which is the statement $R_\psi^T R_\psi = I_2$. (b) Three surrogate trajectories in the laboratory frame ($K = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; the marked point is $k = 0$), coherently circulating. (c) The same trajectories after U_ψ : a random walk anchored on the ring. Because U_ψ is orthogonal the reduction is exact at every diffusion time; the numerical round trip agrees to machine precision (Table 9.3).

with $A := y_0 \sim p_\lambda$ and the empty sum equal to zero. Every frame contains the *same* random anchor, drawn once on the ring, plus a partial sum of independent innovations; the anchor is the only non-Gaussian ingredient of the model.

The clean law factorises over the ring prior and the one-step transitions, so $\log P_0(y) = -(\|y_0\| - 1)^2/2\lambda - (2\sigma^2)^{-1} \sum_{k=0}^{K-2} \|y_{k+1} - y_k\|^2 + \text{const}$. Differentiating is a counting exercise: an interior y_k appears in exactly two summands, the edge arriving at it and the edge leaving it, whose derivatives are $-(y_k - y_{k-1})/\sigma^2$ and $+(y_{k+1} - y_k)/\sigma^2$. Adding them, and using $\nabla_y \|y\| = y/\|y\|$ for the ring term,

$$\begin{aligned} S_0(y, 0) &= \frac{1 - \|y_0\|}{\lambda} \frac{y_0}{\|y_0\|} + \frac{y_1 - y_0}{\sigma^2}, \\ S_k(y, 0) &= \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}, \quad 1 \leq k \leq K-2, \\ S_{K-1}(y, 0) &= \frac{y_{K-2} - y_{K-1}}{\sigma^2}. \end{aligned} \tag{9.28}$$

The interior clean score is a discrete second derivative along internal time, that is a discrete Laplacian: each block measures how far frame k departs from the average of its neighbours. It is strictly local, and it vanishes identically on the model's own noiseless

solution, because a trajectory rotating at radius exactly one maps under the gauge to a constant y_k , for which every second difference vanishes and, with $\|y_0\| = 1$, so does the ring term. That vanishing is verified to machine zero (Table 9.3).

9.4.4 The noised law and the exact score

Conditioning on the anchor, (9.27) gives $\mathbb{E}[y_i | A = \alpha] = \alpha$ and

$$\text{Cov}(y_i, y_j | A) = \sigma^2 \mathbb{E} \left[\left(\sum_{r \leq i} \tilde{\eta}_r \right) \left(\sum_{s \leq j} \tilde{\eta}_s \right)^\top \right] = \sigma^2 \min(i, j) I_2, \quad (9.29)$$

since $\mathbb{E}[\tilde{\eta}_r \tilde{\eta}_s^\top] = \delta_{rs} I_2$ leaves only the $\min(i, j)$ indices common to both sums. Compactly, $y | A = \alpha \sim \mathcal{N}(\mathbf{1} \otimes \alpha, G \otimes I_2)$ with $G_{ij} = \sigma^2 \min(i, j)$, where $\mathbf{1} \in \mathbb{R}^K$ is the all-ones vector. The two Kronecker products separate the temporal index from the spatial one: $\mathbf{1} \otimes \alpha$ repeats the same two-dimensional anchor in every frame, and $(G \otimes I_2)_{(i, \alpha'), (j, \beta')} = G_{ij} \delta_{\alpha' \beta'}$, so G carries all the temporal correlation while I_2 states that the two spatial coordinates are uncorrelated and share the same temporal covariance. Applying the channel and using $I_{2K} = I_K \otimes I_2$ to combine the covariances,

$$x | A = \alpha \sim \mathcal{N}(e^{-t} \mathbf{1} \otimes \alpha, C_t \otimes I_2), \quad C_t = e^{-2t} G + \Delta_t I_K, \quad (9.30)$$

in which C_t does not depend on α : the noised law is a two-parameter location mixture of one fixed Gaussian.

Proposition 9.6 (Exact surrogate score and response). *Let $q = C_t^{-1} \mathbf{1}$, $\beta = e^{-2t} \mathbf{1}^\top C_t^{-1} \mathbf{1}$ and $h = e^{-t} \sum_j q_j x_j \in \mathbb{R}^2$. Then the anchor posterior is*

$$p(\alpha | x) \propto \exp \left[- \frac{(\|\alpha\| - 1)^2}{2\lambda} - \frac{\beta}{2} \|\alpha\|^2 + h^\top \alpha \right], \quad (9.31)$$

and the score and response blocks are

$$S_k(x, t) = - \left(C_t^{-1} x \right)_k + e^{-t} q_k \mathbb{E}[A | x], \quad J_{kj}(x, t) = - \left(C_t^{-1} \right)_{kj} I_2 + e^{-2t} q_k q_j \text{Cov}(A | x). \quad (9.32)$$

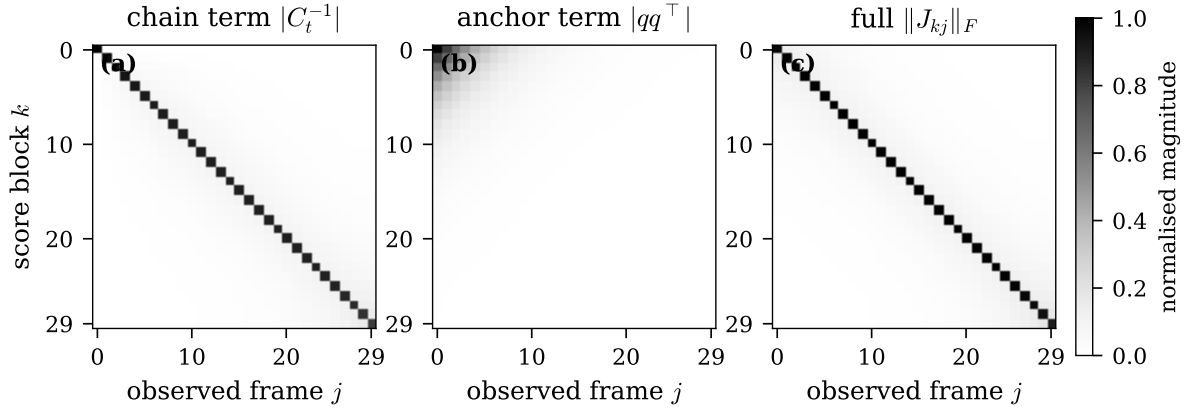


Figure 9.4: Exact decomposition of the surrogate response (9.32) at $t = 0.20$, $K = 30$, $\lambda = 0.05$, $\sigma = 0.15$, averaged over 250 noisy trajectories, each panel normalised by its own maximum. (a) The chain term $|C_t^{-1}|$ is concentrated near the diagonal. (b) The anchor coefficient $|qq^\top|$ is rank one and spread over all lags. (c) The mean norm of the complete 2×2 blocks. The analytic Jacobian agrees with central finite differences to 9.2×10^{-6} (Table 9.3).

Proof. Expanding the exponent of (9.30) in powers of α , as in Step 1 of Toolbox 9.1 but with the correlated covariance $C_t \otimes I_2$, gives (9.31). Writing $\log P_t(x) = \text{const} - \frac{1}{2}x^\top(C_t^{-1} \otimes I_2)x + \log Z(h(x))$, the gradient of the first term is $-(C_t^{-1}x)_k$ in block k , while $\partial h / \partial x_k = e^{-t}q_k I_2$ and $\nabla_h \log Z = \mathbb{E}[A | x]$ by (9.12). Differentiating once more and using (9.13) gives the response. $\square$

The $2K$ -dimensional integral over clean trajectories has collapsed to a two-dimensional posterior over the shared anchor, which is the whole simplification and a direct consequence of (9.27). Its two moments are cheap: the prior and the $\|\alpha\|^2$ term are rotationally symmetric, so the posterior depends on the direction of α only through $h^\top \alpha$, and $\mathbb{E}[A | x] = \mu(\|h\|)\hat{h}$ with $\text{Cov}(A | x) = v_\perp I_2 + (v_\parallel - v_\perp)\hat{h}\hat{h}^\top$, three scalar functions of one scalar argument, tabulated once per diffusion time by one-dimensional quadrature.

The structural content of (9.32) is that the temporal coefficient of the second term is the outer product $qq^\top$, which is of rank one. The nonlocal part of the response is therefore not a chain of local couplings but a single shared latent variable seen by every frame. Figure 9.4 separates the two terms.

Table 9.1: Surrogate diagnostics. “Marginal” is the relative RMS error of replacing the joint score by per-frame marginal scores, I_{off} the intensity (9.17), $\bar{\ell}$ the weighted mean lag (9.18), $\tilde{\Xi}$ the relative weighted reach (9.19) and $L_{5\%}$ the receptive radius (9.20). The last row is the value a flat profile would give.

t	marginal	I_{off}	$\bar{\ell}$	$\tilde{\Xi}$	$L_{5\%}$
0.05	0.959	6.093	2.73	1.467	6
0.20	0.779	2.177	5.34	3.067	10
0.70	0.462	1.171	10.61	6.994	13
1.50	0.154	0.703	13.30	6.430	11
<i>flat</i>	—	—	15.00	—	—

Proposition 9.7 (What the surrogate establishes). *Clean Markov locality does not survive marginalisation through the OU channel as locality of the noised score: by (9.32) the response has nonzero blocks at every lag for every $t > 0$. The nonlocal part is nonetheless structured, decomposing exactly into a near-diagonal chain term and a rank-one correction generated by the shared ring anchor.*

9.4.5 What the surrogate measures

At $K = 30$, $\lambda = 0.05$, $\sigma = 0.15$, $\psi = 2\pi/30$, with 350 clean trajectories corrupted independently at each diffusion time, the profile (9.16) is averaged over all frame pairs at equal lag, so $L_{\text{max}} = 29$ and the structureless ceilings are $\bar{\ell} = 15$ and $\xi^{\text{norm}} = 29$; the window error is measured at the central frame. Table 9.1 reports the result.

Three readings. First, the cost of dropping temporal context is largest at low noise: at $t = 0.02$ the per-frame marginal score has a relative RMS error of 0.977, falling to 0.154 at $t = 1.5$ and 0.011 at $t = 3$. The decline does not mean that the joint structure was unimportant at small t ; it means that at large t the score itself approaches the field $-x$, which the marginal object also reproduces. Second, reach and intensity move in opposite directions: the intensity falls by a factor of 173 between $t = 0.02$ and $t = 3$, while $\bar{\ell}$ rises from 1.94 to 13.9 frames, that is from a genuinely local profile to one close to flat. The composite $\tilde{\Xi}$ resolves the two, rising to 7.7 at $t = 1$ and falling afterwards. Third, the receptive field is non-monotone in t : the radius required for 5% accuracy is 6, 10, 13, 11 at

$t = 0.05, 0.20, 0.70, 1.50$.

9.5 The model as drawn

The polar process of Section 9.1 has no gauge to remove, and its one-step transitions must be computed. Both are exactly available because each of (9.1) and (9.2) is linear.

Toolbox 9.2 — The exact one-step radial law

Step 1: shift to the equilibrium radius. Put $Y_u = R_u - 1$. Since 1 is constant, $dY_u = dR_u$ and (9.1) becomes an Ornstein–Uhlenbeck equation with no offset, $dY_u = -\kappa Y_u du + \sqrt{2D_r} dB_u$.

Step 2: multiply by the integrating factor $e^{\kappa u}$. Take $\phi(y, u) = e^{\kappa u} y$. It is linear in y , so $\partial_y^2 \phi = 0$ and Itô’s lemma (Toolbox 3.2) reduces to the ordinary product rule, with no correction term:

$$\begin{aligned} d(e^{\kappa u} Y_u) &= \kappa e^{\kappa u} Y_u du + e^{\kappa u} dY_u \\ &= \kappa e^{\kappa u} Y_u du + e^{\kappa u} (-\kappa Y_u du + \sqrt{2D_r} dB_u) = \sqrt{2D_r} e^{\kappa u} dB_u. \end{aligned}$$

The two drift terms cancel by construction, which is what the factor is chosen for, and what remains is an exact differential free of Y .

Step 3: integrate over one step. Integrating from u_k to $u_{k+1} = u_k + h$, the left side telescopes to $e^{\kappa u_{k+1}} Y_{k+1} - e^{\kappa u_k} Y_k$. Multiplying throughout by $e^{-\kappa u_{k+1}}$, so that $e^{-\kappa u_{k+1}} e^{\kappa u_k} = e^{-\kappa h}$ on the left, and carrying the constant $e^{-\kappa u_{k+1}}$ inside the integral on the right,

$$Y_{k+1} = e^{-\kappa h} Y_k + \sqrt{2D_r} \int_{u_k}^{u_{k+1}} e^{-\kappa(u_{k+1}-s)} dB_s. \quad (9.33)$$

The kernel is deterministic and weights the noise injected at time s by how much of it survives to u_{k+1} .

Step 4: the stochastic integral has mean zero. In the defining sum of Section 3.5 each term is $f(s_i)(B_{s_{i+1}} - B_{s_i})$ with $f(s) = e^{-\kappa(u_{k+1}-s)}$ a fixed number and

the increment independent of the past with mean zero, so every term has mean zero and so does the limit. Hence $\mathbb{E}[Y_{k+1} | Y_k] = e^{-\kappa h} Y_k$: the conditional mean relaxes geometrically towards zero, that is R relaxes towards one.

Step 5: its variance, by the Itô isometry. With the same deterministic f , and substituting $v = u_{k+1} - s$ so that the limits $s = u_k, u_{k+1}$ become $v = h, 0$,

$$2D_r \int_{u_k}^{u_{k+1}} e^{-2\kappa(u_{k+1}-s)} ds = 2D_r \int_0^h e^{-2\kappa v} dv = \frac{D_r}{\kappa} (1 - e^{-2\kappa h}). \quad (9.34)$$

Because the kernel is deterministic, the integral is moreover exactly Gaussian.

Step 6: name the constants and undo the shift. With $\alpha_r = e^{-\kappa h}$ and $q_r = (D_r/\kappa)(1 - \alpha_r^2)$, returning to $R = 1 + Y$ gives

$$R_{k+1} | R_k = r \sim \mathcal{N}(1 + \alpha_r(r - 1), q_r). \quad (9.35)$$

Two checks: since $\alpha_r \in (0, 1)$ the conditional mean lies strictly between r and 1, so one step moves part of the way home; and iterating (9.35) gives stationary variance $q_r/(1 - \alpha_r^2) = D_r/\kappa$, the Boltzmann value (3.15) for the potential (9.5). $\square$

Toolbox 9.3 — The exact one-step angular law, and why it wraps

Step 1: integrate directly. The drift of (9.2) does not depend on Θ , so no integrating factor is needed. Integrating term by term over $[u_k, u_{k+1}]$, the first integral is elementary and the second has integrand one, so it collapses to a single increment because every partial sum telescopes:

$$\Theta_{k+1} - \Theta_k = \omega h + \sqrt{2D_\theta} (B_{u_{k+1}}^{(\theta)} - B_{u_k}^{(\theta)}).$$

Step 2: the increment. By the definition of Brownian motion an increment over an interval of length h is $\mathcal{N}(0, h)$ and independent of the past, and multiplying by $\sqrt{2D_\theta}$ multiplies the variance by $2D_\theta$. On the real line, $\Theta_{k+1} | \Theta_k = \theta \sim \mathcal{N}(\theta + \omega h, q_\theta)$ with $q_\theta = 2D_\theta h$; the Itô isometry gives the same variance.

Step 3: wrap. What is observed is $a_u = R_u(\cos \Theta_u, \sin \Theta_u)$, which depends on Θ_u only modulo 2π . Collecting all the windings that map to the same observed angle, $\mathbb{P}[\Theta_{k+1} \bmod 2\pi \in d\theta'] = \sum_{n \in \mathbb{Z}} \mathbb{P}[\Theta_{k+1} \in d\theta' + 2\pi n]$, gives the wrapped normal transition

$$M_\theta(\theta' | \theta) = \sum_{n \in \mathbb{Z}} \frac{1}{\sqrt{2\pi q_\theta}} \exp \left[-\frac{(\theta' - \theta - \omega h + 2\pi n)^2}{2q_\theta} \right]. \quad (9.36)$$

When $q_\theta \ll (2\pi)^2$ a single winding dominates and (9.36) is indistinguishable from the Gaussian; the sum matters only when the angular uncertainty approaches a full revolution. At the parameters of this chapter $q_\theta = 3.1 \times 10^{-3}$. $\square$

With $H_k = (R_k, \Theta_k)$ the clean latent law factorises over one-step kernels,

$$P_0(H_{0:K-1}) = P_{\text{init}}(H_0) \prod_{k=0}^{K-2} M_r(r_{k+1} | r_k) M_\theta(\theta_{k+1} | \theta_k), \quad (9.37)$$

which is the precise sense in which the clean trajectory is Markov, and the structure the research question asks us to exploit. We initialise at the stationary radial law and a uniform phase. Differentiating (9.37) gives a clean latent score that is local by the same two-incident-edges counting used for (9.28); converting it to Cartesian coordinates requires the metric factor of the polar area element,

$$\nabla_{a_k} \log P_0 = e_{r,k} \left(\partial_{r_k} \log P_0 - \frac{1}{r_k} \right) + \frac{e_{\theta,k}}{r_k} \partial_{\theta_k} \log P_0, \quad (9.38)$$

the $1/r_k$ on the tangential component expressing that a given change of angle moves the point farther at larger radius.

9.5.1 The noised score has no closed form, and is computed by recursion

At diffusion time t the channel acts on each frame through the likelihood $g_t(x_k | H_k) = \mathcal{N}(x_k; e^{-t} R_k(\cos \Theta_k, \sin \Theta_k)^\top, \Delta_t I_2)$, and the noised density is the integral of (9.37) against those likelihoods over all latent trajectories. That integral has no simple closed form, be-

cause the observation map $(r, \theta) \mapsto r(\cos \theta, \sin \theta)$ is nonlinear and periodic. This is the one structural difference from the surrogate, where the conditional law given the anchor was Gaussian with an anchor-independent covariance and the latent integral collapsed to two dimensions.

The Markov factorisation nonetheless replaces one $2K$ -dimensional integral with $2K$ two-dimensional ones, by the forward-backward recursion of Section 5.5: a forward function $f_{k+1}(H_{k+1}) = g_t(x_{k+1} | H_{k+1}) \int M(H_{k+1} | H_k) f_k(H_k) dH_k$, a backward function $b_k(H_k) = \int M(H_{k+1} | H_k) g_t(x_{k+1} | H_{k+1}) b_{k+1}(H_{k+1}) dH_{k+1}$, and the smoothed one-frame posterior $p(H_k | x) \propto f_k(H_k) b_k(H_k)$, from which $\mathbb{E}[a_k | x]$ and hence (9.9) follow. This is the concrete answer to the first half of the research question: the clean Markov property does not make the noised score local, but it does make the posterior computable by repeated local operations, and it forces any distant influence to be transmitted through a chain of one-step transitions rather than directly.

As in Chapter 8 the recursion is carried on a grid, here polar, with $r \in [0.45, 1.55]$ on 41 points and $\theta \in [0, 2\pi)$ on 128 points; the radial interval spans more than six stationary standard deviations, each kernel is normalised on the grid, and the functions are rescaled at every step to avoid underflow. The result is exact for the discretised latent state space, and is called an exact reference only within the regime where Section 9.8 validates it.

The full score lives in $\mathbb{R}^{2K}$ and cannot be drawn. To see one block we fix every noisy frame except x_k and vary $x_k \in \mathbb{R}^2$; since $\nabla_{x_k} \log P_t(x_k | x_{-k}) = S_k(x, t)$, this is an exact two-dimensional conditional slice. Figure 9.5 shows it next to the corresponding marginal object.

9.6 A local linear model

The grid calculation is exact but opaque. Expanding around a single rotating branch turns the model into a linear Gaussian state-space model, whose response can be computed in closed form.

Take the deterministic reference path $\bar{R}_k = 1$, $\bar{\Theta}_k = \theta_0 + k\omega h$, write $R_k = 1 + \rho_k$

and $\Theta_k = \bar{\Theta}_k + \varphi_k$, and let $e_{r,k}, e_{\theta,k}$ be the moving frame at $\bar{\Theta}_k$. Rotating the reference direction by φ_k gives, exactly, $a_k = (1 + \rho_k)(\cos \varphi_k e_{r,k} + \sin \varphi_k e_{\theta,k})$. Inserting $\cos \varphi = 1 - \varphi^2/2 + O(\varphi^4)$ and $\sin \varphi = \varphi - \varphi^3/6 + O(\varphi^5)$,

$$a_k = \underbrace{e_{r,k} + \rho_k e_{r,k} + \varphi_k e_{\theta,k}}_{\text{first order}} \underbrace{- \frac{1}{2} \varphi_k^2 e_{r,k} + \rho_k \varphi_k e_{\theta,k}}_{\text{second order}} + O\left(\|(\rho_k, \varphi_k)\|^3\right). \quad (9.39)$$

Keeping the first order, radial deviations move perpendicular to the ring and angular deviations along the tangent, and to this order the two are decoupled. The two discarded terms have clear meanings: $-\frac{1}{2}\varphi^2 e_r$ is the curvature of the circle, which one tangent plane cannot represent, and $\rho\varphi e_\theta$ is the fact that at larger radius a given angular deviation moves the point farther. The truncation error is quadratic: for a pure angular deviation at $\rho = 0$, $\|a_k - \hat{a}_k\| = \|(\cos \varphi - 1, \sin \varphi - \varphi)\| = \varphi^2/2 + O(\varphi^3)$, and retaining the second order reduces it to $\varphi^3/6$.

Two of the three ingredients are already exactly linear: the radial deviation follows (9.35) verbatim as an AR(1) chain $\rho_{k+1} = \alpha_r \rho_k + \sqrt{q_r} \xi_k^{(r)}$, and on one unwrapped angular branch the phase is a random walk $\varphi_{k+1} = \varphi_k + \sqrt{q_\theta} \xi_k^{(\theta)}$. Together with the linear observation and the linear channel the trajectory is jointly Gaussian, $x \approx \mathcal{N}(\mu_t, \Sigma_t)$, and its score is $S^{\text{lin}}(x, t) = -\Sigma_t^{-1}(x - \mu_t)$, of exactly the form studied in Chapter 7.

9.6.1 When the expansion may be used

The truncation error makes the condition quantitative: a 5% error in units of the ring radius corresponds to $\varphi \approx 0.32$ radians. Two different quantities must be checked against that number.

The clean spread is small. The angular deviation of the clean process is a random walk, so after $K - 1$ steps its standard deviation is $\sqrt{q_\theta(K - 1)}$, which at the baseline is 0.244 radians. The clean trajectory therefore stays within about a quarter of a radian of one rotating branch over its whole length, and expanding around one branch is reasonable for the prior.

The posterior spread is what actually matters, because the score involves the posterior

of Θ_k given the noisy trajectory, and that widens with t . Measured directly as the circular standard deviation of the smoothed angular posterior at the central frame, averaged over six trajectories, it is 0.085 radians at $t = 0.03$, 0.248 at $t = 0.4$, 0.399 at $t = 0.7$ and 1.62 at $t = 2$. The threshold is crossed near $t \approx 0.55$, and Figure 9.6 shows the measured score error beginning to grow just after it.

Remark 9.8 (The limits of one tangent plane). The expansion is local and phase-conditioned. It cannot represent a posterior spread over a large arc, several wrapped angular branches, or a multimodal phase; beyond the crossing point these are not small effects to be bounded but absent features, and the honest procedure is comparison against the nonlinear grid calculation. A second limitation is independent of t : the reference phase θ_0 must be estimated from the data, and the resulting error does not vanish as $t \rightarrow 0$. It appears in Table 9.4 as a floor of 3 to 4% on the central-block error even at the smallest diffusion times, where the angular spread is negligible.

9.6.2 Reach in closed form

The value of the linear model is that its response can be computed by hand, which turns the statement that the one-step dynamics control the reach into a formula. For an infinite interior chain the AR(1) prior has Fourier-space precision $Q_r(\varpi) = (1 + \alpha_r^2 - 2\alpha_r \cos \varpi)/q_r$, the local observation contributes precision $\lambda_t = e^{-2t}/\Delta_t$ at every frame, and the posterior covariance in Fourier space is $(\lambda_t + Q_r(\varpi))^{-1}$. Defining γ_r by

$$\cosh \gamma_r = \frac{1 + \alpha_r^2 + q_r \lambda_t}{2\alpha_r}, \quad (9.40)$$

the inverse transform is a pure exponential, $C_r(\ell, t) = q_r(2\alpha_r \sinh \gamma_r)^{-1} e^{-\gamma_r |\ell|}$ with correlation length γ_r^{-1} ; the angular channel is the $\alpha_r \rightarrow 1$ case, with $\cosh \gamma_\theta = 1 + q_\theta \lambda_t/2$. By Proposition 9.1 the score responses are these covariances times e^{-2t}/Δ_t^2 . The two factors move in opposite directions: as t grows, $\lambda_t \rightarrow 0$ so $\gamma \rightarrow 0$ and the covariance length increases, while the prefactor drives the score response to zero. This is the mechanism that (9.19) was constructed to summarise, here visible in closed form.

Table 9.2: Polar model: cost of dropping temporal context, reach and weighted reach by channel, and receptive radii. Notation as in Table 9.1; the structureless ceiling for $\bar{\ell}$ is 5.5.

t	marginal	$\bar{\ell}_r$	$\bar{\ell}_\theta$	$\tilde{\Xi}_r$	$\tilde{\Xi}_\theta$	$L_r^{5\%}$	$L_\theta^{5\%}$
0.03	0.732	1.98	4.19	0.208	2.166	2	9
0.10	0.733	2.71	5.01	0.119	2.575	3	9
0.40	0.513	5.04	5.40	0.314	2.672	7	9
0.70	0.349	5.47	5.45	1.292	2.701	8	9
1.00	0.229	5.46	5.46	1.022	1.793	8	9
1.50	0.092	5.48	5.47	0.928	1.042	5	7
2.00	0.037	5.47	5.47	0.428	0.428	0	0

For orientation, the clean process has stationary radial covariance $(D_r/\kappa)e^{-\kappa h|\ell|}$ and first-harmonic angular coherence $e^{-D_\theta h|\ell|}$, hence clean scales $\xi_r^{\text{clean}} = 1/(\kappa h) \approx 1.6$ frames and $\xi_\theta^{\text{clean}} = 1/(D_\theta h) \approx 637$ frames at the baseline. The angular coherence length exceeds the length of the trajectory by more than a factor of 30: within one data point the phase is essentially a single coherent quantity, while the radius forgets itself after a couple of frames. That asymmetry is the prediction tested next.

9.7 Measurements

At the parameters of Figure 9.1, with 28 independent clean trajectories per diffusion time for the response study and 36 for the window study, central frame $k = 10$, so $L_{\text{max}} = 10$ and the structureless ceilings are $\bar{\ell} = 5.5$ and $\xi^{\text{norm}} = 10$. The radial and tangential channels are separated by projecting the response blocks on the moving frame, $J_{kj}^{rr} = e_{r,k}^\top J_{kj} e_{r,j}$ and $J_{kj}^{\theta\theta} = e_{\theta,k}^\top J_{kj} e_{\theta,j}$, using the simulated clean angle as an oracle basis. That is a diagnostic convenience and not an ingredient of the score; Section 9.10 returns to it.

Joint against marginal. Replacing the joint score by per-frame marginal scores costs a relative RMS error of 0.732 at $t = 0.03$. The error stays above 0.69 up to $t = 0.2$, then falls, passing below 0.10 only near $t = 1.5$. The nonlinear process therefore agrees with the surrogate that the effect is not an artefact of the Gaussian construction.

The two channels differ. At $t = 0.03$ the weighted mean lag is 1.98 frames radially against 4.19 tangentially, and the gap widens once intensity is included: $\tilde{\Xi}_r = 0.21$ against $\tilde{\Xi}_\theta = 2.17$, a factor of ten, where the normalised range would have reported the same contrast as a factor of 2.5. This is the clean-scale prediction of Section 9.6.2 confirmed in the tested regime. The linear model is quantitatively right where it should be: at $t = 0.03$ it predicts a radial e-folding length of 1.35 frames against a measured exponential fit of 1.39 frames, an agreement to 3%. Its tangential prediction of 4.45 frames is already comparable to the chain length, and the measured fit of 6.36 frames should be read as finite-size dominated rather than as an estimate of an infinite-system length.

Reach and intensity move in opposite directions. Between $t = 0.03$ and $t = 2$ the intensity I_{off} falls from 8.20 to 0.113, a factor of about 73, while the weighted mean lag rises from 4.10 to 5.48 frames, within 0.4% of the structureless ceiling 5.5. The correct reading is not that high-noise frames are strongly coupled, but that the weak residual coupling is spread evenly along the chain, exactly the situation of Remark 9.2.

The composite diagnostic resolves the two. Figure 9.7(c) reports $\tilde{\Xi}$ rising from 1.55 at $t = 0.03$ to 2.26 at $t = 0.7$ and falling to 0.43 at $t = 2$, while the independently measured receptive radius $L_{5\%}$ takes the values 8, 8, 9, 9, 9, 6, 0 across the same diffusion times. Both are non-monotone and both turn in the same region, whereas I_{off} decreases monotonically and $\bar{\ell}$ and ξ^{norm} increase monotonically, so neither elementary diagnostic can indicate a turn at all. Two cautions belong with this observation: the peaks do not coincide exactly, since in the surrogate $\tilde{\Xi}$ peaks at $t = 1$ and $L_{5\%}$ at $t = 0.7$, and $\tilde{\Xi}$ is a first-moment summary and not a predictor of L_ϵ .

The receptive field. At $t = 0.03$ the radial component of the score reaches 5% accuracy with a window radius of 2, while the tangential component needs 9, nearly the whole trajectory. At intermediate diffusion times both channels need large windows, and by $t = 2$ a single frame is already within 5% of the full score, because the cross-frame correction has become negligible. Read as a possible implication for architecture, in the guarded

Table 9.3: Validation of the rotating-ring computations. The response identity and the analytic surrogate Jacobian are compared with central finite differences of the score; the grid check compares two polar resolutions.

check	discrepancy	tolerance
posterior response identity (9.15) vs finite differences	6.63×10^{-9}	10^{-7}
analytic surrogate Jacobian (9.32) vs finite differences	9.18×10^{-6}	10^{-4}
41×128 against 51×192 polar response grid	3.04×10^{-11}	10^{-7}
rotation gauge round trip (Lemma 9.5)	2.22×10^{-16}	10^{-12}
clean score on an exact co-rotating ring path	0 (exact)	10^{-12}

sense of Section 5.8, this says that the useful temporal receptive field of a denoiser would depend on the noise level and be largest in the middle of the schedule. It is a property of this controlled model, not an established conclusion about trained networks.

The clean dynamics control the low-noise reach. Increasing κ from 0.5 to 4 at $t = 0.1$ shortens the radial normalised range from 4.18 to 2.74 frames and reduces the nearest-neighbour radial intensity by an order of magnitude, from 0.35 to 0.034: stiffer confinement forgets faster. Increasing D_θ from 0.002 to 0.08 at the same diffusion time shortens the tangential normalised range from 8.71 to 2.96 frames: faster dephasing destroys the long tangential channel. Both are what (9.40) predicts, and together they answer the second half of the research question in the tested regime: the reach of the noised score is set by the one-step clean kernel, modulated by the observation precision λ_t .

9.8 Numerical validation

Every identity of this chapter is checked against an independently computed reference, in the protocol of Appendix C.

Beyond Table 9.3 the checks include row normalisation of the transition kernels, global rotation equivariance of the score, stability under halving the finite-difference step, and the exact vanishing of the clean surrogate score on a perfect co-rotating ring path. One caveat on grid convergence: the difference between the 41×128 and 51×192 grids is

Table 9.4: First-order Taylor score against the nonlinear grid score, 32 trajectories per diffusion time. The central-block error has a floor of 3 to 4% from the estimated reference phase; the growth beyond $t \approx 0.7$ follows the angular posterior spread (Section 9.6.1).

t	central RMSE	trajectory RMSE	mean cosine
0.03	0.039	0.122	0.985
0.15	0.035	0.044	0.997
0.40	0.035	0.035	0.998
0.70	0.032	0.035	0.999
1.00	0.061	0.065	0.992
1.50	0.079	0.083	0.990
2.00	0.060	0.067	0.991

small because the baseline likelihoods and kernels are already smooth at those resolutions, and a deliberately coarse 11×24 grid does show visible deviations. The production grid is converged for these parameters rather than trivially grid-independent.

9.9 What this chapter establishes

The clean law is local, and this gives three concrete advantages: the clean score is assembled from adjacent transition innovations (9.28); the noised posterior is computable by repeated one-step operators rather than one integral over $2K$ latent variables; and any distant influence on S_k must propagate through a sequence of local transitions, which is why the parameter sweeps predict the reach from the one-step kernel. What clean Markovity does not give is locality of the noised score: for $t > 0$, $S_k(x, t)$ is not a function of x_{k-1}, x_k, x_{k+1} alone in either model, because integrating out uncertain latent states creates posterior correlations at every lag. In the surrogate that is explicit, the rank-one anchor term of (9.32) touching every pair of frames; in the polar model it is measured.

Both models show the same sequence: a strictly local clean score, then broader posterior dependence after noising, then the weak universal field $-x$. At intermediate diffusion times individual frames are noisy enough that distant context improves the estimate of a clean frame; at large diffusion times the same context remains statistically present but is no longer materially useful, because the prefactor e^{-2t}/Δ_t^2 of (9.15) has gone to zero. The

Table 9.5: Status of the results of this chapter.

result	status
Joint minus marginal gap (9.11)	exact
Posterior response identity (9.15) (Proposition 9.1)	exact
Gauge reduction and score transformation (Lemma 9.5)	exact
Exact surrogate score and rank-one response (Proposition 9.6)	exact
Clean scales $1/(\kappa h)$ and $1/(D_\theta h)$; closed-form reach (9.40)	exact, infinite-chain
First-order linearisation (9.39) and its $\varphi \approx 0.32$ validity threshold	assumption-conditional
Polar joint score on the 41×128 grid	numerical, validated
Reach, intensity, weighted reach and receptive radii of Tables 9.1 and 9.2	numerical, one parameter set
Non-monotonicity of $\tilde{\Xi}$ and $L_{5\%}$ in t	numerical, observed in the tested regime
Receptive field as a reading for architecture	interpretation
Behaviour in K , unknown rotation class, inferred phase	open

polar process adds a feature the Cartesian surrogate cannot express: radial information is damped by mean reversion while tangential information rides a slowly diffusing global phase, so the two score channels have different reaches and different receptive fields.

Table 9.5 classifies the results in the scheme used throughout the thesis.

9.10 Limitations

1. **One parameter set.** Every number reported here comes from a single baseline, with $K = 20$ for the polar model and $K = 30$ for the surrogate, and 28 to 36 trajectories per diffusion time. Only κ and D_θ were swept, and nothing is claimed about scaling in K .
2. **Angular equation.** A constant angular drift was used. A different angular drift changes the transition kernel (9.36) but not the identities of Section 9.2.

3. **Radius positivity.** The shifted process of Toolbox 9.2 is formally supported on all of $\mathbb{R}$. At the baseline the probability of a negative radius is negligible, but a strict model would reflect at the origin or use a positive diffusion.
4. **Known rotation.** Lemma 9.5 assumes ψ known. An unknown sense of rotation would add a discrete latent variable and may produce a speciation phenomenon of the kind discussed in Section 4.7. This is the most interesting single extension.
5. **Finite trajectory.** At large t the reach diagnostics saturate at their structureless ceilings, a finite-size effect and not evidence of a divergent correlation length.
6. **Oracle projections.** The radial and tangential channels were separated using the simulated clean angle; a practical method would infer the phase, and that error would enter the projected profiles.
7. **Grid approximation.** The polar score is numerically converged for these parameters but is not a symbolic closed form.
8. **One tangent plane.** The linear model cannot represent wrapped or multimodal angular posteriors, which by Section 9.6.1 is the regime beyond $t \approx 0.55$.

The immediate extensions are to vary K systematically, to infer the angular phase rather than use an oracle basis, to add an unknown rotation class, and to test whether a time-dependent local architecture with receptive field $L_\epsilon(t)$ can match the full score.

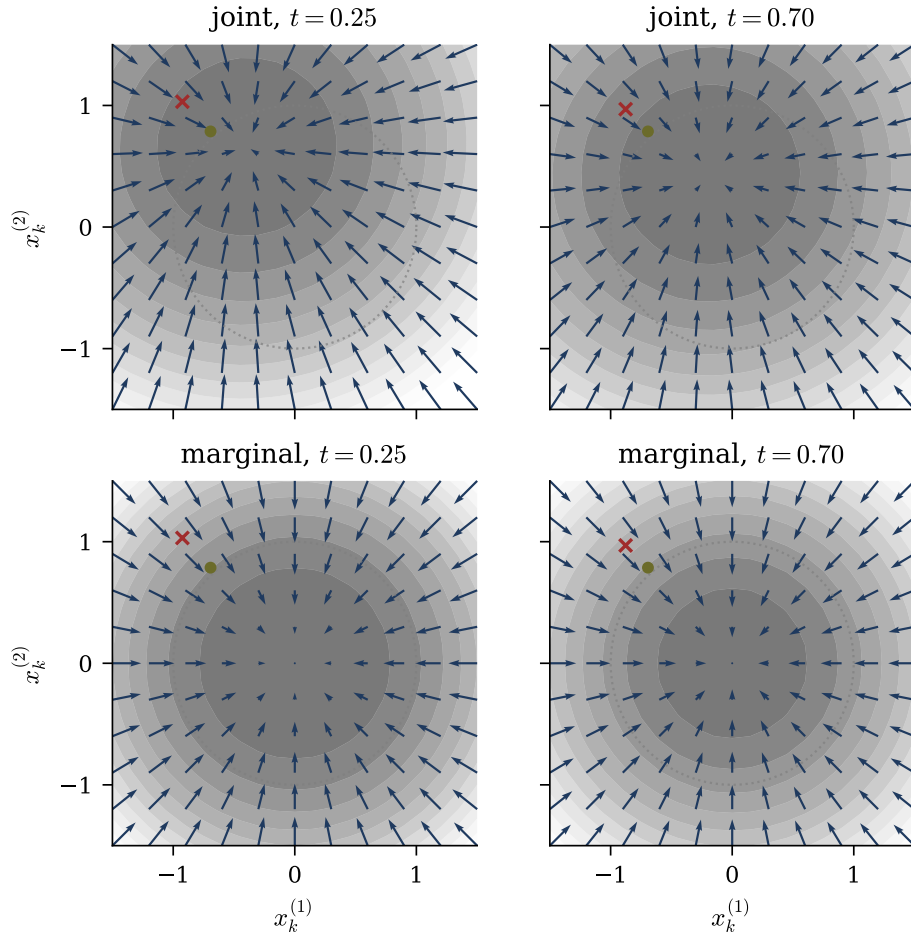


Figure 9.5: Conditional slices of the score block at the central frame, $k = 10$, one fixed clean trajectory and one fixed standardised noise realisation, on a 31×64 polar grid. Shading is $\log P_t(x_k | x_{-k})$, darker for higher density; arrows are the score; the cross is the noisy observation x_k and the disc the clean frame a_k . Top: the joint conditional field is displaced towards the phase inferred from the rest of the trajectory. Bottom: the one-frame marginal field stays ring-symmetric about the origin, because one frame alone does not identify where along the ring the trajectory is. This is the content of (9.11) in two dimensions.

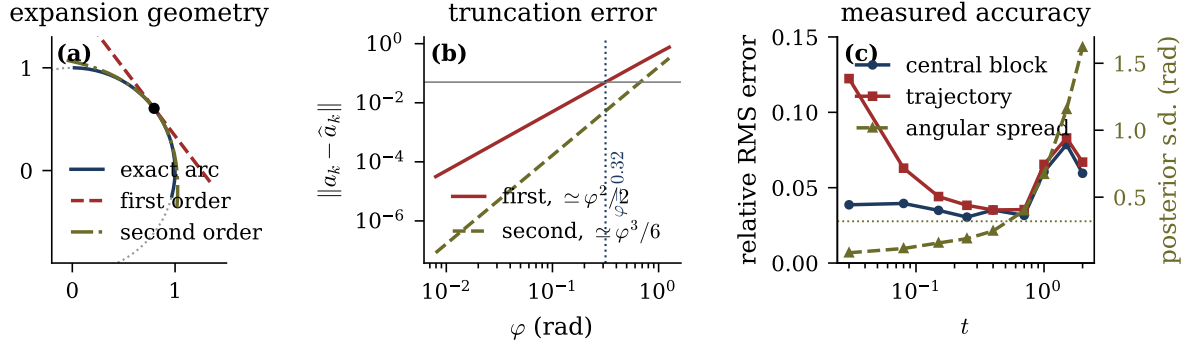


Figure 9.6: The linearisation and its validity. (a) Exact arc, first-order tangent and second-order parabola through the reference point. (b) Truncation error against angular deviation: first order is $O(\varphi^2)$ and second order $O(\varphi^3)$, so a 5% error in units of the ring radius corresponds to $\varphi \approx 0.32$ radians. (c) Measured relative RMS error of S^{lin} against the nonlinear grid score, left axis, with the measured circular standard deviation of the smoothed angular posterior on the right axis; the dotted line is the 0.32 radian threshold. The error grows shortly after the posterior spread crosses it.

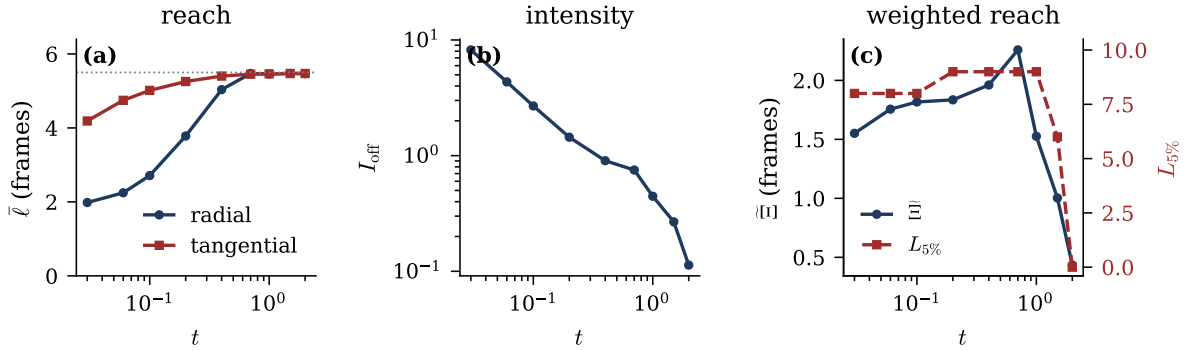


Figure 9.7: Diagnostics of the polar model at the parameters of Figure 9.1, 28 trajectories per diffusion time. (a) Weighted mean lag (9.18) by channel; the dotted line is the structureless ceiling 5.5, which a completely flat profile would produce. (b) Absolute cross-frame intensity (9.17), which falls by a factor of about 73 over the range. (c) The relative weighted reach (9.19), left axis, and the independently measured receptive radius $L_{5\%}$ of (9.20), right axis. Both are non-monotone in t and turn near $t = 0.7$.

Chapter 10

Discussion and Conclusions

10.1 Findings

Derived exactly. For the stationary AR(1) chain under coordinatewise OU corruption, the joint score is $S(x, t) = -Q_t x$ with $Q_t = (e^{-2t}\Sigma_0 + \Delta_t I)^{-1}$, equal by the score-posterior identity to $(e^{-t}\mathbb{E}[a|x] - x)/\Delta_t$ (RQ1). Its coupling structure has a closed-form lifecycle: exactly tridiagonal at $t = 0$, filling one diagonal per order in t by the band-fill law (7.13), and returning to the identity at rate e^{-2t} . In the bulk, the single quantity $q(\alpha, t) \in (0, |\alpha|]$ governs both the posterior correlation length and the geometric decay of evidence influence, with limits $q \rightarrow 0$ as $t \rightarrow 0$ and $q \rightarrow |\alpha|$ as $t \rightarrow \infty$. The posterior of the noisy chain lives on a tree-shaped factor graph, on which one forward and one backward BP sweep compute all smoothing marginals; on the Gaussian chain the messages close algebraically in the Gaussian family, and the recursion is the Kalman filter followed by the RTS smoother (RQ2). The error of a radius- r local score estimator decays exactly as q^r (RQ4). On the same model, the per-node AMP/TAP closure returns the exact score (the mean fixed point is closure-independent) but a different variance, with existence boundary, breakdown time $t_c(\alpha)$, and critical coupling $\alpha_c = \sqrt{2} - 1$ in closed form (Appendix B); all of these formulas were re-verified independently.

Measured numerically. The two Gaussian score routes agree to 10^{-15} – 10^{-14} across the tested (K, α, t) grid, with an independently written implementation reproducing the agreement. On Laplace chains, Gaussian-projected BP measured against validated grid quadrature has median relative score error 0.39 at $t = 0.02$, decaying below 10^{-3} for $t \gtrsim 0.9$; the error is amplified at small t by the exact factor e^{-t}/Δ_t , is worst for bimodal innovation laws, and decreases with stronger chain correlation, while the score direction remains accurate (median cosine ≥ 0.92 throughout) (RQ3). At equal locality budget, truncating the inference is more accurate than truncating the score matrix at small t (12.3% against 21.0% at $t = 0.05$).

The Gaussian-to-Laplace comparison. The structural lesson is where exactness is lost: the tree, the two-sweep schedule, and the score–posterior identity survive any innovation law; what breaks is the representability of messages in a finite-dimensional family. The practical lesson is when it matters: the Gaussian closure is essentially free at moderate and large diffusion times, and expensive precisely in the small- t regime where the score magnitude is amplified, though even there the score direction is preserved much better than its magnitude.

10.2 What the model suggests but does not establish

Within the Gaussian benchmark, local score computations are near-optimal away from intermediate diffusion times, the required receptive field scales as $\xi \log(1/\varepsilon)$ with $\xi = 1/\log(1/q)$, and a local algorithm is preferable to a banded linear readout. These are properties of exact inference in one controlled model. Whether trained score networks on real sequence data inherit them is untested here and would require learning experiments.

10.3 Limitations

(i) The exact solutions concern one-dimensional state per frame and linear AR(1) dynamics. (ii) The non-Gaussian results are numerical: grid BP is a reference only within its validated resolution regime ($\sqrt{2t} \gtrsim 3 dx$), and the Laplace closure error is a measured curve, not a formula. (iii) The locality laws are proved for the Gaussian bulk; their non-Gaussian counterparts are only structurally delimited. (iv) No learning is performed. (v) An earlier conjecture on low-rank structure of the Laplace $K = 2$ posterior Hessian failed outside a narrow regime and is not asserted. (vi) How score errors accumulate along the reverse generation trajectory was not studied.

10.4 Future work

Three directions follow directly from what was measured. First, mixture messages: the single-Gaussian closure fails worst on bimodal beliefs, so a two-component mixture closure on the same sweeps would separate the representation error from the model mismatch, and is the clearest next algorithmic step. Second, discrete chains: with a finite alphabet the BP messages are finite vectors and inference is exact with no closure at all, giving a second solvable benchmark. Third, reverse dynamics: integrating the reverse SDE under exact, Gaussian-projected, and truncated scores would show where along the trajectory the closure and locality errors matter for generation, turning the pointwise error measurements of this thesis into dynamical ones.

Appendix A

Gaussian Identities Used Throughout

This appendix collects, with proofs, the handful of Gaussian identities on which Chapter 7 relies. Notation: a Gaussian density in *moment* parameters is $\mathcal{N}(a; m, v)$; in *natural* parameters it is $\exp(-\frac{\lambda}{2}a^2 + \theta a + \text{const})$ with $\lambda = 1/v$, $\theta = m/v$.

A.1 Quadratic forms and Gaussian normalisation

For $J \succ 0$ (symmetric positive definite) and any h ,

$$\int_{\mathbb{R}^K} \exp\left(-\frac{1}{2}a^\top J a + h^\top a\right) da = \frac{(2\pi)^{K/2}}{\sqrt{\det J}} \exp\left(\frac{1}{2}h^\top J^{-1}h\right), \quad (\text{A.1})$$

by completing the square, $-\frac{1}{2}a^\top J a + h^\top a = -\frac{1}{2}(a - J^{-1}h)^\top J(a - J^{-1}h) + \frac{1}{2}h^\top J^{-1}h$, and translating. Consequently a density $p(a) \propto \exp(-\frac{1}{2}a^\top J a + h^\top a)$ is the Gaussian with mean $m = J^{-1}h$ and covariance J^{-1} : *reading off (J, h) from a log-density is the fastest route to posterior parameters*, used in Proposition 7.5.

A.2 Products of Gaussians in the same variable

$$\mathcal{N}(a; m_1, v_1) \mathcal{N}(a; m_2, v_2) \propto \mathcal{N}(a; m, v), \quad \frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}, \quad \frac{m}{v} = \frac{m_1}{v_1} + \frac{m_2}{v_2}. \quad (\text{A.2})$$

Proof. Add the exponents; the sum of two quadratics in a is a quadratic in a whose a^2 -coefficient is $-\frac{1}{2}(1/v_1 + 1/v_2)$ and whose linear coefficient is $m_1/v_1 + m_2/v_2$; apply (A.1) in one dimension. In natural parameters the rule is addition: $(\lambda, \theta) = (\lambda_1 + \lambda_2, \theta_1 + \theta_2)$, which is why the BP updates of Chapter 7 are two additions per message. $\square$

A.3 Affine propagation (marginalisation)

If $a' = \alpha a + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2)$ independent of $a \sim \mathcal{N}(m, v)$, then

$$\int \mathcal{N}(a'; \alpha a, \sigma^2) \mathcal{N}(a; m, v) da = \mathcal{N}(a'; \alpha m, \alpha^2 v + \sigma^2). \quad (\text{A.3})$$

Proof. a' is a sum of independent Gaussians, hence Gaussian; its mean is αm and its variance $\alpha^2 v + \sigma^2$ by linearity of expectation and independence. Read in the reverse direction (solving for a given a Gaussian summary of a'), the same computation gives $\mathcal{N}(a; m'/\alpha, (v' + \sigma^2)/\alpha^2)$, the backward half-step of the BP sweep. $\square$

A.4 Gaussian conditioning

For a jointly Gaussian pair with precision blocks $\begin{pmatrix} J_{11} & J_{12} \\ J_{12}^T & J_{22} \end{pmatrix}$, the conditional of block 1 given block 2 is Gaussian with precision J_{11} (conditioning *reads off* the precision block) and mean shifted by $-J_{11}^{-1} J_{12} a_2$; marginalisation, dually, is trivial in covariance parameters. This precision/covariance duality (conditioning is local in J , marginalising is local in Σ) is the algebraic engine behind the cavity recursion (B.1): integrating a neighbour out of a tridiagonal precision produces the rank-one correction $-\beta^2/\lambda$ on the adjacent diagonal

entry, by the Schur complement

$$\begin{pmatrix} \lambda & \beta \\ \beta & J_d \end{pmatrix} \longrightarrow J_d - \frac{\beta^2}{\lambda}. \quad (\text{A.4})$$

A.5 Woodbury and the two routes to the score

The Woodbury identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$, specialised to $\Sigma_t = \Delta_t I + e^{-2t}\Sigma_0$, connects the noisy precision Q_t and the posterior precision $J = (e^{-2t}/\Delta_t)I + Q_0$:

$$Q_t = \frac{1}{\Delta_t} \left(I - \frac{e^{-2t}}{\Delta_t} J^{-1} \right), \quad (\text{A.5})$$

which is exactly the operator form of the score-posterior identity: substituting the posterior mean $m = J^{-1}h = J^{-1}(e^{-t}/\Delta_t)x$ into $S = (e^{-t}m - x)/\Delta_t$ gives $S = -Q_t x$ identically. The machine-precision agreement of the two score routes (Section 7.3) is the numerical shadow of (A.5).

Appendix B

AMP, TAP, and the Bulk Fixed Point

This appendix develops the per-node Gaussian closures of statistical physics, AMP and the TAP equations, on the Gaussian chain, and proves the exact breakdown announced in Remark 7.13. The results are complete, audited, and independently re-verified; they are collected here rather than in Chapter 7 because they branch off the thesis’s main line. Summary: on the chain, AMP still returns the exact score, but its variance closure fails past an exact breakdown time $t_c(\alpha)$, with critical coupling $\alpha_c = \sqrt{2} - 1$.

B.1 The bulk cavity fixed point of BP

Chapter 7 solved the bulk covariance by matrix algebra; the same formulas re-emerge from the message recursion, which is worth exhibiting because it localises where BP and AMP differ. In the bulk, parametrise the forward message (evidence absorbed) by its precision λ_k ; the closure lemmas of Section 7.6 give the scalar recursion

$$\lambda_{k+1} = J_d - \frac{\beta^2}{\lambda_k}, \tag{B.1}$$

with J_d, β the bulk parameters (7.20): absorbing one neighbour's summary costs the Gaussian-integration correction $-\beta^2/\lambda$. The fixed points are $\lambda^\pm = (J_d \pm \sqrt{J_d^2 - 4\beta^2})/2$, and since $J_d - 2|\beta| = e^{-2t}/\Delta_t + (1 - |\alpha|)^2/\sigma_\eta^2 > 0$ (Chapter 7), the attractive upper fixed point

$$\lambda^* = \frac{J_d + \sqrt{J_d^2 - 4\beta^2}}{2} \quad (\text{B.2})$$

exists for every $\alpha \in (-1, 1)$ and $t > 0$, with $|f'(\lambda^*)| = \beta^2/\lambda^{*2} < 1$: the sweep converges geometrically from any initialisation. Recombining the two directions and the local term, and using the fixed-point relation $\beta^2/\lambda^* = J_d - \lambda^*$,

$$\Lambda_{\text{bulk}} = J_d - \frac{2\beta^2}{\lambda^*} = 2\lambda^* - J_d = \sqrt{J_d^2 - 4\beta^2} \implies V_{\text{exact}} = \frac{1}{\sqrt{J_d^2 - 4\beta^2}}, \quad (\text{B.3})$$

in exact agreement with Theorem 7.14. Belief propagation on the Gaussian chain has no phase transition: the cavity fixed point exists always, and the discriminant $J_d^2 - 4\beta^2$ is the quantity to watch as the closure is modified.

B.2 AMP/TAP: same score, different variance

AMP, the algorithmic form of the TAP equations (Thouless et al., 1977; Donoho et al., 2009; Zdeborová and Krzakala, 2016), is the message passing one uses when the factor graph is dense: each message is then a sum of many weak contributions, approximately Gaussian by a central-limit argument, and one tracks per-node means and variances instead of per-edge messages, with the Onsager term correcting the self-influence. Rigorous derivations of AMP from BP exist in that dense limit (Genovese and Piana, 2026). The chain is the opposite regime in every respect (diagonal measurement, coupling in the prior, degree two), so it isolates what the per-node closure costs. Viewing the posterior as a Gaussian Markov field with precision J and field h , the three schemes solve, at each node,

$$\text{mean field: } V_i = \frac{1}{J_{ii}}; \quad \text{AMP/TAP: } V_i = \frac{1}{J_{ii} - \sum_{k \neq i} J_{ik}^2 V_k}; \quad \text{BP: } V_i = (J^{-1})_{ii}. \quad (\text{B.4})$$

The AMP closure is the BP cavity with the exclusion of the receiving neighbour dropped ($V_{k \rightarrow i} \approx V_k$): an $O(1/N)$ change on a dense graph, but a change of fixed, order-one size on a chain with two neighbours.

Theorem B.1 (The score is closure-independent). *For a Gaussian posterior, the fixed-point condition for the means is the same linear system $Jm = h$ under mean field, BP, and AMP. All three schemes hence return the exact posterior mean and, via the score-posterior identity, the same exact joint score, for every (K, α, t) , regardless of whether their variance closures are accurate or even solvable.*

The proof is one line: in each scheme the mean update at node i reads $m_i \leftarrow (h_i - \sum_{k \neq i} J_{ik} m_k) / J_{ii}$ at the fixed point; the closures modify the uncertainty bookkeeping, not the linear relation among means, and the unique solution of $Jm = h$ is the exact posterior mean. Numerically, AMP means agree with exact means to 4.3×10^{-12} and AMP scores with BP scores to 1.7×10^{-12} across all tested configurations.

For the variances the story splits. In the bulk the AMP closure (B.4) reads $V = 1/(J_d - 2\beta^2 V)$; the BP cavity is $V_{\text{cav}} = 1/(J_d - \beta^2 V_{\text{cav}})$. The entire difference between BP and AMP on the chain is a factor of two: both neighbours included where BP excludes one.

Theorem B.2 (AMP bulk variance, breakdown time, critical coupling). *The AMP bulk closure is the quadratic $2\beta^2 V^2 - J_d V + 1 = 0$, with physical root*

$$V_{\text{AMP}} = \frac{J_d - \sqrt{J_d^2 - 8\beta^2}}{4\beta^2}, \quad (\text{B.5})$$

which exists iff $J_d \geq 2\sqrt{2}|\beta|$, strictly stronger than BP's always-true $J_d > 2|\beta|$. Since $J_d(t)$ decreases monotonically in t , the existence condition fails, if ever, past a single time; solving it as an equality gives

$$\boxed{t_c(\alpha) = -\frac{1}{2} \log \frac{g}{1+g}, \quad g(\alpha) = \frac{2\sqrt{2}|\alpha| - 1 - \alpha^2}{1 - \alpha^2},} \quad (\text{B.6})$$

finite exactly when $g > 0$, i.e.

$$\boxed{|\alpha| > \alpha_c = \sqrt{2} - 1 \approx 0.4142.} \quad (\text{B.7})$$

For $|\alpha| \leq \alpha_c$ the AMP variance has a fixed point at every diffusion time; for $|\alpha| > \alpha_c$ it ceases to exist for all $t > t_c(\alpha)$. In the weak-coupling regime the two closures agree to second order, with

$$V_{\text{AMP}} - V_{\text{exact}} = \frac{2\beta^4}{J_d^5} + O\left(\frac{\beta^6}{J_d^7}\right) > 0 : \quad (\text{B.8})$$

AMP always overestimates the bulk variance, at fourth order in the coupling.

Toolbox B.1 — Roots, branch, and the phase boundary

Branch selection. The quadratic has roots $V_{\pm} = (J_d \pm \sqrt{J_d^2 - 8\beta^2})/(4\beta^2)$; as $\beta \rightarrow 0$, $V_- \rightarrow 1/J_d$ (the decoupled answer) while V_+ diverges, so V_- is the physical branch, and the attractive fixed point of the damped iteration used in practice. *Critical coupling.* $g > 0 \iff \alpha^2 - 2\sqrt{2}|\alpha| + 1 < 0 \iff |\alpha| \in (\sqrt{2} - 1, \sqrt{2} + 1)$; intersecting with $|\alpha| < 1$ leaves $|\alpha| > \sqrt{2} - 1$. Consistency check at $t = \infty$: there the posterior is the bare prior, and the existence condition applied to Q_0 itself reads $(1 + \alpha^2) \geq 2\sqrt{2}|\alpha|$, i.e. AMP applied to a bare AR(1) chain has a variance fixed point iff $|\alpha| \leq \sqrt{2} - 1$, which is (B.7) read backwards. *Weak coupling.* With $\epsilon := \beta^2/J_d^2$: $(1 - 4\epsilon)^{-1/2} = 1 + 2\epsilon + 6\epsilon^2 + O(\epsilon^3)$ for the exact variance; $1 - \sqrt{1 - 8\epsilon} = 4\epsilon + 8\epsilon^2 + O(\epsilon^3)$, divided by $4\epsilon J_d$, gives $V_{\text{AMP}} = (1 + 2\epsilon + 8\epsilon^2 + O(\epsilon^3))/J_d$; subtract. $\square$

Numerically, on a $K = 12$ chain at $\alpha = 0.8$: the AMP score error stays at 10^{-13} – 10^{-12} for every t , while the variance error is 1.3×10^{-4} at $t = 0.05$, 3.5×10^{-2} at $t = 0.2$, and past $t \approx 0.23$ the variance update has no positive fixed point at all. The audit further verifies the closed form (B.5) against the converged iteration to 2.5×10^{-12} ; the existence condition at all 180 points of an (α, t) scan; t_c against the bisected flip point of the iteration to 6.6×10^{-6} relative; convergence below α_c even at $t = 50$; and the weak-coupling ratio $(V_{\text{AMP}} - V_{\text{exact}})/(2\beta^4/J_d^5) \rightarrow 1$ within 0.4%. The interpretation is not that AMP is wrong; on a dense benchmark (random $N = 600$ symmetric positive-definite

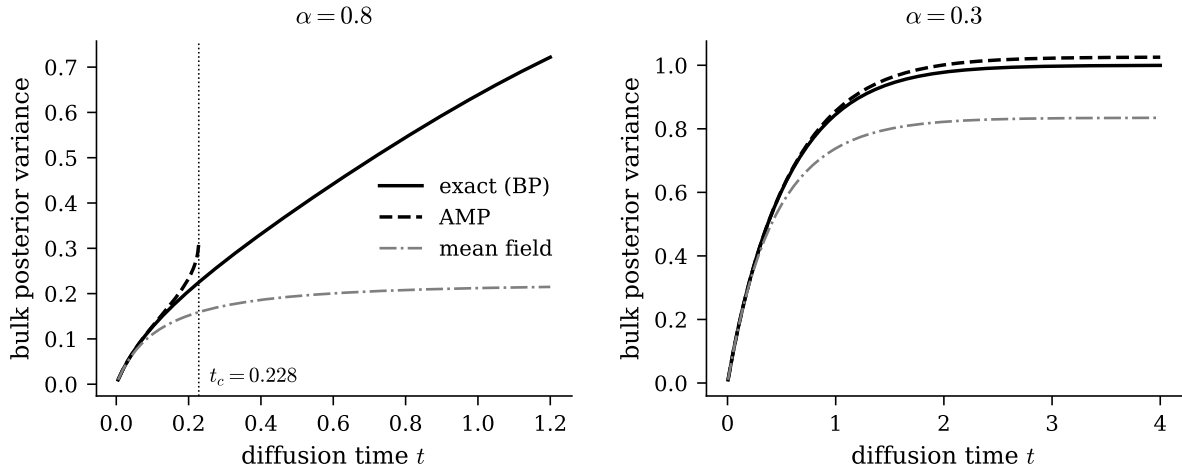


Figure B.1: The three closures in closed form. Left ($\alpha = 0.8 > \alpha_c$): exact = BP, AMP (B.5), and mean field; the AMP branch terminates with a square-root singularity at the breakdown time $t_c(0.8) = 0.228$ from (B.6) (dotted line); beyond it AMP has no variance fixed point. Right ($\alpha = 0.3 < \alpha_c$): AMP exists at every t and slightly overestimates the variance, as predicted by (B.8).

precision) the same closure gets variances to 1.7×10^{-3} where mean field errs at 4×10^{-2} . It is built for a different graph: the central-limit justification of the per-node closure needs many incoming messages, and a chain provides two.

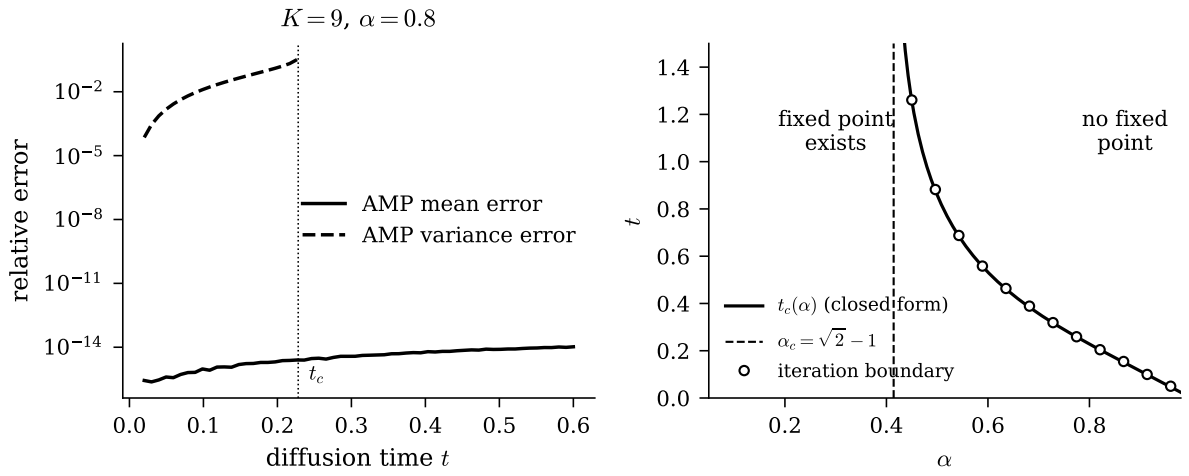


Figure B.2: BP against AMP on a finite chain ($K = 9$, $\alpha = 0.8$). Left: the AMP mean error (about 10^{-12} , flat in t : Theorem B.1) against the AMP variance error, which grows with t up to the breakdown time t_c (dotted line). Right: the phase boundary of the AMP variance fixed point in the (α, t) plane: the solid curve is the closed form $t_c(\alpha)$ of (B.6), the dashed line is $\alpha_c = \sqrt{2} - 1$, and the open circles mark the breakdown as located independently by damped fixed-point iteration; the audit suite performs the same comparison over 180 points of this plane.

Appendix C

Reproducibility

C.1 Repository layout

All source material lives in a single repository; the thesis directory is self-contained (L^AT_EX source, figures, and the figure script are committed together).

Path	Content
thesis/	this document: L ^A T _E X source, figures, figure script
research/gaussian-bp/	Gaussian chain note, code, 72-check audit
research/bp-from-scratch/	independent BP derivation and implementation
research/unified-note/	consolidated working note, code, 55-check audit
research/nongaussian-bp/	grid BP vs Gaussian-projected BP experiments
research/gaussian-ar1-bp/	Gaussian AR(1) package (BP = matrix score)
research/initial-experiments/	early experiments and toy-model material
research/experiment1-rotating-ring/	rotating-ring package: code, data, validation

C.2 Audit protocol

Every closed-form claim is verified by a deterministic check against an independently computed reference (brute-force linear algebra, dense grid quadrature, or finite differences), collected in audit drivers that exit non-zero on any failure. The audit driver of

`research/gaussian-bp` runs 72 checks: assembly and inversion of Σ_t ; the score-posterior route against the matrix score; the $K = 2$ closed forms; Convention-A BP against the matrix posterior (means, variances, scores); bulk variance, covariance decay q^d , and the recombination identity; AMP mean and score closure-independence; the AMP variance formula, existence boundary, breakdown time, and weak-coupling law; the locality slopes; the band-fill coefficients (with the factorial variant explicitly rejected); spectral preservation and the large- t return. The audit driver of `research/unified-note` runs 55 checks on the consolidated note, including the Laplace $K = 1$ quadrature checks and the one-step boundary-message identity. `research/nongaussian-bp/` contains experiment drivers with logged master-error-identity residuals (10^{-15} – 10^{-12} throughout), grid self-convergence gates, and per-trial CSV outputs; the grid calibration of Section 8.4 is experiment 01, the closure measurement of Section 8.5 experiment 02. `research/experiment1-rotating-ring` carries the package behind Chapter 9: the exact surrogate score and Jacobian, the polar-grid recursion, the response diagnostics in a single module, and a validation driver that writes `validation.json` and exits non-zero on any failure. Its five checks are the ones reported in Table 9.3, and every number quoted in Chapter 9 is regenerated from the curated CSVs of that package rather than entered by hand. All checks were re-executed on the build machine at the time of writing: 72/72, 55/55 and 5/5 pass, and all experiment gates are green.

C.3 Figure-to-code map

The figures of Chapters 7 to 8 and of the appendices are regenerated by one script, `figures/make_figures.py`, which recomputes each panel from the closed forms or quadrature of the corresponding chapter; the only one of those that reads stored experimental data is Figure 8.2, built from the per-trial CSV of `nongaussian-bp` experiment 02. The figures of Chapter 9 are regenerated by `figures/make_ring_figures.py`, in the same style; their closed-form panels are recomputed at build time and their measured panels read the curated CSVs under `research/experiment1-rotating-ring/data/`. All figures

are vector PDF with white background; randomness, where used, is seeded.

Figure	File	Source of the numbers
7.1	<code>fig_band_fill.pdf</code>	closed forms, Ch. 7
7.2	<code>fig_tridiag_loss.pdf</code>	closed forms, Ch. 7
7.3	<code>fig_local_vs_full.pdf</code>	closed forms, Ch. 7
8.1	<code>fig_laplace_k1.pdf</code>	quadrature, Ch. 8
8.2	<code>fig_laplace_closure.pdf</code>	nongaussian-bp exp. 02 CSV
B.1	<code>fig_bulk_variance.pdf</code>	closed forms, App. B
B.2	<code>fig_bp_vs_amp.pdf</code>	closed forms + iteration, App. B
9.1	<code>fig_ring_model.pdf</code>	closed forms + simulation, Ch. 9
9.2	<code>fig_ring_joint_vs_marginal.pdf</code>	schematic, Ch. 9
9.3	<code>fig_ring_gauge.pdf</code>	closed forms + simulation, Ch. 9
9.4	<code>fig_ring_surrogate_response.pdf</code>	exact surrogate score, Ch. 9
9.7	<code>fig_ring_diagnostics.pdf</code>	experiment1-rotating-ring CSVs
9.5	<code>fig_ring_score_field.pdf</code>	polar-grid recursion, Ch. 9
9.6	<code>fig_ring_taylor.pdf</code>	closed forms + CSVs, Ch. 9

C.4 Environment

Python ≥ 3.10 with NumPy, SciPy, and Matplotlib; no GPU, no neural network framework, no stochastic training. All experiments are deterministic given the recorded seeds; the error measures of the locality and truncation studies are seed-free by construction (closed-form traces). The document compiles with a single run of `tectonic main.tex`.

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